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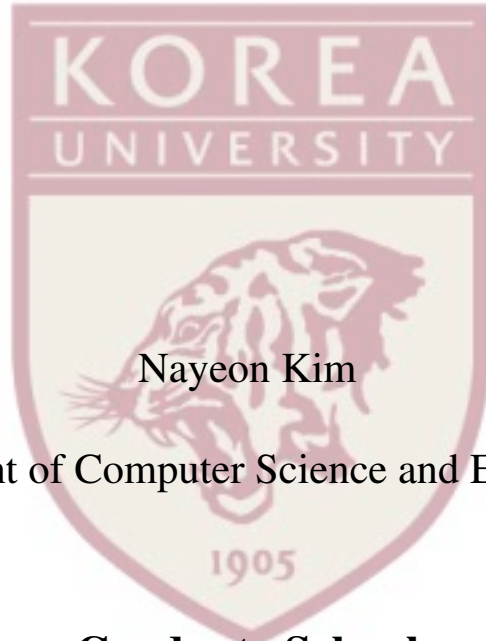
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Doctoral Dissertation

**Incorporating Linguistic Knowledge into
Korean Natural Language Processing**



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Department of Computer Science and Engineering

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Korea University

August 2026

Incorporating Linguistic Knowledge into Korean Natural Language Processing

by
Nayeon Kim

under the supervision of Professor SangKeun Lee

A dissertation submitted in partial fulfillment of
the requirements for the degree of
Doctor of Philosophy

Department of Computer Science and Engineering

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The dissertation of Nayeon Kim has been approved
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Incorporating Linguistic Knowledge into Korean Natural Language Processing

by Nayeon Kim

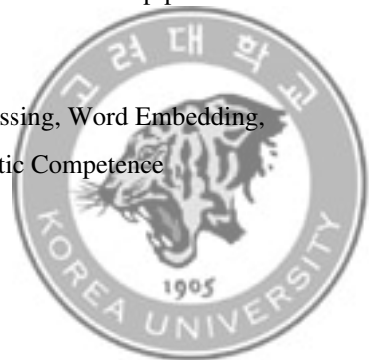
Department of Computer Science and Engineering
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Abstract

The field of natural language processing (NLP) has achieved remarkable progress with the advent of large language models (LLMs). Looking back at the history of NLP, the role of linguistics has been indispensable, and this remains true for the development of language-specific NLP systems. This dissertation discusses a series of Korean NLP studies based on linguistic grounding, which we define as the systematic incorporation of the unique linguistic knowledge of a given language into NLP system design. We argue that systematically leveraging the unique linguistic structure of Korean is essential for building better Korean NLP systems. Specifically, we present three lines of research on linguistically grounded Korean NLP.

First, we propose novel subword units derived from the creation principles of Hangeul. Word representations based on these units consistently outperform existing methods across both intrinsic and extrinsic tasks. Second, we propose a framework for out-of-vocabulary (OOV) word representation learning that leverages the phoneme information of Korean words. Through experiments, we demonstrate that OOV representations incorporating phoneme information effectively improve performance on downstream tasks. Finally, to evaluate the Korean linguistic competence of large language models, we construct a Korean grammar evaluation benchmark organized around theoretical linguistic categories. Through evaluation of a diverse set of publicly available LLMs, we reveal that current models struggle with grammar tasks requiring experiential knowledge such as pronunciation and contextual understanding. Through these three lines of research, we demonstrate that linguistic knowledge provides practical and consistent benefits across the full pipeline of Korean NLP.

Keywords: Linguistic Knowledge, Korean Natural Language Processing, Word Embedding, Out-of-Vocabulary words, Large Language Models, Evaluating Linguistic Competence



언어학 지식 기반의 한국어 자연어처리

김 나 연

컴퓨터학과

지도교수: 이상근

국문 초록

최근 자연어처리 분야는 대규모 언어 모델의 등장과 함께 눈부신 발전을 이루고 있다. 이러한 발전 역사에서 언어학은 중요한 역할을 하였다. 수많은 자연어처리 태스크와 방법론이 언어학 지식을 근간으로 했으며, 개별 언어에 특화된 자연어처리 시스템 개발에도 언어학 지식 기반의 접근이 이루어지고 있다. 본 학위 논문에서는 언어학 지식을 자연어처리 시스템 설계에 체계적으로 반영하는 언어학 기반의 한국어 자연어처리에 대해 논의한다. 한국어의 고유한 언어 구조를 체계적으로 활용하는 것은 더 나은 한국어 자연어처리 시스템 개발에 필수적이다. 구체적으로, 우리는 한국어 고유 지식을 활용한 세 가지 연구를 제시한다.

첫째, 훈민정음의 창제 원리에 기반한 새로운 서브워드 단위를 제안한다. 새로운 서브워드 기반의 단어 표현은 내재적 및 외재적 태스크 전반에서 기존 방법론을 일관되게 능가한다. 둘째, 한국어 단어와 음소 간의 상관 관계를 활용한 미등록어 표현 학습 프레임워크를 제안한다. 또한, 음소 정보를 결합한 미등록어 표현이 다운스트림 태스크의 성능 향상에 효과가 있음을 실험을 통해 입증한다. 마지막으로, 대규모 언어모델의 한국어 언어 능력 평가를 위해, 이론언어학 범주를 기반으로 한 한국어 문법 평가 벤치마크를 구축한다. 현재 공개된 다양한 대규모 언어모델 평가를 통해, 최신 모델들이 발음이나 맥락 등의 경험적 지식을 요구하는 문법 태스크를 수행하는데 어려움이 있음을 밝힌다. 이러한 연구들은 언어학 지식이 한국어 자연어처리의 전 과정에 걸쳐 실질적이고 일관된 이점을 제공함을 입증한다.

중심어: 언어학 지식, 한국어 자연어처리, 단어 임베딩, 미등록어, 대규모 언어 모델, 언어 능력 평가



Dedicated to the memory of my grandmother, 정분남 (Jeong BunNam).
I know you are proud of me from there.



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Over the past five years, he has taught and supported me in countless ways. In truth, my doctoral journey may have been one long effort to fully understand what he was telling me. His vocabulary and mine were worlds apart. To a student like me, who majored in the humanities, his words often felt sharp and cold, and honestly, there were many times when I cried. As I entered my final semester, I came to see that his words were not shards of glass, but jewels shining with love for his students. I hold deep respect for his dedication, passion, expertise, and leadership.

I would like to carry everything he taught me for the rest of my life. Among his many lessons, what I remember most is that he showed me how to live strategically, and he taught me that effort and diligence can sometimes surpass even innate talent. I am profoundly grateful that he waited for me with patience as I struggled to absorb and internalize what I was learning, and that he gave me courage every time I wanted to give up. I will engrave his teachings into my DNA and carry them with me throughout my academic journey. It is my greatest regret that the only words I can find to express what I feel are *gratitude* and *admiration*. I wish I could make explicit what remains only implicit in my heart.



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Now, I turn to those who have been part of my personal life. I am deeply grateful to Prof. Deokho Kim, who has guided my life in ways distinct from my academic advisor. He taught me that it is only when we let go of everything that we become genuinely full, and that insight has proven valuable in every area of my life. He is the most humble and compassionate person I have ever encountered, and it saddens me deeply that he is retiring. Prof. Kilim Nam first introduced me to the computational approach to linguistics during my undergraduate years. Her advice to focus on the fundamentals has always proven right. Prof. Miae Ahn consistently has never failed to demonstrate excellence and professionalism, and she generously shared invaluable advice when I was struggling to write this dissertation.

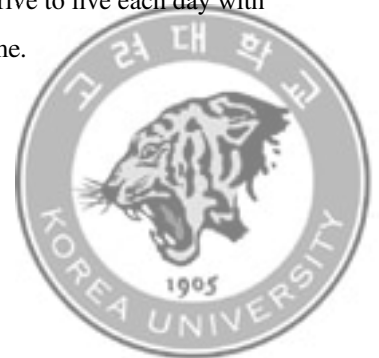
To all my friends, especially the members of the Animal Farm group — Youngmi (the sole human among us), Jihyun (Cow), Jeonghyun (Rabbit), and Yujin (Touille) — we share so many memories, and you have always encouraged me. Even when we cannot meet often, you are never far from my thoughts. Jina, your calm presence has always tempered my impatience. I am always rooting for the path you walk with your plants! Hyeon, you have been a strong supporter throughout the long and difficult years of my doctoral journey, and the Christmas party you held last year was one I will not forget. Jinyeong, Junhyung, Hyeonjin, and Cheolhui, thank you for cheering me on despite my many shortcomings. Seuli and Garam, the time we spent dreaming the same dreams, worrying, laughing, and talking through our twenties together is something I will carry with me long after. Finally, I send infinite gratitude and love to my godmother, Veronica Jeong-ok Kwon, who has always prayed for me.



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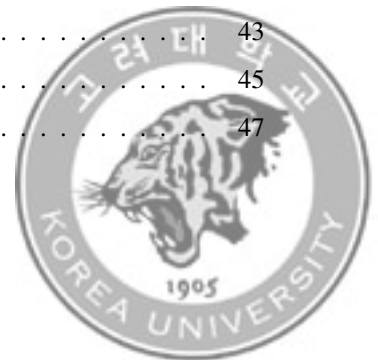


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Chapter 1

Introduction

How can we build better Korean natural language processing (NLP) systems? This dissertation addresses this fundamental question in Korean NLP. We argue that the answer lies in linguistic grounding, which is defined as the systematic integration of the unique linguistic properties of a given language into NLP system design. Linguistic grounding can improve the performance of NLP systems we build. Since the fundamental goal of NLP is to handle language, understanding language itself should be the starting point. This is especially true for Korean, whose linguistic properties differ fundamentally from English. This dissertation demonstrates this argument through three lines of research, covering word representation grounded in the creation principles of Hangeul, handling out-of-vocabulary word with phoneme representations, and the evaluation of linguistic competence in language models through the lens of Korean linguistic structure.

1.1 Motivation

NLP is a field of artificial intelligence (AI) that makes machines understand human language and carry out natural language tasks. Recently, there have been significant strides in NLP, driven by the emergence of Large Language Models (LLMs). Despite being trained solely to predict the next token, LLMs have demonstrated surprisingly broad task coverage, spanning reasoning, summarization, question answering, story generation, and code generation. Behind this progress, it



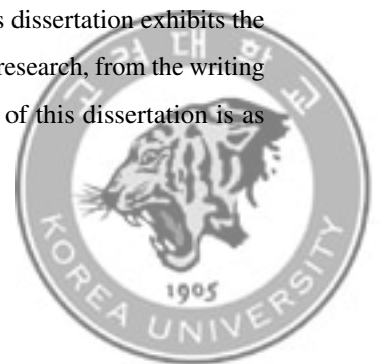
is undeniable that linguistics has played a crucial role in bringing the field to where it is today. Traditional NLP tasks including dependency parsing (DP), part-of-speech tagging (POS), and machine translation [1], as well as probing methods and evaluation benchmarks, all reflect the deep contributions of linguistic knowledge to NLP. Although current LLMs perform various natural language tasks without explicit linguistic modeling, this does not mean that the role of linguistics in NLP is no longer relevant. Language itself is the fundamental object of NLP, and developing systems for language requires not merely the pursuit of performance gains, but a genuine understanding of language through linguistic knowledge.

Moreover, language understanding of machines should not be limited to a single major language such as English, but should encompass the diverse languages of humanity. Yet, most existing NLP systems have been developed with English as the primary target language, which leads to clear limitations in multilingual understanding [2, 3]. Languages differ fundamentally in many aspects, from their typology ranging from agglutinative to inflectional, to their writing systems and phonological properties. These differences should be reflected in NLP system design, and prior works across various languages have consistently shown that approaches using language-specific properties outperform methods naively adapting English-centric methods [4, 5, 6, 7].

Such efforts exist for Korean as well, including morphology-aware tokenizers [8, 9], word embedding models [10, 11], and linguistically motivated evaluation benchmarks [12, 13], all of which have shown better results than simply using methods that are focused on English and trying to adapt them to other languages. However, systematic efforts to ground Korean NLP systems in the linguistic properties of Korean, from its writing system to its linguistic structure, remain limited. For this reason, this dissertation aims to build better Korean NLP systems by demonstrating that incorporating linguistic knowledge leads to consistent and meaningful improvements.

1.2 Outline

To build better Korean NLP systems with linguistic knowledge, this dissertation exhibits the importance of linguistic grounding as a framework through three lines of research, from the writing system to linguistic properties and systematic evaluation. The outline of this dissertation is as follows.



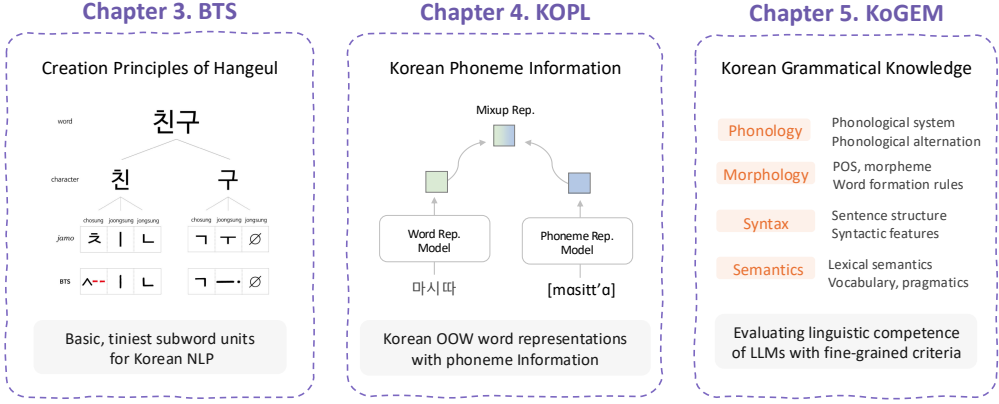


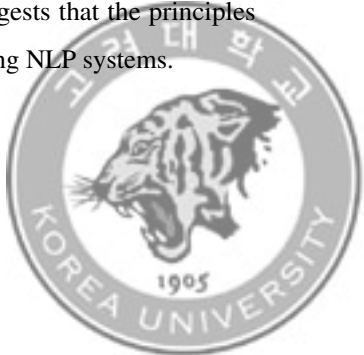
Figure 1.1: Overview of Dissertation. In this dissertation, Korean linguistic knowledge expands from its writing system to phonemic properties to linguistic structure, and is applied to word representation, Out-of-Vocabulary word handling, and language model evaluation, respectively.

Chapter 2. Foundations of Korean Language

This chapter provides the foundations of the Korean language for this dissertation. We describe the core linguistic characteristics of Korean that motivate the linguistic grounding approach taken in this work.

Chapter 3. Subword Informed Word Representations for Korean

In this chapter, we introduce BTS (Basic, Tiniest Subword) units, a novel subword decomposition derived from these creation principles. Hangeul, the Korean writing system, was created with unique principles. This background gives us a chance to design NLP systems based on the structure of the writing system. While prior work on Korean word embeddings had decomposed words into jamo letters, we take a step further by examining the underlying design of Hangeul. Our results show that using BTS units to represent words is more effective than previous methods, capturing both semantic and syntactic characteristics of Korean words. This suggests that the principles behind a writing system can serve as a promising foundation for designing NLP systems.



Chapter 4. Phoneme-Guided Out-of-Vocabulary Word Representations for Korean

In this chapter, we leverage the phonemic properties of Korean in the development of out-of-vocabulary (OOV) word processing systems. While prior approaches treat out-of-vocabulary words purely as subword sequences, we propose KOPL (Korean OOV words with Phoneme representation Learning), a framework that explicitly models the relationship between orthography and phonological realization. KOPL not only demonstrates consistent performance improvements across five Korean downstream tasks, but can also be readily combined with diverse word representation models. This study highlights the significance of phonological knowledge as an underexplored resource in Korean NLP.

Chapter 5. Evaluating Linguistic Competence of Large Language Models in Korean

In this chapter, we evaluate the linguistic competence of Large Language Models (LLMs) by applying knowledge of the linguistic structure of Korean. Linguistic competence refers to the underlying principles and fundamental knowledge of a language. Exploring the linguistic competence of LLMs can potentially provide important guidance for improving their ability to perform language tasks. However, given the implicit nature of linguistic competence itself, we employ grammatical knowledge as an explicit, measurable proxy. In doing so, we construct KoGEM (Korean Grammar Evaluation Benchmark), a benchmark grounded in theoretical linguistics, designed to systematically assess the grammatical knowledge of LLMs, and we evaluate 27 LLMs against human performance. Our findings reveal that current LLMs fall short of human performance in language areas that depend on experiential grounding such as actual pronunciation and contextual interactions.

1.3 Contributions

This dissertation argues that linguistically grounded, language-specific approaches are not merely engineering optimizations; they are principled and necessary for achieving robust natural language processing. The main contributions of this dissertation are summarized as follows:



- We introduce BTS, subword units for Korean word representations based on the creation principles of Hangeul, and empirically demonstrate consistent performance improvements across diverse Korean NLP tasks.
- We propose KOPL, a framework for Korean OOV word representation that leverages the phonemic properties of Korean, achieving consistent improvements across Korean downstream tasks.
- We construct KoGEM, a benchmark organized around theoretical linguistic categories, and reveal substantial performance gaps between LLMs and humans, demonstrating the necessity of granular evaluation frameworks for Korean.



Related Publications.

The dissertation is heavily based on my following works (**Equal Contributions*):

- **Nayeon Kim***, Jun-Hyung Park*, Joon-Young Choi, Eojin Jeon, Youjin Kang, and SangKeun Lee. "Break It Down into BTS: Basic, Tiniest Subword Units for Korean." In: *Proceedings of EMNLP*. 2022. — In Chapter 3
- **Nayeon Kim***, Eojin Jeon*, Jun-Hyung Park, and SangKeun Lee. "Handling Korean Out-of-Vocabulary Words with Phoneme Representation Learning." In: *Proceedings of PAKDD*. 2025. — In Chapter 4
- SungHo Kim*, **Nayeon Kim***, Taehee Jeon, and SangKeun Lee. "Polishing Every Facet of the GEM: Testing Linguistic Competence of LLMs and Humans in Korean." In: *Proceedings of ACL*. 2025. — In Chapter 5

Claude was used for English translation, grammatical refinement, and language polishing in this dissertation.



Chapter 2

Foundations of Korean Language

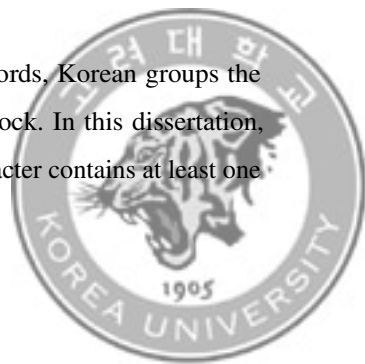
This chapter outlines the background necessary for understanding this dissertation. We describe basic knowledge of the Korean language. Specifically, we provide the Korean writing system (Section 2.1) and the linguistic structure of Korean (Section 2.2).

2.1 Writing System

Prior to the Middle Ages, Korean was written in Classical Chinese, as no dedicated writing system existed for the language. Hangeul was created by the fourth king of the Joseon Dynasty, King Sejong (1397-1450), as a solution to this problem. Hangeul was completed in 1443, and *Hunminjeongeum*, a book documenting the principles underlying Hangeul, appeared in 1446. In contrast to most writing systems, whose origins and design principles are not fully documented, those of Hangeul are thoroughly recorded in *Hunminjeongeum*. It is worth noting that *Hunminjeongeum* also serves as the original name for the Hangeul script itself.

2.1.1 Syllable Block

Unlike English, where letters are arranged sequentially to form words, Korean groups the consonants and vowels of a syllable together into a single syllable block. In this dissertation, syllable blocks are treated as equivalent to characters. Each Korean character contains at least one



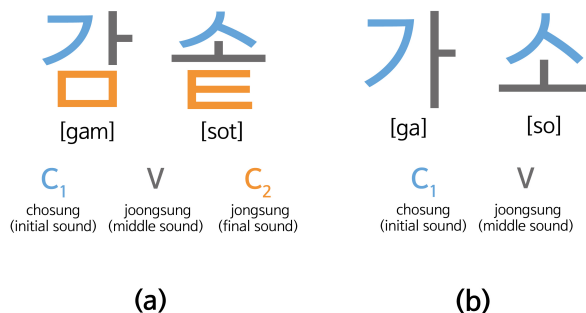


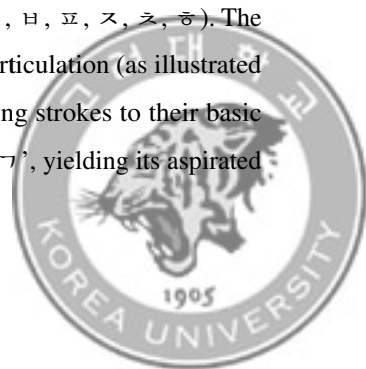
Figure 2.1: Structure of Korean characters. C_1 and C_2 denote consonants and V denotes a vowel, corresponding to chosing (first sound), joongsung (middle sound), and jongsung (final sound), respectively. As illustrated in (a), a Korean character generally adheres to the C_1VC_2 structure, though as shown in (b), jongsung is not mandatory.

consonant and one vowel, and syllable blocks generally follow the C_1VC_2 order [14], where C_1 and C_2 represent consonants and V represents a vowel. Each slot in this structure has a dedicated linguistic label: C_1 is the initial consonant, known as *chosing*; V is the medial vowel, known as *joongsung*; and C_2 is the final consonant, known as *jongsung*. *Joongsung* appears either to the right of or below *chosing*, and *jongsung*, when present, is placed underneath both *chosing* and *joongsung*. As shown in Figure 2.1 (b), *jongsung* is not a required component of a Korean character.

2.1.2 Consonants and Vowels

Korean uses the term ‘*jaeum*’ for consonants and ‘*moeum*’ for vowels, which together are collectively referred to as ‘*jamo*’. Below, we describe how each type of jamo is constructed according to the foundational principles of Hangeul.

Consonant extension via stroke addition. There are fourteen consonants in Korean, organized into five basic (ㄱ, ㄴ, ㄷ, ㄹ, ㅁ) and nine extended forms (ㅋ, ㆁ, ㆅ, ㆆ, ㆊ, ㆋ, ㆌ, ㆍ, ㆎ). The basic consonants take their shapes from the vocal organs used in their articulation (as illustrated in Figure 3.1 (a)), while the extended consonants are obtained by adding strokes to their basic forms. For examples, ‘ㅋ [k^h]’ is produced by adding a single stroke to ‘ㄱ’, yielding its aspirated



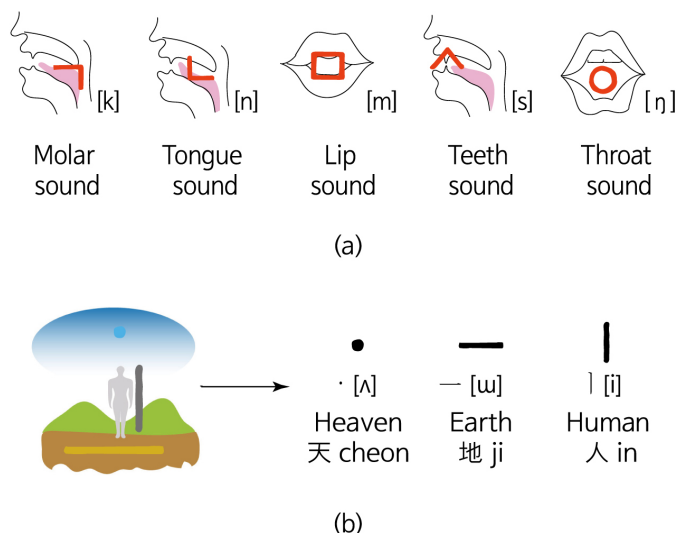
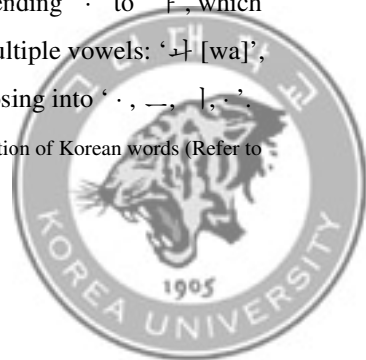


Figure 2.2: Creation principles of the basic letters of Hangeul. (a) The five basic consonants were each shaped to reflect the articulation posture of the vocal organs producing them. (b) The three basic vowels represent a round sky, a flat earth, and an upright human figure.

counterpart. Figure 3.1 (b) shows how the remaining consonants are derived from their basic forms. When identical consonants are combined, double consonants result (e.g., ㅅ + ㅅ → ㅆ), whereas combining two different consonants produces consonant clusters (e.g., ㄷ + ㅈ → ㄸ). Both types are used in character construction.

Vowel creation via combination. Hangeul vowels are built from three basic vowels (·, —, l), each carrying a symbolic meaning: the round sky ‘· [ʌ]’, the flat ground ‘—[u]’, and the standing human ‘l [i]’¹, rendered in Sino-Korean as ‘*cheon*’ (天_{heaven}), ‘*ji*’ (地_{earth}), and ‘*in*’ (人_{human}). See Figure 2.2. Further vowels are obtained by placing ‘·’ above, below, or to either side of the elemental vowels, producing ‘ㅏ, ㅑ, ㅓ, ㅕ’, and adding another ‘·’ yields ‘ㅗ, ㅛ, ㅜ, ㅠ’ [15]. For instance, ‘ㅑ’ decomposes into ‘ㅓ’ and ‘·’, since it results from appending ‘·’ to ‘ㅓ’, which itself consists of ‘ㅓ’ and ‘·’. Compound vowels arise from joining multiple vowels: ‘ㅗㅓ [wa]’, for example, is formed by combining ‘ㅏ’ and ‘ㅓ’, ultimately decomposing into ‘·, —, l, ·’.

¹We adopt International Phonetic Alphabet (IPA) notation to transcribe the pronunciation of Korean words (Refer to <https://www.internationalphoneticassociation.org/>).



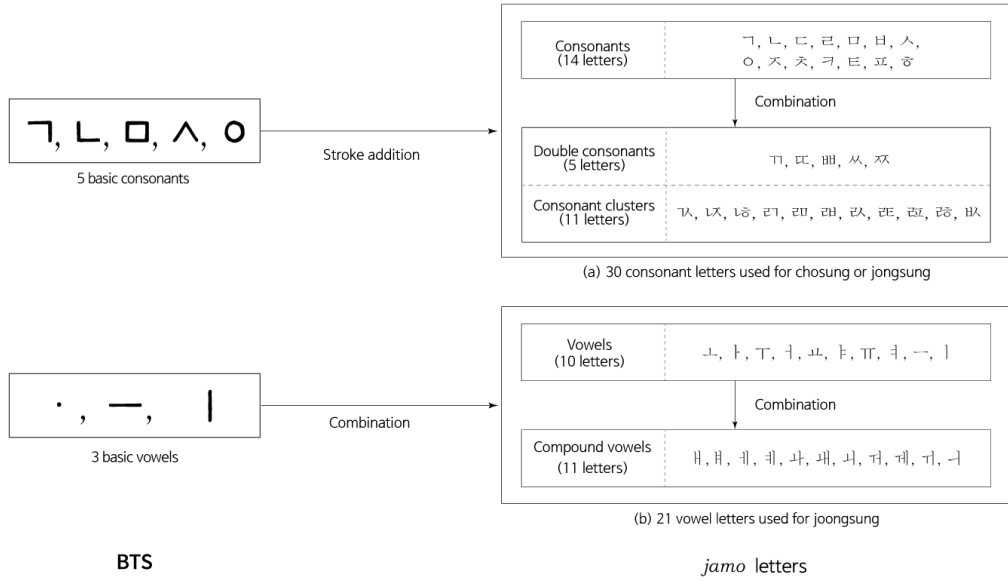


Figure 2.3: The 51 jamo letters constituting Korean characters, comprising (a) 30 consonant letters and (b) 21 vowel letters. BTS units account for all jamo letters through stroke addition or letter combination.

Consonant and vowel letters. Hangeul consists of a total of 51 jamo letters², of which 30 are consonants and 21 are vowels. The 30 consonant letters include 5 basic consonants, 9 extended consonants, 5 double consonants, and 11 consonant clusters (all formed by combining basic and extended consonants). Of these, all basic consonants, extended consonants, and double consonants can be used as *chosung*, initial sound. For *jongsung*, final sound, a total of 27 consonants are permitted, including all basic consonants, extended consonants, two double consonants (‘ㄲ’ and ‘ㄸ’), and all consonant clusters. The 21 vowel letters comprise 10 vowels formed by combining basic vowels and 11 compound vowels, all of which may serve as *joongsung*, the middle sound. All letters are presented in Figure 2.3

²Jamo is defined phonologically and distinguished from the corresponding letters used in writing. We indicate them as ‘jamo letters’ in this dissertation.



2.2 Linguistic Structure

Linguistics is the academic discipline devoted to the study of natural language, and it is divided into subfields according to the aspect of language under investigation. When examining a language, the first aspect to consider is its structure: how words are ordered, how consonants and vowels are organized, and how inflection operates, among other examples. The fields that deal with such linguistic structure, namely phonology, morphology, and syntax, constitute the core of linguistics. Beyond these, semantics and pragmatics further examine how meaning is constructed and interpreted in context. We briefly introduce each of these subfields, highlighting the distinctive characteristics of Korean relevant to the NLP tasks addressed in this dissertation.

Phonology. Phonology investigates the sound system of a language, governing the distribution and alternation of consonants and vowels. The consonant and vowel inventory of Korean differs substantially from that of English, and Korean is characterized by a rich array of phonological alternation phenomena. Representative examples include nasalization, lateralization, and palatalization. For instance, 국물_{soup} is orthographically represented as /국물/ but is pronounced as [궁물] due to nasalization, and 신라_{Silla} is written as /신라/ but realized as [실라] through lateralization. Such alternations create a systematic divergence between written and spoken form, posing important challenges for Korean NLP systems that must bridge orthographic and phonological representations.

Morphology. Morphology examines how words are internally structured and how they are formed. As an agglutinative language, Korean exhibits highly complex morphological structure, with a single root combining with a wide range of endings and affixes to produce rich morphological variation. Korean morphemes fall into two broad classes: lexical morphemes, which bear core semantic content (e.g., “먹_{-eat}”, “꽃_{flower}”) and grammatical morphemes, which fulfill syntactic and functional roles. Grammatical morphemes such as particles (e.g., “은_{is}”, “에_{at}”) and verbal endings (e.g., “-다_{verb ending}”) are particularly well-developed in Korean. For example, “학교에_{at}서_{school}” is formed by attaching the locative particle “에_{at}” to the noun “학교_{school}”, marking its role as a locative in the sentence. This morphological complexity poses significant



challenges for obtaining OOV word representations.

Syntax. Syntax investigates sentence structure and the principles by which words combine to produce grammatically well-formed sentences. Korean sentences adopt the subject-object-verb (SOV) order, with particles marking the grammatical role of each sentence constituent rather than relying on fixed word order. Beyond the basic subject, object, and predicate structure, Korean sentences may include a range of adverbials and modifiers whose ordering is relatively flexible. Furthermore, subjects and objects are frequently omitted when they can be inferred from context, making contextual understanding an essential component of syntactic analysis in Korean.

Semantics and Pragmatics. Semantics is the theory of meaning in language, investigating how the meanings of words and sentences are shaped. Lexical semantics, a representative subfield, focuses on semantic relationships among words, such as hypernymy and hyponymy. Pragmatics extends this inquiry beyond linguistic form to examine how meaning is shaped by context, speaker intention, and the relationship between speakers.



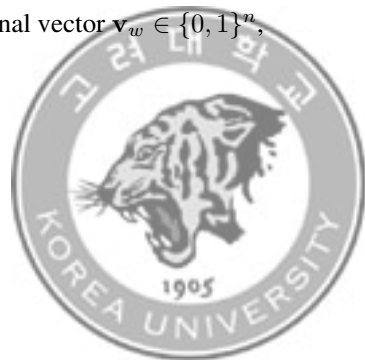
Chapter 3

Subword Informed Word Representations for Korean

3.1 Overview

“Mama”, “papa”, “no”, “go”, and so on. Words are the first units of language that humans speak. Language acquisition begins with individual words, gradually expanding to phrases and sentences, developing the capacity to understand and produce increasingly complex linguistic structures. This trajectory of human language acquisition resembles the development of NLP systems. The remarkable capabilities of current LLMs were made possible by the success of word representations. This chapter focuses on word representations for Korean, and before proceeding, we first clarify what *word representation* is.

What is Word Representation? For a computer to process language, words should be converted into a numerical format. Early approaches employ one-hot-vectors to represent individual words. Given a vocabulary of size n , each word w is assigned an n -dimensional vector $\mathbf{v}_w \in \{0, 1\}^n$,



defined as:

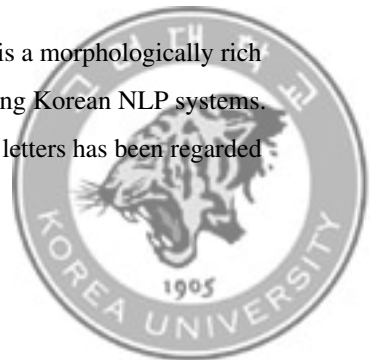
$$[\mathbf{v}_w]_i = \begin{cases} 1 & \text{if } i = i_w \\ 0 & \text{otherwise} \end{cases} \quad (3.1)$$

where i_w denotes the index of word w in the vocabulary. For example, given a vocabulary of $\{\textit{apple}, \textit{orange}, \textit{car}\}$, the word *orange* would be represented as $[0, 1, 0]$. However, this representation cannot encode any semantic relationship between words. Despite “apple” and “orange” being semantically related, their one-hot vectors are entirely orthogonal, making it hard to reflect the degree of relatedness between them.

To address this limitation, the concept of distributed representation was introduced. Rather than encoding the meaning of a word as a single scalar, each word is embedded into a continuous vector space such that semantically similar words are placed closer together. Word2Vec [16, 17] is one of the most influential instantiations of this approach. Based on the distributional hypothesis [18], which holds that the meaning of a word is determined by the words surrounding it, Word2Vec acquires dense vector representations for individual words through training on large-scale corpora. As a result, well-encoded word vectors place semantically similar words such as “apple” and “orange” close together in the embedding space, while semantically distant words such as “apple” and “car” are placed far apart.

Why is Subword Information Needed? Although distributed word embeddings [16, 17, 19] demonstrated remarkable effectiveness across various NLP tasks, treating each word form as the smallest unit proved to be a fundamental limitation, as these approaches lack any awareness of the internal structure of words. To solve this problem, many approaches have considered the internal structure of words with subword information [20, 21, 22, 23, 24]. In addition, for non-English languages with distinct linguistic structures, such as Chinese and Korean, prior studies have demonstrated the effectiveness of word embeddings that incorporate language-specific subword information [25, 26].

How We Obtain Better Word Representations for Korean Korean is a morphologically rich and an agglutinative language [27], posing distinct challenges for building Korean NLP systems. Within Korean word embedding methods, decomposing words into jamo letters has been regarded



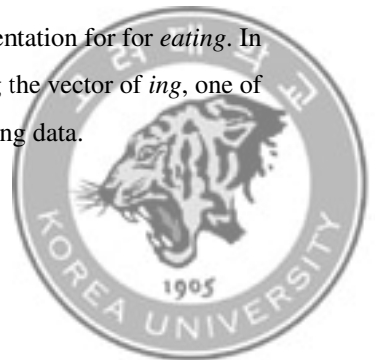
as the most effective approach for encoding the syntactic and semantic properties of Korean words, as empirically demonstrated by [25]. However, prior work has disregarded the theoretical foundations of Hangeul, the Korean writing system.

As described in Chapter 2, Hangeul is built upon five basic consonants (ㄱ, ㄴ, ㄷ, ㄹ, ㅎ) and three basic vowels (ㅏ, ㅗ, ㅜ), from which all other letters are systematically derived through stroke addition or combination [15]. This design principle, though widely acknowledged in Korean linguistics, remains underexplored in the context of NLP. To bridge this gap, we introduce **Basic, Tiniest Subword (BTS)** units, novel subword units grounded in the creation principles of Hangeul. BTS units capture the fundamental consonants and vowels of Hangeul and cover all jamo letters either directly or through systematic expansion, forming the basis of a simple yet effective word embedding framework.

In Chapter 3, Section 3.2 briefly review prior works on subword representations. Section 3.3 introduces BTS, our proposed subword decomposition method grounded in the creation principles of Hangeul, covering BTS-level decomposition, n-gram extraction, and the resulting word representation framework. Section 3.4 covers the experimental settings and implementation details, while Section 3.5 reports the results and analysis alongside a discussion of limitations. The chapter concludes in Section 3.6.

3.2 Related Work

Subword-based Word Embeddings As mentioned in Section 3.1, word-level embedding approaches have limitations to handle out-of-vocabulary (OOV) words especially in morphologically rich languages. To address this, subword-based approaches have emerged as a key strategy for handling the OOV words. A representative example is FastText [22], which represents each word as a bag of character n-grams and computes the word vector as the sum of its n-gram vectors. For instance, if the training data contains {*eat*, *get*, *getting*} and *eating* appears for the first time in a downstream task, a word-level embedding would have no vector representation for *eating*. In contrast, FastText can construct a representation for *eating* by leveraging the vector of *ing*, one of the trigrams extracted from *getting* (i.e., *get*, *eti*, *titi*, *tin*, *ing*) in the training data.



Subsequently, a number of studies proposed word embeddings that incorporate language-specific knowledge for individual languages. In particular, research on Chinese and Japanese has explored the use of the unique writing systems of these languages as subword information. For Chinese, various approaches have been proposed that reflect the visual and structural properties of characters, including embeddings based on radical information [28], stroke n-grams in *cw2vec* [29], and glyph-based representations [30, 26]. For Japanese, methods leveraging the sub-components of characters have also been proposed [31]. Collectively, these studies demonstrate that incorporating the unique linguistic structures of individual languages meaningfully contributes to improving the quality of word representations.

Subword Information in Korean NLP Korean is an agglutinative language with a rich morphological structure, in which a single root can combine with a wide variety of affixes to produce new word forms. This gives rise to a large number of surface forms sharing the same root, which leads to severe data sparsity problems for word-level embeddings. A line of research has therefore explored subword-based embeddings that reflect the linguistic characteristics of Korean. Choi et al. [32] explored Korean word representations using syllable-level information, where a syllable corresponds to a single Korean character. Park et al. [10] extended the syllable-based approach by incorporating jamo-level information within each syllable, achieving enriched word representations compared to word- and syllable-level approaches.

While these studies effectively leverage the structural elements of Korean words, they do not attend to the systematic letter construction rooted in the creation principles of Hangeul as documented in *Hunminjeongeum*. Hangeul is a writing system built upon systematic principles, in which five basic consonants and three basic vowels give rise to all other letters through stroke addition or combination. Despite being well recognized in the Korean linguistics community, these creation principles remain largely unexplored in Korean NLP. This study aims to fill that gap by proposing to decompose Korean words into Basic, Tiniest Subword (BTS) units grounded in the creation principles of Hangeul.



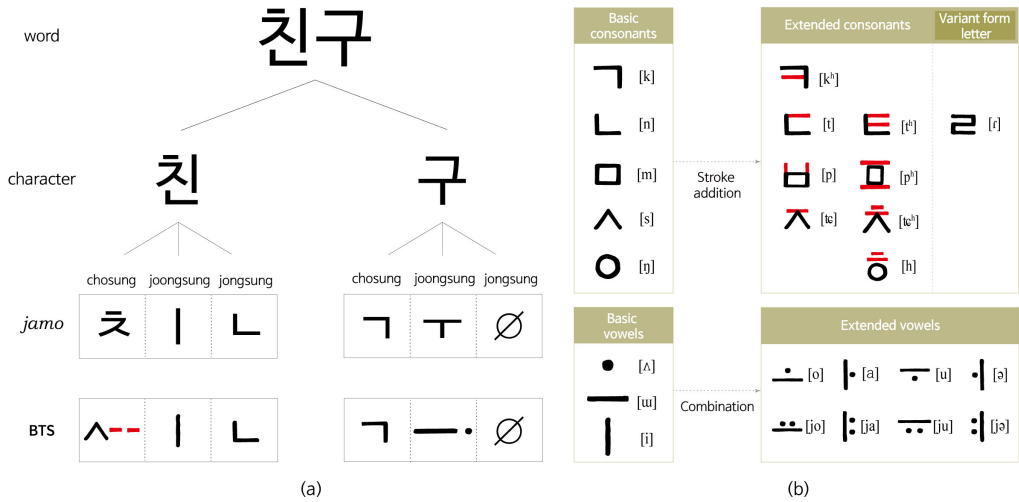


Figure 3.1: (a) Decomposition of the Korean word “친구_{friend}” at the character-, jamo-, and BTS levels, with each jamo letter resolved into its underlying basic consonants and vowels. (b) The two principles underlying Hangeul letter construction: **stroke addition** for consonants and **combinations** for vowels.

3.3 Methodology: BTS

In this section, we introduce **BTS** (Basic, Tiniest Subword) units, a novel subword unit for better Korean word embeddings.¹ Prior work on Korean word embeddings has treated jamo letters as the smallest meaningful unit of decomposition [10]. However, from a linguistic perspective, we ask: *is jamo truly the smallest unit of Hangeul?* According to *Hunminjeongeum*, Hangeul is built from eight basic units, from which all other jamo letters are derived through stroke addition or combination. This suggests that decomposing Korean words into these basic units could yield richer and more linguistically grounded word representations.

¹Note that in this dissertation, the term “subword” broadly refers to units smaller than a word, including subcharacter-level units such as **BTS** units.



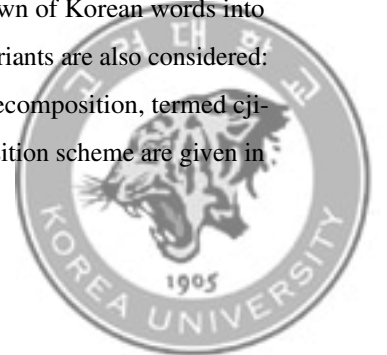
3.3.1 BTS-Level Decomposition

We now describe how BTS units are constructed. Drawing on the creation principles of Hangeul, we decompose Korean words into BTS units. These units are built around eight atomic elements (ㄱ, ㄴ, ㄷ, ㄹ, ㅁ, ㅂ, ㅅ, ㅇ), each of which is defined as an atomic unit in *Hunminjeongeum*.

Consonants. Hangeul consonants are organized into five basic forms (ㄱ, ㄴ, ㄷ, ㄹ, ㅁ) and nine extended forms (ㅋ, ㆁ, ㆅ, ㆆ, ㆊ, ㆋ, ㆌ, ㆍ, ㆎ), as illustrated in Figure 3.1 (b). The extended consonants are systematically constructed from their basic counterparts through stroke addition: a single additional stroke produces first-level extensions (ㅋ, ㆁ, ㆅ, ㆆ), while two additional strokes yield second-level extensions (ㆊ, ㆋ, ㆌ, ㆍ), with each level corresponding to a progressively stronger degree of articulation [15]. In BTS-level decomposition, stroke addition is encoded using a dedicated separator ‘-’, so that, for instance, ‘ㅋ’ is represented as ‘ㄱ-’. For *chosung* and *jongsung* positions, 30 consonant letters are employed, each written either as one of the 14 base consonants or as a combination of multiple consonants. One exception is ‘ㆅ’: although it is historically linked to ‘ㄴ’, it does not pattern as a stronger articulation thereof [15], and we therefore treat it as a basic consonant in this work.

Vowels. Hangeul vowels originate from three elemental vowels, which carry symbolic meanings rooted in Sino-Korean pronunciation: *cheon* (天, heaven), *ji* (地, earth), and *in* (人, human). From these three elements, first-level vowels (ㅏ, ㅑ, ㅓ, ㅕ) are constructed through pairwise combination, and second-level vowels (ㅗ, ㅛ, ㅜ, ㅠ) arise from further layering. To illustrate, ‘ㅑ’ breaks down into ‘ㅏ’ and ‘ㅓ’; since ‘ㅏ’ itself resolves into ‘ㅗ’ and ‘ㅓ’, the full BTS representation of ‘ㅑ’ is ‘ㅗㅓㅓ’ [15]. It should be noted that the basic vowel ‘ㅓ’ is distinct from the stroke separator ‘—’ introduced for consonants. For the *joongsung* position, 21 vowel letters are available, each expressed using 10 vowels (excluding ‘ㅓ’) either in isolation or in combination.

In this dissertation, BTS-level decomposition refers to the breakdown of Korean words into basic consonants, stroke separators, and basic vowels. Two additional variants are also considered: consonant-only decomposition, termed stroke-level, and vowel-only decomposition, termed cji-level (short for *cheon*, *ji*, *in*). Representative examples of each decomposition scheme are given in



Decomposition Level	Subword Sequence
word	친구
character	친, 구
jamo [25]	ㅈ, ㅅ, ㅅ, ㅍ, ㅍ
stroke (ours)	ㅈ, ㅅ, ㅅ, ㅍ, ㅍ, ㅍ, ㅍ
cji (ours)	ㅈ, ㅅ, ㅅ, ㅍ, ㅍ, ㅍ, ㅍ
BTS (ours)	ㅈ, ㅅ, ㅅ, ㅍ, ㅍ, ㅍ, ㅍ, ㅍ

Table 3.1: Examples of subword decomposition for the word ‘친구_{friend}’ sorted by level of decomposition units

Table 3.1, and a complete mapping of all 51 jamo letters to their BTS representations is provided in Appendix A.1.

3.3.2 BTS-Level N-gram Extraction

BTS-level decomposition enables n-gram extraction from Korean words at five distinct levels: BTS-, stroke-, cji-, jamo-, and character-level. Subword n-grams across these levels are subsequently aggregated to construct word representations. In contrast to the jamo-level decomposition of [25], which guarantees exactly three jamo letters per character, BTS-level decomposition produces variable-length sequences per character. For example, ‘가’ yields the trigram ‘ㅏ ㅓ ㅓ’, whereas ‘ㅋ’ yields the 4-gram ‘ㅏ ㅓ ㅓ ㅓ’. To mark character boundaries explicitly, a character delimiter ‘/’ is appended at the end of each character, and word boundaries are indicated by ‘<’ and ‘>’ at the beginning and end of each word. As an concrete example, the BTS-level subword sequence of ‘친구_{friend}’ is as follows:

$$\{<, ㅈ, ㅅ, ㅅ, ㅍ, ㅍ, ㅍ, ㅍ, /, >\}.$$

BTS-level n-grams. BTS-level subword n-grams are extracted directly from the decomposed sequence above. For example, the BTS-level trigrams of ‘친구_{friend}’ are extracted as follows:

$$\begin{aligned} &\{<, ㅈ, ㅅ\}, \{ㅈ, ㅅ, ㅅ\}, \{ㅅ, ㅅ, ㅍ\}, \{ㅅ, ㅍ, ㅍ\}, \\ &\{ㅍ, ㅍ, ㅍ\}, \{ㅍ, ㅍ, ㅍ\}, \{ㅍ, ㅍ, ㅍ\}, \{ㅍ, ㅍ, ㅍ\}, \\ &\{ㅍ, ㅍ, ㅍ\}, \{ㅍ, ㅍ, ㅍ\}. \end{aligned}$$



$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_{t+j}|w_t). \quad (3.2)$$

Although $p(w_{t+j}|w_t)$ can be formulated as a softmax function, computing exact softmax outputs is computationally intractable. We therefore adopt negative sampling [17], which approximates the log probability using the following binary logistic loss:

$$\log(1 + e^{-s(w_t, w_{t+j})}) + \sum_{n \in \mathcal{N}_{t,j}} \log(1 + e^{s(w_t, n)}), \quad (3.3)$$

where $\mathcal{N}_{t,j}$ denotes the set of negative samples, and s is a scoring function that maps each word pair to a scalar value. In the Skip-gram model, each word w is assigned a unique vector, and the score $s(w_t, w_{t+j})$ is computed as the dot product between the input vector of w_t and the output vector of w_{t+j} .

Building on this, the subword information Skip-gram model [22] represents a word as the average of the vectors of its constituent n-grams. The scoring function s is accordingly defined as:

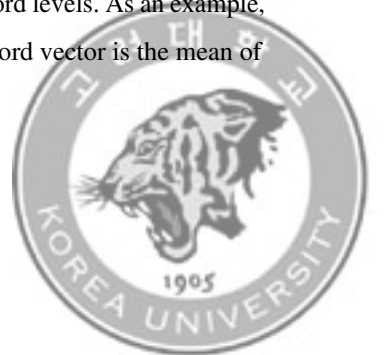
$$s(w_t, w_{t+j}) = \frac{1}{|G_t|} \sum_{g_t \in G_t} \mathbf{z}_{g_t}^\top \mathbf{v}_{t+j}, \quad (3.4)$$

where G_t is the set of n-grams extracted from word w_t , $\mathbf{z}_{g_t}$ is the vector corresponding to each n-gram $g_t \in G_t$, and $\mathbf{v}_{t+j}$ is the output vector of word w_{t+j} .

For each Korean word, we extract n-grams at the character, jamo, and one of the stroke-, cji-, or BTS-levels. The resulting word representation is obtained by averaging the vectors of the extracted n-grams. To handle varying n-gram level combinations across different models, scoring function s is generalized as:

$$s(w_t, w_{t+j}) = \frac{1}{|U_t|} \sum_{g_t \in U_t} \mathbf{z}_{g_t}^\top \mathbf{v}_{t+j}. \quad (3.5)$$

Here, U_t refers to the set of all n-grams drawn from the selected subword levels. As an example, when character- and BTS-level n-grams are jointly used, the resulting word vector is the mean of whole word, character-, and BTS-level n-gram vectors.



3.4 Experiments

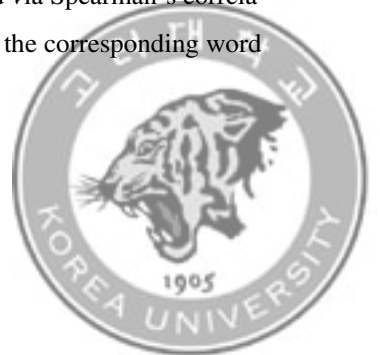
In this section, we describe the evaluation tasks and experimental setup. Detailed information on the evaluation datasets is provided in Appendix A.2

3.4.1 Evaluation Tasks

In line with [10], we assess SISG_{BTS}, its variants, and the baselines on word analogy and word similarity as intrinsic tasks, and sentiment analysis as an extrinsic task. The intrinsic tasks directly assess the semantic and syntactic properties of the learned representations, while the extrinsic task examines whether the representational improvements translate into gains on a downstream Korean NLP task. To ensure reliability, we average the results over five runs with distinct random seeds.

(1) Word Analogy. We assess the semantic and syntactic properties of word vectors using the Korean word analogy dataset [10], where each question follows the form A is to B as C is to D , with D as the target word. For example, a semantic question might take the form “서울_{Seoul} is to 한국_{Korea} as 도쿄_{Tokyo} is to X ”, where the expected answer X is 일본_{Japan}. Following [10], we measure the distance between predicted and target vectors using 3COSADD and report results at the category level, including averages over semantic and syntactic categories as well as the overall average. Furthermore, during our experiments, we observe that syntactic evaluation results exhibit overfitting, showing trends inconsistent with those of other tasks. As a result, selecting the optimal n -gram range based on syntactic performance may lead to a skewed model selection. To address this, we select the optimal hyperparameter setting based solely on semantic performance, specifically the setting that achieves the lowest average distance across the five semantic categories.

(2) Word Similarity. We assess pre-trained word representations on the Korean WS-353 dataset [33], a collection of word pairs annotated with human similarity scores on a scale of 0 to 10 (e.g., 호랑이_{tiger} and 고양이_{cat} = 7.17). Performance is measured via Spearman’s correlation between the human similarity scores and the cosine similarities of the corresponding word representations, with higher coefficients indicating greater alignment.



(3) Sentiment Analysis. Sentiment analysis requires models to accurately capture the semantic properties of words, making it a natural choice for evaluating the quality of word representations. To evaluate the efficacy of our word vectors on downstream task, we evaluate them on the Naver Sentiment Movie Corpus (NSMC) ², reporting accuracy, precision, recall, and F1 score. We train a single-layer LSTM with 300 hidden dimensions and a sigmoid activation function applied to the final hidden state. Training is conducted for 10 epochs with a dropout rate of 0.5, a batch size of 32, cross-entropy loss, and the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and 0.001 learning rate. The embedding layer is initialized with our trained word vectors and kept frozen throughout training.

3.4.2 Experimental Setup

(1) Training Corpus We construct training corpus from three sources to ensure rich and diverse domain coverage: 1) 2020 newspaper corpus provided by the MODU corpus³, 2) Korean Wikipedia ⁴, and 3) 21st century Sejong corpus [34]. Sentences from all three sources are randomly mixed to form the final training corpus, which comprises 19M sentences and 1.13M unique vocabulary words.

(2) Models. We evaluate three proposed Subword Information Skip-Gram (SISG) models, each corresponding to a different level of BTS decomposition: **stroke-level**, **cji-level**, and **BTS-level**, referred to as $SISG_{\text{stroke}}$, $SISG_{\text{cji}}$, and $SISG_{\text{BTS}}$, respectively. Their performance is compared against those of the baseline models. We further experiment with models that combine subword n-grams across multiple levels, specifically by integrating jamo- or character-level n-grams with one of **stroke-**, **cji-**, or **BTS-level** n-grams.

(3) Baseline. We adopt the word-level Skip-Gram (SG) model [16, 17] and character- and jamo-level SISG models ($SISG_{\text{ch}}$ and $SISG_{\text{jm}}$, respectively) [22, 10] as our baselines. We additionally include $SISG_{\text{ch4+jm}}$ and $SISG_{\text{ch6+jm}}$ from [10], which are trained on combined character- and jamo-level n-grams. As our training corpus differs from that of prior work, all baselines are re-

²<https://github.com/e9t/nsmc>

³<https://corpus.korean.go.kr/>

⁴<https://dumps.wikimedia.org/kowiki/>



implemented and evaluated using code released by [10]. We note that several bugs in the original code have been identified and corrected.

(4) N-gram range. Baseline models adopt the character- and jamo-level n-gram settings of [10], while the optimal n-gram range for our proposed models is determined via a search over values from 2 or 3 to 20 for each task. Detailed results for each hyperparameter setting are provided in Appendix A.3.

(5) Hyperparameter settings. All models are trained with 300-dimensional embeddings over 5 epochs, using a context window of size 5, a learning rate of 0.025 and 5 negative samples. Words appearing fewer than 10 times in the corpus are filtered out, and subsampling [17] is applied with a threshold of $t = 10^{-4}$. N-grams are mapped to integers in the range $[1, 10^7]$ using the hash function from [22].

3.5 Results and Discussion

In this section, we present the experimental results and analysis, and discuss the effectiveness of the proposed BTS units in Section 3.5.1 and 3.5.2. We further discuss the limitations of this work and directions for future research in Section 3.5.3.

3.5.1 Effectiveness of BTS units Across Tasks

(1) Word Analogy

The word analogy results are presented in Table 3.2. As shown in the upper half of the table, SISG_{BTS} outperforms all baseline models across every category, demonstrating that BTS units are effective in capturing the linguistic information of Korean words. $\text{SISG}_{\text{stroke}}$ and SISG_{cji} also surpass all baselines, yet SISG_{BTS} achieves the greatest improvements in most categories. This suggests that BTS-level consonants and BTS-level vowels each contribute complementary information, and that their combination yields the best overall performance.

We further examine whether combining BTS units with other-level n-grams brings additional gains. As shown in the lower half of Table 3.2, the combined models perform comparably to



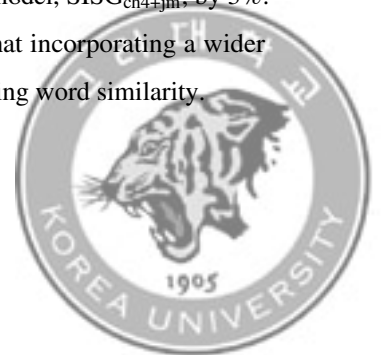
Model	Semantic					Syntactic					Sem.	Syn.	Avg.
	Capt	Gend	Name	Lang	Misc	Case	Tense	Voice	Form	Honr	Avg.	Avg.	
SG	0.463	0.531	0.585	0.435	0.644	0.533	0.612	0.543	0.677	0.538	0.532	0.581	0.556
SISG _{ch}	0.417	0.460	0.554	0.374	0.561	0.234	0.472	0.456	0.545	0.357	0.473	0.413	0.443
SISG _{jnm}	0.413	0.430	0.510	0.346	0.557	0.164	0.351	0.364	0.415	0.297	0.451	0.318	0.385
SISG _{ch4+jnm}	0.402	0.432	0.506	0.337	0.556	0.152	0.346	0.361	0.404	0.294	0.447	0.311	0.379
SISG _{ch6+jnm}	0.404	0.430	0.502	0.337	0.556	0.151	0.345	0.364	0.400	0.295	0.446	0.311	0.378
SISG _{stroke}	0.347	0.368	0.448	0.309	0.481	0.154	0.324	0.352	0.380	0.260	0.391	0.294	0.342
SISG _{cji}	0.347	0.368	0.447	0.312	0.485	0.156	0.321	0.355	0.374	0.268	0.392	0.295	0.343
SISG _{BTS}	0.342	0.360	0.440	0.306	0.473	0.151	0.319	0.348	0.370	0.267	0.384	0.291	0.338
SISG _{jnm+stroke}	0.343	0.360	0.446	0.303	0.476	0.151	0.316	0.351	0.363	0.256	0.386	0.287	0.337
SISG _{jnm+cji}	0.350	0.383	0.444	0.312	0.497	0.152	0.319	0.349	0.387	0.265	0.397	0.294	0.346
SISG _{jnm+BTS}	0.339	0.362	0.443	0.305	0.475	0.153	0.327	0.355	0.369	0.264	0.385	0.294	0.339
SISG _{ch4+stroke}	0.346	0.358	0.451	0.303	0.477	0.153	0.319	0.352	0.372	0.260	0.387	0.291	0.339
SISG _{ch4+cji}	0.352	0.382	0.445	0.316	0.502	0.153	0.322	0.347	0.389	0.266	0.399	0.296	0.347
SISG _{ch4+BTS}	0.346	0.389	0.444	0.311	0.499	0.159	0.338	0.358	0.407	0.269	0.398	0.306	0.352
SISG _{ch6+stroke}	0.348	0.381	0.447	0.309	0.492	0.153	0.328	0.352	0.391	0.266	0.395	0.298	0.347
SISG _{ch6+cji}	0.349	0.376	0.452	0.312	0.486	0.158	0.328	0.362	0.384	0.273	0.395	0.301	0.348
SISG _{ch6+BTS}	0.348	0.372	0.447	0.307	0.490	0.146	0.314	0.337	0.378	0.253	0.393	0.286	0.339

Table 3.2: Word analogy results for baseline models and single-decomposition models (upper half) and multi-decomposition models (lower half).

or worse than SISG_{BTS} alone. This indicates that BTS units already capture the syntactic and semantic information encoded by character- and jamo-level n-grams, making additional levels redundant for this task. Regarding the baseline results, we note that our findings do not fully align with those reported in prior work [25]. For instance, unlike the prior work, we find that SISG_{ch} outperforms SG on semantic categories. We attribute this discrepancy to differences in the training corpus and bugs identified in the original code.

(2) Word Similarity

The word similarity results are presented in Table 3.3. In line with the word analogy results, SISG_{stroke}, SISG_{cji}, and SISG_{BTS} all outperform the baseline models. Notably, SISG_{cji} and SISG_{BTS} surpass the previous state-of-the-art Korean word embedding model, SISG_{ch4+jnm}, by 3%. Furthermore, SISG_{ch6+cji} achieves the best overall result, suggesting that incorporating a wider range of character-level n-grams provides additional benefits for capturing word similarity.



Model	Similarity
SG	0.591
SISG _{ch}	0.665
SISG _{jm}	0.675
SISG _{stroke}	0.703
SISG _{cji}	0.707
SISG _{BTS}	0.707
SISG _{jm+stroke}	0.708
SISG _{jm+cji}	0.703
SISG _{jm+BTS}	0.707
SISG _{ch4+jm}	0.687
SISG _{ch4+stroke}	0.704
SISG _{ch4+cji}	0.700
SISG _{ch4+BTS}	0.704
SISG _{ch6+jm}	0.684
SISG _{ch6+stroke}	0.703
SISG _{ch6+cji}	0.714
SISG _{ch6+BTS}	0.709

Table 3.3: Word similarity evaluation results for single and multiple decomposition models.

(3) Sentiment Analysis

The sentiment analysis results on the Naver Sentiment Movie Corpus (NSMC) are presented in Table 3.4. SISG_{BTS} consistently outperforms all baseline models across all evaluation metrics. Furthermore, incorporating character-level subword n-grams alongside BTS units leads to additional improvements in accuracy and F1 score. These results confirm that BTS-level decomposition not only improves intrinsic word representation quality, but also translates into tangible gains on downstream Korean NLP tasks.

3.5.2 Further Insights into BTS units

(1) N-gram Range Analysis

We analyze how the performance of SISG_{BTS} on the word analogy task varies with the subword n-gram range. The overall average performance is maximized at n-gram ranges of 2-11 and 2-15,



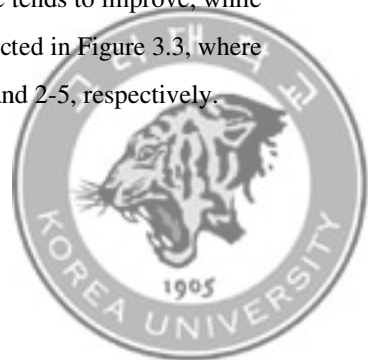
Model	Acc.	Prc.	Rec.	F1
SG	78.07	0.818	0.738	0.776
SISG _{ch}	81.03	0.876	0.732	0.797
SISG _{jm}	81.83	0.865	0.762	0.810
SISG _{ch4+jm}	81.67	0.879	0.740	0.804
SISG _{ch6+jm}	81.66	0.861	0.763	0.809
SISG _{stroke}	82.44	0.878	0.758	0.814
SISG _{cji}	82.50	0.862	0.781	0.820
SISG _{BTS}	82.18	0.843	0.798	0.820
SISG _{jm+stroke}	82.46	0.867	0.773	0.817
SISG _{jm+cji}	82.23	0.870	0.764	0.813
SISG _{jm+BTS}	82.21	0.861	0.775	0.815
SISG _{ch4+stroke}	82.64	0.869	0.773	0.818
SISG _{ch4+cji}	82.35	0.866	0.772	0.816
SISG _{ch4+BTS}	82.33	0.854	0.786	0.818
SISG _{ch6+stroke}	82.86	0.865	0.785	0.823
SISG _{ch6+cji}	82.42	0.855	0.789	0.821
SISG _{ch6+BTS}	83.80	0.857	0.819	0.838

Table 3.4: Sentiment Analysis Evaluation Results

which are notably wider than the optimal range of 3-6 for SISG_{jm}. This is attributable to the fact that BTS-level decomposition produces longer sequences than jamo-level decomposition. Since each jamo letter is represented as multiple BTS units, wider n-gram ranges are needed to capture the full representations of individual jamo letters.

Word vectors incorporating BTS bigrams consistently outperform those without them across all evaluation criteria, as shown in Figure 3.3. Given that every extended consonant and vowel letter decomposes into at least two BTS units, we conjecture that BTS bigrams provide richer information for representing individual jamo letters in the word analogy task.

Of particular interest is the divergence between semantic and syntactic analogy performance. As longer BTS n-grams are incorporated, semantic analogy performance tends to improve, while syntactic analogy performance tends to decline. This trend is clearly reflected in Figure 3.3, where the optimal n-gram ranges for semantic and syntactic analogy are 2-15 and 2-5, respectively.



Query	SG		SISG _{ch}		SISG _{jm}		SISG _{BTS}	
	NN	Eq	NN	Eq	NN	Eq	NN	Eq
케익	버터크림	✗	프랄린	✗	케익	✓	케익	✓
	버터크림을	✗	태슬	✗	케일	✗	케익을	✓
	카나페	✗	피넛	✗	케인	✗	케이크 _{cake}	✓
찐한	사랑법이	✗	틀막	✗	찐득한	✗	찐득한	✗
	애절함	✗	3MC의	✗	찐찌	✗	진한 _{strong}	✓
	애뜻하면서도	✗	잔망	✗	찐	✗	찐	✗
웬지	괜스레	✗	어쩐지	✗	어쩐지	✗	웬일	✗
	허전하고	✗	웬일인지	✗	웬지 _{for some reason}	✓	웬일이지	✗
	초조하고	✗	웬일이지	✗	웬일인지	✗	웬일이니	✗
뒤치닥거리	<i>n/a</i>	<i>n/a</i>	푸닥거리	✗	푸닥거리	✗	뒤치다꺼리 _{cover for}	✓
	<i>n/a</i>	<i>n/a</i>	복닥거리는	✗	푸닥거리를	✗	뒤치다꺼리를	✓
	<i>n/a</i>	<i>n/a</i>	노닥거리는	✗	노닥거리고	✗	푸닥거리	✗

Table 3.5: Nearest neighbor results for queries with common typos, showing the top-3 candidates from each model. ✓ in the Equiv column indicates that the query and neighbor word are semantically equivalent based on human annotation.

(2) Nearest Neighbor Words Analysis

We further analyze the unique advantages of BTS units to gain deeper insight. For a set of Korean query words containing common typos, we examine the top-3 nearest neighbors retrieved by each model and report the results in Table 3.5, along with human judgments on semantic equivalence between query and neighbor words.⁵ As the examples in Table 3.5 illustrate, BTS units exhibit a notable strength in handling both typos and Out-of-Vocabulary (OOV) words, where OOV refers to words absent from the model’s vocabulary that were never encountered during training.

Consider the query ‘케익’, a misspelling of ‘cake’. SISG_{BTS} successfully retrieves the standard form ‘케이크_{cake}’, alongside semantically equivalent variants such as ‘케익’ (an alternative misspelling) and ‘케익을’ (formed by appending the objective case marker ‘을’). The baseline models SG and SISG_{ch}, by contrast, retrieve semantically related but non-equivalent words such as ‘버터크림_{butter cream}’ and ‘프랄린_{praline}’, failing to identify the intended word.

⁵Annotations are determined by majority vote among three Korean native-speaking graduate students, excluding authors.



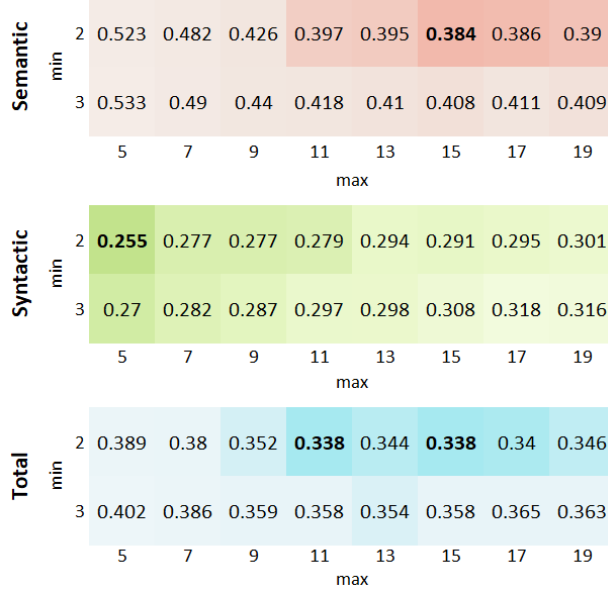


Figure 3.3: Word analogy performance of SISG_{BTS} models across subword n-gram ranges. Darker cells indicate higher performance.

BTS units also prove effective for intentional typos, a common phenomenon in Korean where speakers deliberately alter spellings for expressive emphasis. In the case of ‘쥼한’, formed by substituting ‘ㅈ’ with stronger consonant ‘ㅉ’, SISG_{BTS} correctly retrieves the standard form ‘진한_{strong}’. That said, SISG_{BTS} is not without limitations. Like other SISG models, it tends to over-prioritize surface-level character overlap. This is evident in the case of ‘웬지’, where all retrieved neighbors contain the character ‘웬’, despite the intended target ‘웬지_{for some reason}’ not containing it. Mitigating this over-reliance on overlapping characters remains an open challenge for subword-based word embeddings.

On the OOV front, SISG_{BTS} demonstrates a particularly compelling result. Given the OOV query ‘뒤치닥거리’, it successfully retrieves the standard form ‘뒤치다꺼리_{cover for}’, whereas all baseline models fail to do so. This finding points to a promising avenue for future work on BTS-based representations.



3.5.3 Limitations and Future Work

While the BTS units proposed in this dissertation demonstrate clear effectiveness for static word embeddings, their application to Transformer-based pre-trained language models remains an important open question. The two-dimensional structure of Korean characters suggests that a vanilla Transformer architecture, which processes input as a one-dimensional sequence, may not be well-suited to fully exploit the linguistic structure of Hangeul. Investigating and potentially designing the transformer architecture and tokenization units to better accommodate Korean subcharacter structure is therefore a compelling direction for future work.

At the time of writing this dissertation, Kim et al. [35] have taken a first step in this direction by proposing KOMBO, a framework that incorporates the design and combination principles of Hangeul subcharacters into pre-trained language models. In KOMBO, subcharacter-level units serve as the initial representations, which are then progressively merged into character-level representations in accordance with the combination rules of Hangeul. Their findings indicate that grounding representations in the linguistic principles of Hangeul is a promising direction not only for static word embeddings, but also for pre-trained language models, demonstrating the potential of extending the creation principles of Hangeul to transformer-based architectures.



3.6 Summary

In this chapter, we proposed **BTS** units grounded in the creation principles of Hangeul. Unlike existing approaches that treat jamo as the smallest unit of decomposition, **BTS** units go one level deeper by decomposing jamo letters into their most fundamental building blocks, namely basic consonants, stroke symbols, and basic vowels. Experiments across intrinsic and extrinsic tasks demonstrated that SISG_{BTS} consistently outperforms character- and jamo-level baselines, confirming that grounding subword representations in the creation principles of Hangeul yields richer Korean word embeddings.

Beyond quantitative gains, qualitative analysis revealed that **BTS** units are particularly effective in handling typos and OOV words. This strength stems from the fact that many common Korean typos operate at the level of basic consonants and vowels, making **BTS**-level representations a natural fit for capturing such lexical variation. Furthermore, n -gram range analysis revealed an interesting tradeoff between semantic and syntactic performance. Incorporating longer **BTS** n -grams tends to improve semantic analogy performance, while syntactic analogy performance declines. This suggests that semantic and syntactic information in Korean words are encoded at different levels of subword granularity, a finding that may inform future work on Korean word representation.



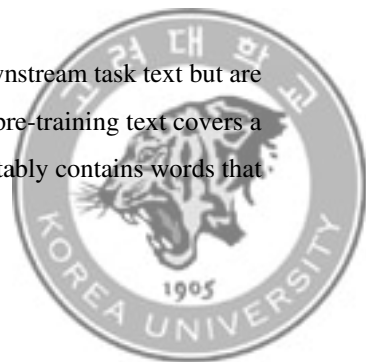
Chapter 4

Phoneme-Guided Out-of-Vocabulary Word Representations for Korean

4.1 Overview

Chapter 3 demonstrated that grounding Korean word representations in the subcharacter structure of Hangeul leads to consistent improvements in embedding quality. Such embeddings serve as input vectors for NLP systems and directly shape performance across a range of downstream tasks, including sentiment analysis, named entity recognition, and machine translation. In practice, however, NLP systems inevitably encounter words that were absent from the training corpus, commonly referred to as out-of-vocabulary (OOV) words. Since the lack of a reliable representation for such words can degrade downstream task performance, developing effective methods for OOV word handling remains an important challenge. In this chapter, we approach this problem from the perspective of Korean as a phonemic script, and propose a framework for learning high-quality OOV word representations by leveraging the phonological properties of Korean.

What is Out-of-Vocabulary? OOV refers to words that appear in downstream task text but are absent from the pre-training corpus. As illustrated in Figure 4.1, while pre-training text covers a large portion of possible text, the supervised downstream task text inevitably contains words that



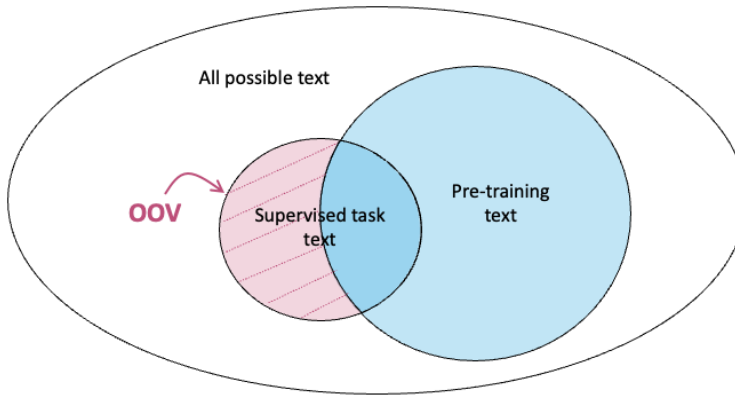
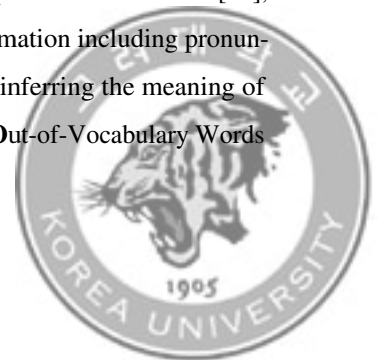


Figure 4.1: Scope of OOV words

fall outside this coverage. Since all possible text extends far beyond what any pre-training corpus can cover, such unseen words are an inevitable occurrence in real-world NLP systems.

Why Do OOV Words Raise Problems in Downstream Tasks? When a pre-trained word embedding does not contain a vector representation for such a word, the downstream model cannot process it appropriately, leading to degraded performance. For instance, as illustrated in Figure 4.2, a pre-trained embedding trained on “I love this movie” contains a representation for “love”, but lacks one for the unseen word “hate”. Consequently, when a sentence such as “I hate this movie” is given as input to a downstream task such as sentiment analysis, the model cannot adequately represent “hate”, which may negatively affect performance.

How We Handle This Problem. In this chapter, we go beyond purely subword-based approaches and explore the potential of leveraging the linguistic characteristics of Korean as a phonemic script, specifically its phonological information, for more effective OOV processing. This direction is motivated by two observations: the high correlation between letters and phonemes in Korean [36], and the psycholinguistic finding that humans draw on multimodal information including pronunciation and imagery, rather than orthographic knowledge alone, when inferring the meaning of words [37]. Building on these observations, we propose KOPL (**K**orean **O**ut-of-Vocabulary **W**ords



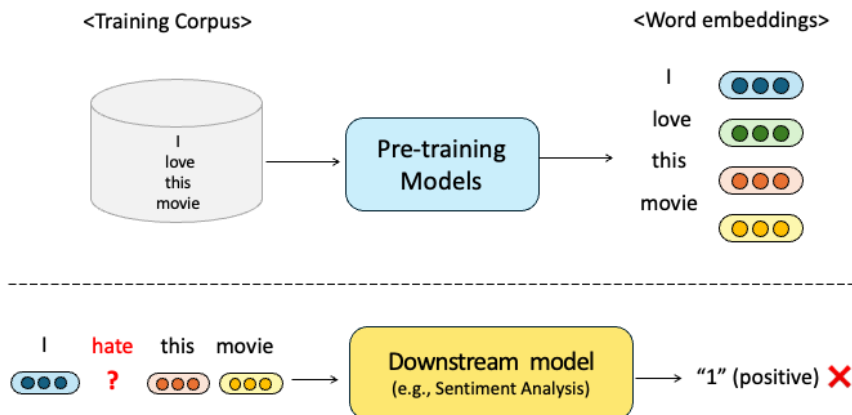


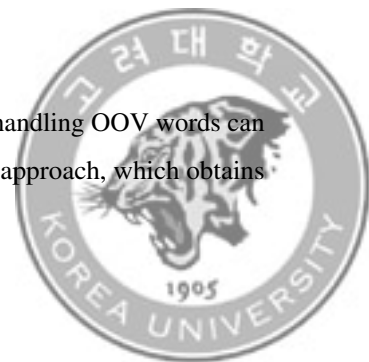
Figure 4.2: Illustration of the OOV problem in word embeddings. Suppose the training corpus contains only four words: ‘I’, ‘love’, ‘this’, and ‘movie’. At inference time on a downstream task, the word ‘hate’, which did not appear in the training corpus, has no corresponding vector representation, leading to degraded performance in downstream tasks.

with **Phoneme representation Learning**), a framework for Korean OOV word processing that explicitly incorporates phonological knowledge.

In Chapter 4, Section 4.2 reviews related work and background on OOV word representations. Section 4.3 introduces KOPL, a framework for generating OOV word representations. We present KOPL in two phases: we first describe how phoneme information is integrated into the process of deriving OOV word representations, and then propose two strategies for incorporating the resulting phoneme-enriched representations during the fine-tuning stage. Section 4.4 describes the experimental setup, and Section 4.5 presents the experimental results and analysis, along with a discussion of the limitations of this work. Finally, Section 4.6 concludes the chapter.

4.2 Related Work

Out-of-Vocabulary Word Representation. Existing approaches to handling OOV words can be broadly divided into two categories. The first is the subword-based approach, which obtains



representations by decomposing words into subword units. Representative examples include FastText [22], which learns word representations based on character n-grams, and subword embedding methods for Korean based on jamo decomposition [10, 11], both of which have demonstrated effectiveness in handling OOV words. Chapter 3 of this dissertation also falls into this category. The second approach obtains OOV word embeddings not by directly learning subword or character-level representations, but through a framework that mimics pre-trained word embeddings. MIMICK [38] trains an RNN to predict the pre-trained embedding of a word from its character sequence, learning from words in the training vocabulary and their corresponding embeddings. In short, the model learns to predict a pre-trained word embedding from the spelling of a word. LOVE [39] proposes a contrastive learning framework using pairs of original and corrupted words to generate more robust OOV representations. However, these approaches are primarily designed around English and do not explicitly leverage phonological properties for OOV word representation learning. To fill this gap, we propose KOPL, a framework for learning OOV word representations that incorporates the phonemic information of Korean words based on the mimicking approach.

Phoneme Information for NLP. Prior work on incorporating phoneme information into NLP models falls broadly into two lines. The first line aims to build models that are robust to textual perturbations. RoCBert [40] jointly pre-trains on text, phoneme, and visual information to produce a BERT model that is robust to Chinese textual perturbations, while PhonemeBERT [41] jointly models text and phoneme sequences to handle errors in ASR transcriptions. A shared limitation of these approaches is that they require pre-training dedicated Transformers on large-scale corpora from scratch, incurring considerable computational cost. The second line applies phoneme information to spelling correction. [42] and [43] incorporate phoneme cues into Chinese spelling correction systems, [44] proposes phoneme-guided pre-training for correcting Chinese character errors, and [45] explores phonological similarity for correcting errors made by language learners. Although this line of work is tangentially related to ours, since misspelled words may surface as OOV words, the underlying goal is to detect and correct spelling errors, not to learn representations for OOV words directly. In contrast, we use phoneme information specifically to improve the quality of Korean OOV word representations, without relying on large-scale pre-training.



4.3 Methodology: KOPL

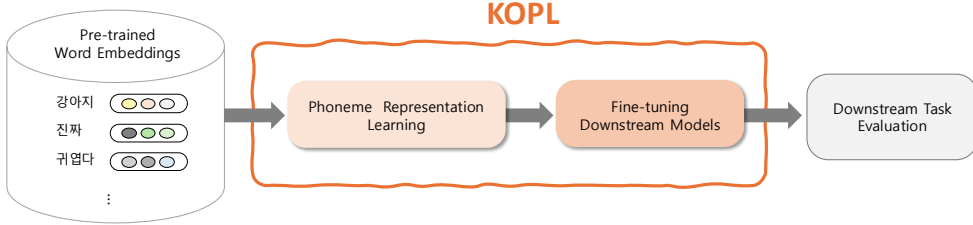


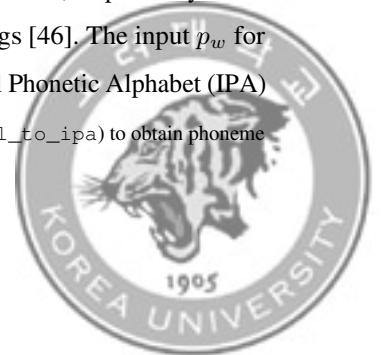
Figure 4.3: Overview of Our Framework.

In this section, we describe how phonological information is leveraged to derive representations for OOV words. As discussed in Section 4.2, KOPL follows the line of mimicking-based OOV word representation frameworks, operating between pre-trained word embeddings and downstream task models (See Figure 4.3). KOPL consists of two stages: phoneme representation learning (Section 4.3.1) and fine-tuning downstream models (Section 4.3.2). In the first stage, KOPL learns to generate word representations by leveraging phonological information, using words present in the pre-training vocabulary as training targets. In the second stage, two strategies are employed to effectively incorporate the learned phoneme representations into downstream model fine-tuning: phoneme-aware multimodal learning and cross-modal ensemble.

4.3.1 Phoneme Representation Learning

We begin by considering a vocabulary $\mathcal{V}$ of fixed size, along with a pre-trained word embedding $\mathbf{v}_w \in \mathbb{R}^d$ for each word $w \in \mathcal{V}$. Our goal is to learn representations that capture the distinct properties of both phonemes and words, while also encoding the complementary information shared between the two modalities. To this end, we jointly train two neural networks, F_P and F_W , each producing a d -dimensional representation vector for phonemes and words, respectively. Both networks are built on a self-attention module with positional embeddings [46]. The input p_w for F_P is a sequence of phonemes for word w ,¹ expressed using International Phonetic Alphabet (IPA)

¹We use the hangeul_to_ipa program (https://github.com/stannam/hangeul_to_ipa) to obtain phoneme sequence.



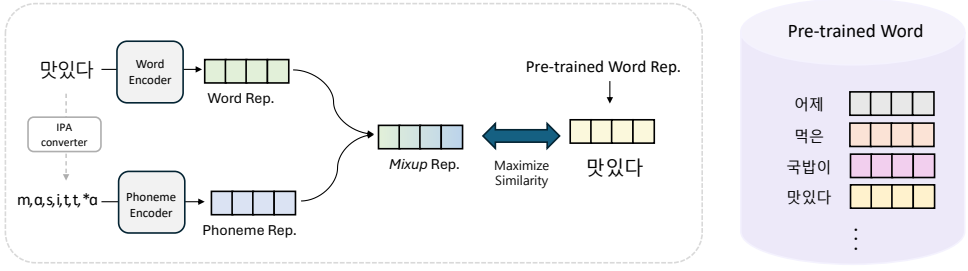


Figure 4.4: Phoneme Representation Learning of KOPL. “Rep.” refers to “representations”.

symbols [47]. F_W takes as input a mixed sequence of subcharacters and morphemes constituting w . Specifically, we adapt the input method of LOVE [39] to Korean by combining subcharacter and morpheme sequences, as detailed in Table 4.1.

To encourage the two representations to carry complementary information, we fuse them during pre-training via linear interpolation:

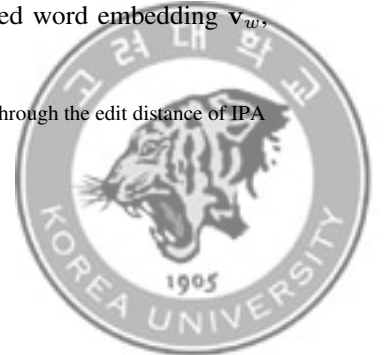
$$F(w) = \lambda F_P(p_w) + (1 - \lambda) F_W(w), \quad (4.1)$$

where $0 \leq \lambda \leq 1$ is a hyperparameter controlling the mixing ratio. Interpolation allows the model to balance the contribution of phoneme and word representations, enabling their effective combination.

This is particularly important because phoneme representations alone can be insufficient for distinguishing semantically distinct words that share similar pronunciations². For instance, “맛있다_{delicious}” and “*마싯다_{dilisious}”, both pronounced as [masitt’a], are phonetically identical yet differ in spelling. Conversely, “찾다_{find}” and “찼다_{kicked}”, both pronounced as [tɕ^hatt’a], share the same pronunciation but carry entirely different meanings. In the latter case, relying solely on phoneme representations risks conflating semantically distinct words; interpolation with word representations helps preserve these distinctions.

The fused representation F_w is then used to predict the pre-trained word embedding $\mathbf{v}_w$, training F_P and F_W jointly via the following contrastive loss [39, 48]:

²In this work, the term ‘similar pronunciation’ can be defined quantitatively, such as through the edit distance of IPA sequences or acoustic similarity of sound waves.



$$\mathcal{L}_{\text{cl}} = -\frac{1}{|\mathcal{V}|} \sum_{w \in \mathcal{V}} \log \frac{e^{s(F(w), \mathbf{v}_w)/\tau}}{e^{s(F(w), \mathbf{v}_w)/\tau} + \sum e^{s(F(w), \mathbf{v}^-)/\tau}}, \quad (4.2)$$

where $s(\cdot, \cdot)$ is a scoring function such as the dot product, τ is a temperature parameter, $(F(w), \mathbf{v}^-)$ denotes a negative sample drawn from other words in the same mini-batch. Through this process, F_P and F_W learn to produce representations that together approximate the pre-trained word embedding. Once training is complete, both phoneme and word representations can be obtained for any arbitrary word w , including those absent from the pre-training vocabulary.

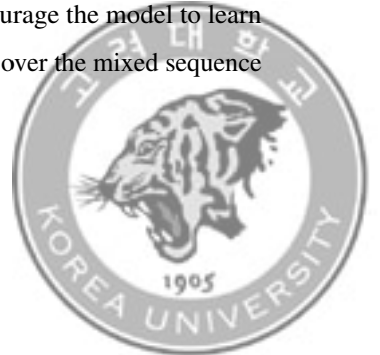
4.3.2 Fine-tuning Strategies

To effectively incorporate phoneme representations into downstream tasks, we propose two complementary fine-tuning strategies: Phoneme-aware Multimodal Learning and Cross-modal Ensemble.

Phoneme-aware Multimodal Learning Consider a downstream classification task where a sequence of words $\{w_i\}_{i=1}^n \in \mathcal{X}$ is to be mapped to a label $y \in \mathcal{Y}$, using the learned modules F_P and F_W to obtain phoneme and word representations, respectively. The core idea is to supply the downstream model with multiple types of representations as separate inputs, allowing it to learn modality-specific features from both phonemic and textual information. Concretely, we derive a sequence of phoneme representations $\mathbf{p} = \{F_P(p_{w_i})\}_{i=1}^n$ and a sequence of word representations $\mathbf{e} = \{F_W(w_i)\}_{i=1}^n$ using F_P and F_W , respectively. Let G denote the downstream model that maps a sequence of vectors to prediction scores (e.g., logits). We define two losses as follows:

$$\mathcal{L}_P = \ell(Z_P, y), \quad \mathcal{L}_W = \ell(Z_W, y), \quad (4.3)$$

where $Z_P = G(\mathbf{p})$, $Z_W = G(\mathbf{e})$, and ℓ denotes a loss function such as the cross-entropy loss. Through $\mathcal{L}_P$ and $\mathcal{L}_W$, the downstream model is encouraged to capture phonemic information from $\mathbf{p}$ and morphological information from $\mathbf{e}$, respectively. To further encourage the model to learn the connections between two modalities, we introduce an additional loss over the mixed sequence $\mathbf{m} = \{F(w_i)\}_{i=1}^n$:



$$\mathcal{L}_M = \ell(Z_M, y), \quad (4.4)$$

where $Z_M = G(\mathbf{m})$. The overall training loss is the weighted sum of the three losses:

$$\mathcal{L} = \alpha_1 \mathcal{L}_P + \alpha_2 \mathcal{L}_W + \alpha_3 \mathcal{L}_M, \quad (4.5)$$

where α_1 , α_2 , and α_3 are hyperparameters that balance the contribution of each loss term.

Cross-modal Ensemble Since the downstream model is trained with phoneme, word, and mixed representations as parallel inputs, it learns to extract complementary information from each modality. At inference time, we leverage this diversity by adopting a cross-modal ensemble strategy, aggregating the prediction scores from all three input types via a weighted average:

$$Z = \beta_1 Z_P + \beta_2 Z_W + \beta_3 Z_M, \quad (4.6)$$

where β_1 , β_2 , and β_3 are fixed weights for each prediction score.

4.4 Experiments

This section describes the experimental setup for evaluating KOPL. Section 4.4.1 introduces the Korean downstream datasets used for evaluation, and Section 4.4.2 describes the baseline models, implementation details, and the construction of OOV subsets.

4.4.1 Evaluation Tasks

To evaluate KOPL on downstream tasks, we select five Korean datasets that contain a substantial proportion of real-world OOV words: KOLD [49], KLUE-TC, KLUE-DP, KLUE-NER [12], and NSMC³. The first three are used for sentence classification, and the latter two for sequence tagging.

³<https://github.com/e9t/nsmc>



Model	Input	Example
MIMICK [38]	subcharacter	{ □, ㅏ, ㅓ, ㅇ, ㅣ, ㅕ, ㅛ, ㅜ }
LOVE [39]	subcharacter + morpheme	{ □, ㅏ, ㅓ, ㅇ, ㅣ, ㅕ, ㅛ, ㅜ } ∪ { 맛있, ##다 }
GRM [48]	morpheme	{ 맛있, ##다 }

Table 4.1: Korean-specific input methods of different baselines originally tailored to English OOV words, with the Korean word `맛있다`_{delicious} as an example.

KOLD. KOLD [49] is a publicly available offensiveness detection dataset comprising 40K examples. Following [49], we reserve 10% of the examples each for validation and test. The dataset contains 2 classes, and we report binary F1 as the evaluation metric.

KLUE-TC. KLUE-TC [12] is a topic classification dataset with 45K training, 9K validation, and 9K test examples. Since the test set labels are not publicly available, we evaluate on the official validation set and hold out 10% of the training set for validation. The dataset contains 7 classes, and we report macro F1.

NAVER Sentiment Movie Corpus (NSMC). NSMC is a sentiment analysis dataset consisting of 150K training and 50K test examples. We hold out 25K training samples for validation. The dataset contains 2 classes, and we report accuracy as the evaluation metric.

KLUE-DP. KLUE-DP [12] is a dependency parsing dataset with 10K training, 2K validation, and 2.5K test examples. As the test set labels are unavailable, we evaluate on the official validation set and hold out 10% of the training set for validation. The dataset contains 38 dependency relation classes, and we report labeled attachment score (LAS).

KLUE-NER. KLUE-NER [12] is a named entity recognition dataset comprising 21K training, 5K validation, and 5K test examples. As with KLUE-DP, the test set labels are unavailable, so we evaluate on the official validation set and hold out 10% of the training set for validation. The dataset contains 12 entity classes, and we report entity-level macro F1.

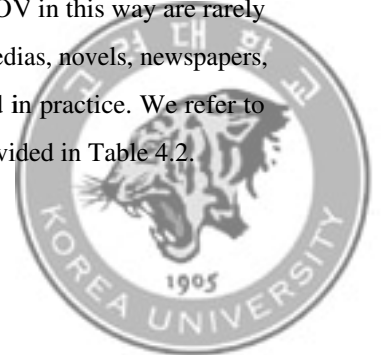


4.4.2 Experimental Setup

(1) Baselines. To ensure a comprehensive and fair comparison, we evaluate KOPL against two categories of baseline models. The first category comprises methods specifically designed for Korean OOV words, including SISG_{jm} [10], misK [50], and SISG_{BTS} [11]. The second category consists of methods originally developed for English OOV words, namely MIMICK [38], LOVE [39], and GRM [48], which we adapt to Korean by applying the language-specific input methods described in Table 4.1. We also include an additional baseline, Ensemble, which aggregates predictions from three LOVE checkpoints trained with different random seeds, to verify that the gains of KOPL stem from phoneme representations rather than the ensemble strategy alone. Further details are provided in Appendix B.1

(2) Implementation Detail. We implement F_P and F_W as 2-layer self-attention models [46] with a hidden dimension of 300. The input to F_P is a sequence of IPA symbols for each word (e.g., {m,a,s,i,t',a} for the word “맛있다_{delicious}”), while F_W takes a mixed sequence of subcharacters and morphemes, following the Korean input method of LOVE [39] detailed in Table 4.1. For the interpolation in Eq. 4.1, we set λ to 0.1. The pre-trained word embeddings from SISG_{jm} [10] serve as target embedding ($\mathbf{v}_w$ in Section 4.3.1), and the temperature for the contrastive loss in Eq. 4.2 is set to $\tau = 0.07$. We train the model for 10 epochs using AdamW as the optimizer. During downstream fine-tuning, the input vectors produced by KOPL are frozen, following [39]. For downstream model architectures, we use a 1-layer CNN for KOLD, KLUE-TC, and NSMC, a BiLSTM for KLUE-DP, and a BiLSTM with CRF [51] for KLUE-NER. We use $\alpha_1 = \alpha_2 = \alpha_3 = 1$ and $\beta_1 = \beta_2 = \beta_3 = 1/3$ during fine-tuning.

(3) OOV Subset Construction. To demonstrate the robustness of KOPL against OOV words, we construct additional evaluation sets from each dataset by selecting samples that contain at least one OOV word. A word is treated as OOV if it does not appear in the vocabulary of SISG_{jm} [10], which contains 1.13M words. As shown in [11], words identified as OOV in this way are rarely observed across a broad range of real-world corpora, including encyclopedias, novels, newspapers, and dialogue, making them representative of OOV words encountered in practice. We refer to these subsets as OOV subsets in what follows. Dataset statistics are provided in Table 4.2.



Dataset	Task	Train	Dev	Test	OOV	# Class
KOLD	Offensive Detection	32.3k	4k	4k	0.5k	2
KLUE-TC	Topic Classification	45.7k	9.1k	9.1k	2.4k	7
NSMC	Sentiment Analysis	125k	25k	50k	8.5k	2
KLUE-DP	Dependency Parsing	10k	2k	2.5k	0.5k	38
KLUE-NER	Named Entity Recognition	21k	5k	5k	1.8k	12

Table 4.2: Statistics of each dataset. ‘‘OOV’’ denotes the number of examples in the OOV subset of the test set.

4.5 Results and Discussion

Section 4.5.1 reports the experimental results, followed by Section 4.5.2, which presents several analyses to validate the effectiveness of KOPL. Finally, Section 4.5.3 concludes the section with a discussion of limitations and directions for future research.

4.5.1 Performance on Downstream Tasks

Table 4.3 presents results on the OOV subsets and their original counterparts, where KOPL demonstrates superior performance across both dataset types. Notably, KOPL consistently yields larger performance gains compared to the ensemble baseline on all datasets, suggesting that combining phoneme and word representations effectively captures the relationship between Korean letters and phonemes, thereby improving OOV word handling. The Korean adaptations of English OOV methods such as MIMICK, LOVE, and GRM generally underperform or merely match the state-of-the-art Korean OOV method SISG_{BTS} on average, highlighting the inherent difficulty of addressing Korean OOV words even with recently proposed methods originally designed for English. In contrast, KOPL achieves a substantial improvement of 2.6% over SISG_{BTS} on the OOV subsets, demonstrating that explicitly modeling the linguistic characteristics of Korean, namely the correlation between letters and phonemes, leads to more robust downstream performance on Korean OOV words.



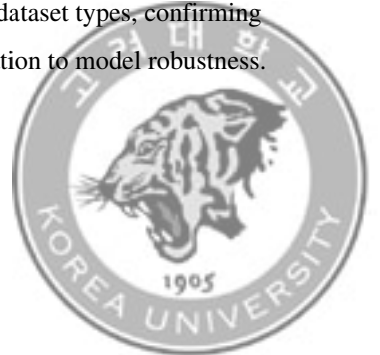
Model	KOLD (F1 $\uparrow$)		KLUE-TC (F1 $\uparrow$)		NSMC (Acc. $\uparrow$)		KLUE-DP (LAS $\uparrow$)		KLUE-NER (F1 $\uparrow$)		Avg.	
	OOV	Orig.	OOV	Orig.	OOV	Orig.	OOV	Orig.	OOV	Orig.	OOV	Orig.
SISG _{jm} [10]	70.6	76.5	77.9	79.2	78.4	83.7	78.1	81.7	76.8	80.8	76.0	80.1
misK [50]	73.1	74.5	65.8	66.1	79.2	82.7	77.9	80.2	74.3	75.5	74.0	75.8
SISG _{BTS} [11]	71.4	75.4	76.3	78.0	78.9	84.0	78.5	82.1	81.9	83.8	77.4	80.7
MIMICK [38]	64.9	70.1	51.7	51.7	69.1	74.2	77.3	80.7	69.1	73.4	66.4	70.0
LOVE [39]	72.2	75.1	73.9	75.8	78.3	83.1	81.1	83.1	82.7	83.6	77.6	80.2
GRM [48]	69.8	74.5	72.0	75.5	74.9	81.6	75.9	78.7	76.0	78.9	73.7	77.8
Ensemble	72.8	75.5	74.7	76.7	78.5	83.3	81.5	83.3	83.0	83.7	78.1	80.5
KOPL (Ours)	76.5	77.0	76.8	78.4	79.8	84.0	82.9	84.2	83.8	84.4	80.0	81.6

Table 4.3: Performances on the downstream tasks. ‘‘OOV’’ and ‘‘Orig.’’ denote the OOV subset of the dataset and its original counterpart, respectively.

4.5.2 Efficacy of KOPL

Ablation Study. As our approach fine-tunes the model with three types of input representations (phoneme, word, and mixed), we conduct an ablation study to verify the contribution of each type, with results shown in Table 4.4. Phoneme representations prove critical for handling Korean OOV words: adding them to word-only representations (row 1 to row 4) improves performance on both original datasets and OOV subsets, while removing them from the full model (row 6) causes performance degradation. Word representations are equally indispensable, as excluding them (row 5) also leads to a consistent performance drop compared to the full model (row 7).

Furthermore, omitting mixed representations (row 6) results in the largest performance degradation on most datasets, suggesting that capturing the interaction between phoneme and word representations is particularly important. This is especially relevant for Korean given its phonemic script where letters and phonemes are strongly correlated [36]. Overall, the full model using all three representations (row 7) achieves the best performance across both dataset types, confirming that each representation contributes distinct and complementary information to model robustness.

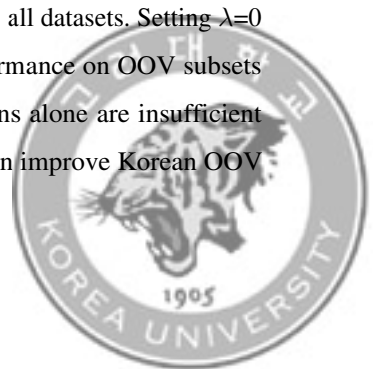


Input Representation				KOLD (F1 $\uparrow$)		KLUE-TC (F1 $\uparrow$)		NSMC (Acc. $\uparrow$)		KLUE-DP (LAS $\uparrow$)		KLUE-NER (F1 $\uparrow$)	
	Phoneme	Word	Mixed	OOV	Orig.	OOV	Orig.	OOV	Orig.	OOV	Orig.	OOV	Orig.
1	✓			68.5	70.8	29.0	27.6	66.7	69.4	75.0	78.5	54.2	60.1
2		✓		72.2	75.1	73.9	75.8	78.3	83.1	81.1	83.1	82.7	83.6
3			✓	73.7	75.9	74.6	76.5	78.5	83.4	81.3	83.2	82.9	83.7
4	✓	✓		74.2	76.0	74.0	76.0	78.2	83.1	81.1	83.2	82.9	83.7
5	✓		✓	75.5	76.5	75.3	76.9	79.1	83.4	82.4	83.7	83.4	84.0
6		✓	✓	74.9	76.2	74.6	76.6	78.5	83.4	81.5	83.3	83.2	83.8
7	✓	✓	✓	76.5	77.0	76.8	78.4	79.8	84.0	82.9	84.2	83.8	84.4

Table 4.4: Performances on the downstream tasks with different types of input representations used to fine-tune downstream models.

Model Adaptability. To demonstrate the adaptability of KOPL, we combine it with three existing Korean embedding models: SISG_{jm} [10], SISG_{BTS} [11], and $\text{BERT}_{\text{Morph.}}$ [8]. The first two are Korean variants of FastText [22] that produce static word embeddings, while the last generates contextual embeddings dynamically. The combination strategy during fine-tuning differs depending on whether the baseline uses static or contextual embeddings. For static embedding models, we use the word representations from the baseline alongside the phoneme representations from KOPL. For the contextual embedding model, we follow [39] by identifying OOV words and replacing their sub-word embeddings with those generated by KOPL. As shown in Table 4.5, KOPL consistently improves performance across all datasets regardless of the baseline, demonstrating its plug-and-play applicability to existing Korean embedding models. We attribute these gains to the explicit modelling of the correlation between Korean letters and phonemes, a linguistic property that prior Korean word embedding works [11, 10] have largely overlooked.

Mixup Ratio. To analyze the effect of the mixup ratio λ when interpolating between word and phoneme representations, we experiment with varying values of λ across all datasets. Setting $\lambda=0$ corresponding to using word representations alone, yields lower performance on OOV subsets compared to other values of λ . This indicates that word representations alone are insufficient for handling Korean OOV words, and that phoneme representations can improve Korean OOV



Model	KOLD (F1 $\uparrow$)		KLUE-TC (F1 $\uparrow$)		NSMC (Acc. $\uparrow$)		KLUE-DP (LAS $\uparrow$)		KLUE-NER (F1 $\uparrow$)		Avg.	
	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.	OOV Orig.
SISG _{jm} [10]	70.6	76.5	76.1	77.9	78.4	83.7	78.1	81.7	76.8	80.8	76.0	80.1
SISG _{jm} + KOPL	75.6	77.4	79.3	80.0	79.7	83.9	80.9	83.2	79.1	82.2	78.9	81.3
SISG _{BTS} [11]	71.4	75.4	76.3	78.0	78.9	84.0	78.5	82.1	81.9	83.8	77.4	80.7
SISG _{BTS} + KOPL	73.3	76.2	79.5	80.1	81.0	84.3	81.1	83.0	83.2	84.4	79.6	81.6
BERT _{Morph.} [8]	79.4	79.6	82.9	84.2	87.5	89.5	85.1	86.6	84.5	86.1	83.8	85.2
BERT _{Morph.} + KOPL	81.6	80.3	83.8	84.5	88.1	89.7	86.0	87.0	86.0	86.4	85.1	85.6

Table 4.5: Downstream task results when existing Korean embedding models are combined with KOPL.

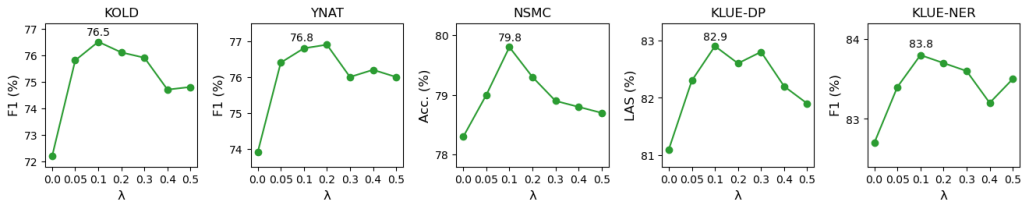


Figure 4.5: Performances on the OOV subsets of the datasets with different mixup ratio λ .

representations. On the other hand, performance tends to decline when λ exceeds 0.1. A plausible explanation is that placing greater weight on phoneme representations increases the similarity between words that share similar pronunciations but carry different meanings (e.g., “ 찾다_{find} ” and “ $\text{쫓다}_{\text{kicked}}$ ”), thereby hindering the model from capturing semantic distinctions between words.

4.5.3 Limitations and Future Work

In this section, we have shown how phoneme information can improve the quality of OOV word representations in Korean. We conclude by discussing the limitations of KOPL and directions for future research.

Language Generalization. In this work, we implement our phoneme-integrated OOV representation learning approach exclusively for Korean. While the method demonstrates consistent



performance improvements across several Korean downstream tasks, whether it generalizes to other languages remains unexplored. Although Korean letters exhibit a particularly high correlation with phonemes, this does not imply that other languages lack such correlations. Phonological information plays an important role across all languages, and exploring the multilingual generalizability of our framework is therefore an important direction for future work.

Extending to Acoustic Modalities. The present study relies solely on IPA symbols as a means of representing phonological information, which limits its consideration of other modalities such as acoustic signals. Incorporating actual speech information, such as mel spectrograms, could facilitate the learning of richer, multimodal word representations. Furthermore, given recent advances in speech language models that process text and speech in a mutually complementary manner, exploring a broader range of modalities for embedding learning represents a promising avenue for future research.

Integration with Large Language Models. Finally, integrating the proposed framework with recent LLMs remains an important direction for future research. Since KOPL is designed to be easily combined with existing word embedding models, it has the potential to enhance the OOV handling capabilities of LLMs in real-world applications, and empirical validation of this integration remains an open research question.



4.6 Summary

In this chapter, we introduced KOPL, a framework for Korean OOV word processing that leverages both the character sequence and the phoneme sequence of a word to mimic its pre-trained embedding. Existing approaches to OOV handling fall into two broad categories: subword-based pre-training methods that incorporate subword information, and mimicking-based methods that predict pre-trained word vectors from the spelling of a word. KOPL builds on the latter, extending it with phoneme information motivated by psycholinguistic theory and the phonemic nature of Korean as a writing system. KOPL outperformed existing models across five Korean downstream tasks, including both methods originally developed for English and directly applied to Korean (MIMICK, LOVE, GRM) and subword-based pre-trained embedding models (SISG_{jm}, SISG_{BTS}). The gains were particularly pronounced on the OOV subset. Ablation results further confirm that phoneme information contributes meaningfully to performance improvements.

Above all, the key takeaway from this chapter is not that KOPL is the definitive solution, but that it represents a further case in which grounding a system in language-specific linguistic knowledge leads to meaningful improvements in Korean NLP. So far, we have explored ways of directly integrating linguistic knowledge into representation models. The next chapter examines the extent to which models trained without explicit linguistic knowledge nonetheless acquire an understanding of such knowledge.



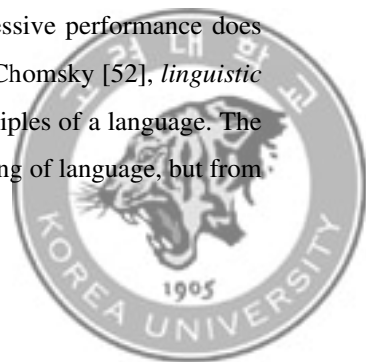
Chapter 5

Evaluating Linguistic Competence of Large Language Models in Korean

5.1 Overview

Thus far, in Chapters 3 and 4, we have explored methods for incorporating linguistic structure into word representation and OOV word processing in Korean. These approaches demonstrate that explicitly grounding models in linguistic structure leads to consistent improvements. At the same time, recent advances in contextualized representation learning and large-scale pre-training have led to the emergence of large language models (LLMs), which achieve strong performance across a wide range of tasks without explicit linguistic modeling. This raises a natural question: do LLMs that demonstrate strong language performance truly understand the underlying structure and knowledge of language?

What is Linguistic Competence? LLMs have shown remarkable performance across a wide range of natural language processing tasks. Nevertheless, such impressive performance does not necessarily guarantee strong linguistic competence. According to Chomsky [52], *linguistic competence* is defined as the ability to internalize the underlying principles of a language. The strong performance of LLMs may not stem from a genuine understanding of language, but from



probabilistic inference over large-scale training data [53].

Why Understanding Linguistic Competence of LLM is Crucial and Challenging? Investigating the linguistic competence of language models is an important endeavor for better understanding and utilizing LLMs. However, since linguistic competence is not an explicitly observable property, directly assessing it is challenging. Then, how can we evaluate the linguistic competence of LLMs and Humans? One promising adopted approach is to evaluate grammatical knowledge, as grammar serves as a measurable proxy for linguistic competence.

How We Evaluate Linguistic Competence. To evaluate linguistic competence of LLMs, we introduce KoGEM, the **K**orean **G**rammar **E**valuation **B**ench**M**ark. Since individual languages possess their own unique linguistic properties, evaluating the linguistic competence of language models should reflect the characteristics of each target language. Given that Korean is the target language of this dissertation, this chapter examines how the linguistic competence of language models can be assessed with a focus on Korean. Beyond comparing LLMs against one another, we further benchmark LLMs against native Korean speakers, providing human evaluation scores for investigating model performance.

In Chapter 5, Section 5.2 reviews how linguistic competence has been evaluated in prior work, and survey the current landscape of Korean linguistic competence and grammatical knowledge evaluation datasets. Section 5.3 describes the construction process of KoGEM in detail. Since transparency in dataset construction is essential for the reproducibility and reliability of benchmark research, we therefore provide a thorough account of the grammar taxonomy definition and data construction pipeline. Section 5.4 introduces the overall experimental settings, describing which language models are evaluated, as well as the human evaluation for comparison. The experimental results and discussions are presented in Section 5.5. Finally, after reflecting on the limitations of this work and directions for future research, we conclude the chapter in Section 5.6



5.2 Related Work

Linguistic Competence Evaluation In the field of NLP, there have been steady efforts to evaluate the linguistic knowledge of models. Probing methods have explored the linguistic information encoded in the hidden representations of pre-trained models [54, 55, 56, 57, 58]. These works have primarily focused on morphological or syntactic knowledge in English. More recent works have also investigated the linguistic competence of language models by broadening the scope to cover additional dimensions such as phonology and pragmatics [59, 60, 61]. While these probing techniques provide insights into the linguistic knowledge encoded in model representations, they have limitations in directly measuring systematic grammatical competence. In particular, since these works use English as the target language, it is difficult to apply their approach directly to Korean. Therefore, we propose a comprehensive benchmark that can systematically evaluate the Korean linguistic competence of LLMs.

Korean Grammatical Knowledge Evaluation Previous work on Korean has produced benchmarks for evaluating the linguistic knowledge of models, though these efforts have largely focused on limited aspects of linguistic knowledge, and systematic benchmarks to explore comprehensive linguistic knowledge have remained scarce. Son et al. [62] concentrated on lexical knowledge, such as identifying native Korean equivalents for loanwords, while Kim et al. [63] conducted a broader evaluation of Korean grammar. However, the scope of grammatical knowledge covered remained underspecified. Yoon et al. [64] examined Korean grammatical error types, focusing on morphosyntactic and orthographic correctness. In this work, we provide a systematic framework and benchmark for evaluating the Korean linguistic competence of LLMs, with particular emphasis on grammatical knowledge. Establishing such systematic criteria could serve as an important guide to improve the Korean linguistic capabilities of language models.

5.3 Benchmark: KoGEM

We first define the scope and types of grammar we aim to evaluate in Section 5.3.1. We then describe the dataset construction process in Section 5.3.2.



5.3.1 Taxonomy of Grammar

To evaluate the Korean linguistic competence of LLMs, we first define the taxonomy of Korean grammar. Specifically, the term *grammar* in this study is based on prescriptive grammar. Prescriptive grammar differs from descriptive grammar, which analyzes and describes actual linguistic phenomena, in that its purpose is to establish clear answers regarding correct language usage. Since descriptive grammar covers an extensive range of linguistic phenomena, we focus on prescriptive grammar to evaluate the Korean grammatical knowledge of LLMs. In addition, as prescriptive grammar provides precise answers, it has the advantage of allowing for a quantitative assessment of the linguistic competence of LLMs.

Building on this definition, we define the main categories grounded in theoretical linguistics: phonology, morphology, syntax, and semantics. For convenience, pragmatics is not treated as an independent category but is instead subsumed under semantics, given its close relationship with semantics in a broader sense. In addition to these four categories, we include norms as a separate main category. Norms encompass rules such as spelling conventions, standard language norms, and standard pronunciation, and constitute a set of guidelines for correct Korean language use. Beyond the main categories, we further define subcategories to capture more fine-grained distinctions within each domain of linguistic phenomena. The main categories and their corresponding subcategories are presented below:

- **Phonology:** Phonological System, Phonological Alternation
- **Morphology:** Part-of-Speech, Morpheme, Word Formation
- **Syntax:** Sentence Structure, Syntactic Features
- **Semantics:** Vocabulary, Lexical Semantics, Pragmatics
- **Norms:** Orthography, Standard Language, Standard Pronunciation, Loanword Orthography, Romanization, Cross-Category

Detailed descriptions of each category are as follows. First, the Phonology category consists of two subcategories. Phonological System covers the inventory of Korean phonemes, while Phonological Alternation addresses phenomena in which phonemes are inserted,



deleted, or replaced in specific phonological environments. A representative example is the word ‘국물_{soup}’, which is composed of the syllables ‘국/guk/’ and ‘물/mul/’ but is pronounced as [gung.mul] rather than /guk.mul/ due to nasalization. Such alternations are pervasive in Korean, where phonemes frequently undergo changes conditioned by their surrounding environment.

The Morphology category covers Part-of-Speech, Morpheme, and Word Formation. Notably, Korean distinguishes nine parts of speech, in contrast to the eight recognized in English (see Table C.2). Questions in the Morpheme subcategory require familiarity with morpheme types, including lexical morphemes that carry core semantic content (e.g., ‘꽃_{flower}’, ‘해_{sun}’) and functional morphemes that serve grammatical roles (e.g., ‘의_{of}’, ‘고_{and}’). The Word Formation subcategory requires knowledge of how words are constructed in Korean, including compound words (e.g., ‘식기+세척기_{dish+washer}’) and derived words (e.g., ‘씻기_{washing}’, formed from ‘씻_{wash}’ and ‘-기_{-ing}’), and the ability to identify the constituent morphemes of a given word.

The Syntax category examines the structural organization of Korean sentences, with a dedicated subcategory for syntactic features covering grammatical elements such as tense markers (e.g., the prefinal ending ‘겠_{future tense}’) and negative adverbs (e.g., ‘안_{not}’ and ‘못_{unable}’). The Semantics category probes the meanings of words, sentences, and conversational contexts. The Vocabulary subcategory targets rote knowledge of word meanings as defined in dictionaries, while Lexical Semantics assesses the ability to recognize semantic relationships between words, such as the hypernym-hyponym relation between ‘orange’ and ‘fruit’.

The Norms category encompasses Orthography, Standard Language, Standard Pronunciation, Loanword Orthography, and Romanization, collectively covering prescriptive rules for accurate language use. The Cross-Category subcategory captures questions that cut across multiple subcategories within Norms, such as those asking which option conforms to both Korean orthographic and standard language conventions simultaneously.

5.3.2 Construction Pipeline

We provide a detailed explanation of source data, data format, collection, categorization, and data statistics.



1) Source Data In constructing KoGEM, we collect Korean grammar QA pairs from four official examinations administered to native Korean speakers: (1) the College Scholastic Ability Test (CSAT); (2) the National United Achievement Test (NUAT); (3) the High School Qualification Exam (HSQE); and (4) the Civil Service Exam (CSE). These examinations were selected on the grounds that they are designed to assess linguistic competence in both academic and professional contexts, and together provide broad coverage of the Korean grammar taxonomy introduced in Section 5.3.1. Further details on the characteristics of each source exam are given in Appendix C.3.1.

2) Data Format Since prescriptive grammar centers on correctness, we formulate the task as multiple-choice QA, where each item has a single clear answer. Each QA pair comprises up to four components: `passage`, `question`, `paragraph`, and `choices`. The `passage` situates the question in context, and the `paragraph` supplies concise grammatical explanations or illustrative examples. Answer options consist of either four or five choices as shown in Figures C.4 to C.8.

3) Data Collection The source exams, publicly available in PDF format, are processed through OCR¹ to extract raw text. Three authors then manually review each grammar item and apply preprocessing steps including underlining, segmentation, and box separation in HTML format. Items whose interpretation depends on images are excluded given our text-only evaluation setting, and tables are converted into sequential text following predefined rules. HTML formatting rules applied during preprocessing are described to facilitate reproducibility.

¹<https://clova.ai/hyperclova>



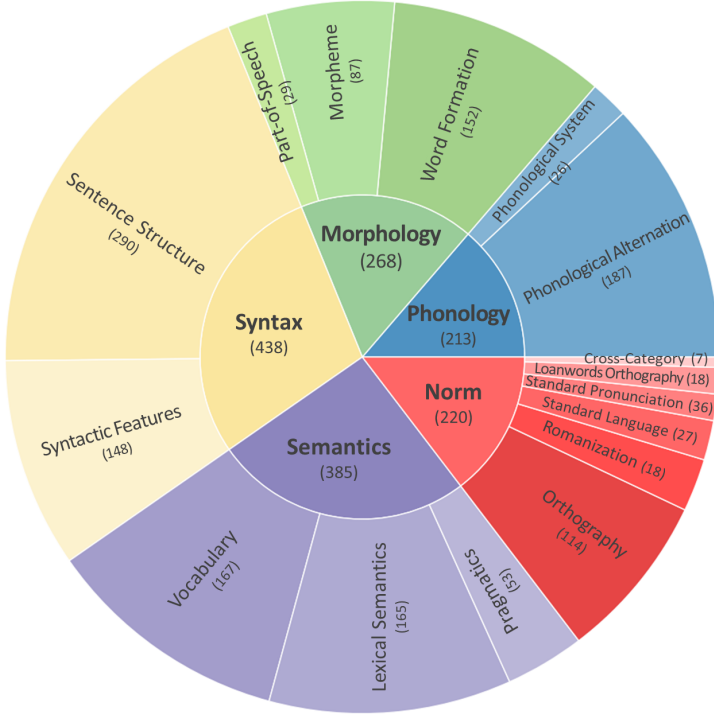
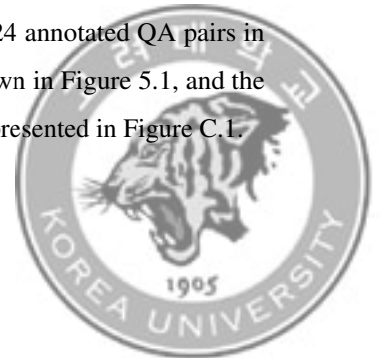


Figure 5.1: KoGEM data distribution organized into five main categories and 16 subcategories. Numbers in parentheses indicate the number of QA samples in each subcategory.

4) Data Categorization Three annotators with backgrounds in Korean language and literature independently assign each preprocessed item to one of 16 subcategories, proceeding in two steps: first determining the main linguistic category, then selecting the appropriate subcategory. Final labels are assigned based on majority vote, with any disagreements addressed through group discussion. Items spanning two categories are assigned to the one most central to the correct answer, while those that draw on more than three categories are excluded to maintain subcategory-level focus.

5) Data Statistics The resulting KoGEM benchmark comprises 1,524 annotated QA pairs in total. The distribution across main categories and subcategories is shown in Figure 5.1, and the number of samples per main category broken down by source exam is presented in Figure C.1.



5.4 Experiments

We assess the linguistic competence of LLMs with KoGEM. By conducting a comprehensive analysis of 27 LLMs, we measure their linguistic competence and compare the performance of current state-of-the-art LLMs to that of humans. Section 5.4.1 describes the baseline LLMs we evaluate, and Section 5.4.2 describes the metrics and human evaluation details.

5.4.1 Baseline Models

We evaluate a total of 27 LLMs under zero-shot and few-shot settings: 8 models known for strong Korean capabilities, including the EXAONE and HyperCLOVA X series, and 19 general LLMs spanning a broad range of architectures and scales, including GPT, Claude, Gemini, Llama, and Qwen series. Detailed descriptions of all evaluated models are provided in Appendix C.1.

5.4.2 Evaluations

Metrics. We assess LLMs on KoGEM using accuracy as the primary metric, under both zero-shot and few-shot settings. The prompt follows the same structure as the test presented to human participants, consisting of `passage`, `question`, `paragraph`, and `choices`, delivered as a single input. Each model is instructed to select the correct answer and provide a brief explanation for its choice. Further details on prompt design and experimental settings, including hyperparameters and hardware configurations, are provided in Appendix C.2.

Human Evaluation. Since we draw our source data from official Korean competency examinations, we begin by analyzing publicly available empirical data on human performance. For CSAT and NUAT, we obtain question-level average accuracy based on 10,000+ responses.² For HSQE and CSE, however, we contact the relevant institutions directly, only to find that such data are not publicly disclosed. We therefore conduct a human evaluation via a data research platform to obtain human scores for these exams.³

²The Korean private education company Megastudy has published question-level accuracy rates since 2016 (<https://www.megastudy.net/Entinfo/correctRate/main.asp>).

³Macromill Embrain is a Korean company specializing in online research (<https://embrain.com>).



To align with the distinctive nature of each examination, we employ tailored recruitment strategies for our crowdworkers. For the CSE, human evaluation involves gathering responses from active civil servants ranging from their teens to their 60s. For the HSQE, the target group comprises first-year undergraduate students who have completed the relevant curriculum within the past year. In total, we obtain reliable human scores by collecting at least 10 responses per question from 352 participants across 583 QA samples. To ensure sustained attention and minimize bias, each worker is restricted to a maximum of 20 questions, with a standard per-response compensation of approximately \$0.24.

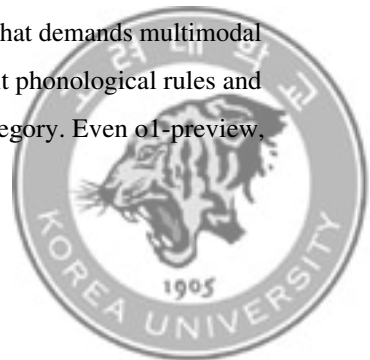
5.5 Results and Discussions

5.5.1 Overall LLM Performance in Main Categories

Zero-Shot Results. Table 5.1 presents the performance of LLMs and humans across the five main linguistic categories of KoGEM. Overall, LLM performance broadly follows the scaling law [65]. Among all evaluated models, o1-preview is the only one to surpass human performance, doing so by an average margin of 18%. All other models fall short of human-level performance: Claude-3.5-Sonnet and GPT-4o trail humans by 3.07% and 5.17%, respectively.

We further distinguish between models known for strong Korean capabilities and other LLMs. Many of the former exhibit relatively lower performance, likely attributable to their smaller sizes and less recent architectures. Notably, several models such as s1-32B, DeepSeek-R1-Distill-Qwen-32B, and Qwen2.5-32B, despite their moderate sizes, match or outperform all Korean-centric models. This may be partly explained by their multilingual training across more than 29 languages [66], which enables the transfer of shared linguistic features [67, 68, 69]. Furthermore, the strong results of o1-preview, s1-32B, and the two DeepSeek-R1 models suggest that test-time scaling [70, 71] is an effective strategy for enhancing linguistic competence in Korean.

A notable pattern emerges when comparing LLM and human performance across linguistic categories. Humans perform particularly well in Phonology, a category that demands multimodal reasoning, specifically the ability to integrate textual input with implicit phonological rules and pronunciation knowledge. LLMs, by contrast, struggle most in this category. Even o1-preview,



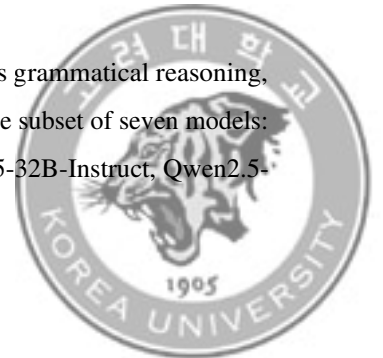
Language	Type	Model	Phonology	Morphology	Syntax	Semantics	Norm	Avg.
Korean	Open	Bllossom-8B	24.41	25.00	22.15	34.55	29.09	27.10
		SOLAR-v1.0-10.7B-Instruct	24.88	25.75	26.71	31.69	25.45	27.36
		KULLM-3-10.7B	21.60	26.49	26.03	29.35	28.18	26.64
		EEVE-v1.0-10.8B-Instruct	22.54	27.24	27.85	40.78	27.73	30.25
		EXAONE-3.5-7.8B-Instruct	24.88	30.22	32.19	43.64	31.36	33.60
		EXAONE-3.5-32B-Instruct	27.23	37.31	36.30	50.65	37.27	38.98
	Closed	HyperCLOVA-HCX-DASH-001	23.94	31.34	25.57	39.74	31.36	30.77
		HyperCLOVA-HCX-003	32.39	41.79	41.10	55.32	48.18	44.62
English	Open	Gemma-2-9B-Instruct	24.41	31.34	31.05	42.08	29.09	32.68
		Gemma-2-27B-Instruct	27.33	29.48	36.76	47.27	29.09	35.70
		Qwen2.5-7B-Instruct	23.00	30.97	34.02	44.42	25.00	33.27
		Qwen2.5-14B-Instruct	29.58	38.43	39.27	52.21	33.64	40.22
		Qwen2.5-32B-Instruct	26.29	36.19	41.78	62.86	35.45	43.04
		DeepSeek-R1-Distill-Qwen-14B	29.11	32.09	39.27	47.79	32.27	37.73
		DeepSeek-R1-Distill-Qwen-32B	36.15	45.52	40.41	61.56	30.00	44.55
		s1-32B	39.91	43.28	46.12	62.60	40.00	48.03
		Llama-3.1-8B-Instruct	24.41	24.25	23.97	36.36	26.82	27.62
		Llama-3-70B	24.41	32.46	34.47	45.71	27.73	34.58
	Closed	Llama-3.1-405B	37.56	43.28	47.03	62.34	35.00	47.18
		Gemini-1.5-flash	37.56	38.81	44.52	60.78	35.45	45.34
		Gemini-2.0-flash-exp	46.01	49.63	56.85	70.91	45.91	56.04
		Claude-3-haiku	21.13	34.33	35.62	44.68	30.91	34.97
		Claude-3.5-Sonnet	7.42	2.61	4.38	4.55	46.82	9.97
		GPT-3.5-turbo	21.13	27.61	27.40	31.43	25.45	27.30
		GPT-4o-mini	32.39	34.70	36.53	52.21	39.09	39.96
		GPT-4o	44.60	51.49	55.48	71.95	8.64	57.87
		o1-preview	71.83	79.48	80.14	89.35	79.09	81.04
	LLMs Avg.		31.33	37.08	39.00	51.36	35.71	40.24
	Human		66.70	56.95	64.75	70.84	54.34	63.04

Table 5.1: Zero-shot accuracy evaluation results on our benchmark.

despite its overall superiority, exceeds human performance in Phonology by only 5.13%, a substantially smaller margin than the 18% average gap observed across all categories. On the other end of the spectrum, the Norms category shows the smallest gap between humans and LLMs, likely because it relies heavily on rote knowledge such as correct spelling and loanword conventions.

Whether trends at the main category level are sufficient to fully characterize the linguistic differences between LLMs and humans remains an open question. To gain finer-grained insights, Sections 5.5.2 and 5.5.3 decompose the five main categories into 16 subcategories, offering a more nuanced picture of where LLMs excel and where they fall short.

Few-Shot Results. To examine whether exemplar-based context aids grammatical reasoning, we extend our evaluation to 1-shot and 5-shot settings for a representative subset of seven models: EEVE-v1.0-10.8B-Instruct, EXAONE-3.5-7.8B-Instruct, EXAONE-3.5-32B-Instruct, Qwen2.5-



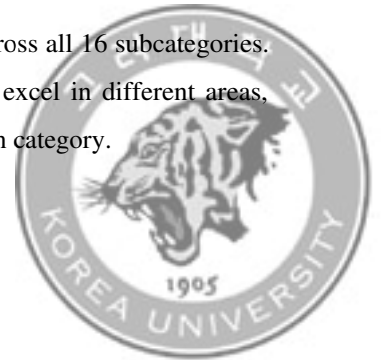
Language	Model		Phonology	Morphology	Syntax	Semantics	Norm	Avg.	Δ
Korean	EEVE-Instruct-10.8B-v1.0	0-shot	22.54	27.24	27.85	40.78	27.73	30.25	-
		1-shot	21.23	29.70	26.96	38.52	29.95	29.97	-0.28
		5-shot	23.15	30.23	25.48	37.95	24.64	29.00	-1.25
	EXAONE-3.5-7.8B-Instruct	0-shot	24.88	30.22	32.19	43.64	31.36	33.60	-
		1-shot	28.30	31.20	32.41	43.06	33.64	34.49	0.89
		5-shot	27.72	30.87	33.84	41.11	30.82	33.86	0.26
	EXAONE-3.5-32B-Instruct	0-shot	27.23	37.31	36.30	50.65	37.27	38.98	-
		1-shot	29.58	37.07	37.03	50.97	40.87	39.97	0.99
		5-shot	28.70	37.13	38.44	52.44	41.13	40.52	1.54
English	Qwen2.5-14B-Instruct	0-shot	29.58	38.43	39.27	52.21	33.64	40.22	-
		1-shot	26.42	36.84	37.10	53.30	34.56	39.26	-0.96
		5-shot	27.09	40.70	37.98	51.94	32.37	39.61	-0.61
	Qwen2.5-32B-Instruct	0-shot	26.29	36.19	41.78	62.86	35.45	43.04	-
		1-shot	28.30	38.72	43.55	63.85	33.18	44.16	1.12
		5-shot	30.05	39.53	47.36	64.72	32.85	45.78	2.74
	Claude-3.5-Sonnet	0-shot	47.42	52.61	64.38	74.55	46.82	59.97	-
		1-shot	48.45	54.50	65.22	75.12	48.58	60.99	1.02
		5-shot	49.01	56.37	66.99	75.54	49.51	62.23	2.26
	GPT-4o	0-shot	44.60	51.49	55.48	71.95	58.64	57.87	-
		1-shot	45.02	53.56	59.30	73.56	58.26	59.75	1.88
		5-shot	51.24	50.20	61.35	74.03	59.91	60.94	3.07

Table 5.2: Few-shot accuracy results on KoGEM. The table is organized into two groups: Korean-centric LLMs trained primarily on Korean data and English-centric LLMs trained primarily on English data. Δ denotes the difference between 0-shot and n-shot accuracy.

14B-Instruct, Qwen2.5-32B-Instruct, Claude-3.5-Sonnet, and GPT-4o. The results, summarized in Table 5.2, reveal a clear capacity threshold: models with at least 32B parameters consistently improved with more examples, whereas smaller models largely failed to capitalize on the additional context. This pattern suggests that the ability to leverage exemplar-based context for nuanced linguistic tasks is closely tied to model scale.

5.5.2 Further Analysis on Subcategory Results

Figure 5.2 breaks down the performance of LLMs and humans across all 16 subcategories. Examining individual subcategories reveals that LLMs and humans excel in different areas, underscoring the importance of fine-grained evaluation beyond the main category.



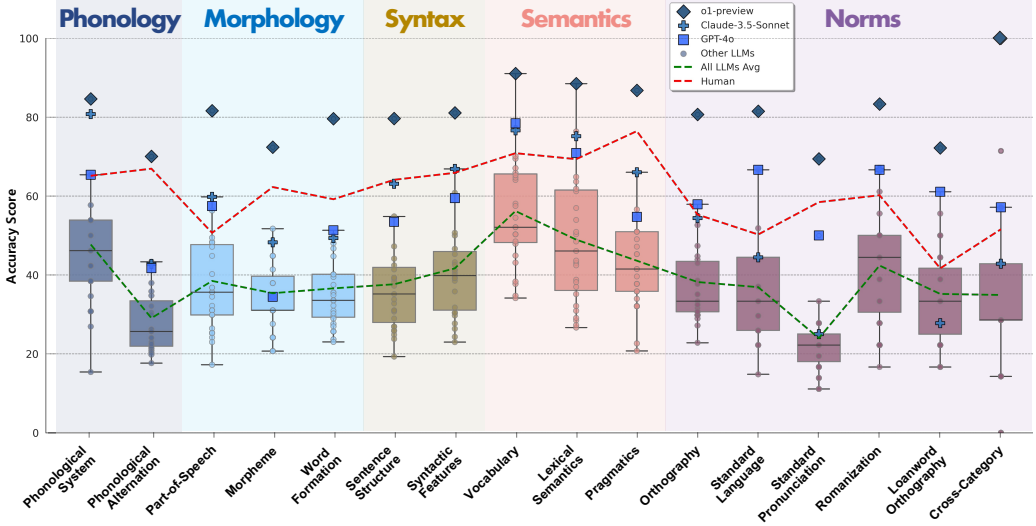


Figure 5.2: Performance comparison between LLMs and humans across 16 subcategories. Box plots show the score distributions of all LLMs, with the top three models highlighted individually as scatter plots. The green dashed line denotes the mean performance across all LLMs, and the red dashed line indicates human performance.

Phonology. Phonology is the category where the performance gap between LLMs and humans is most pronounced. In particular, the `Phonological Alternation` subcategory shows the largest disparity of any subcategory in KoGEM, with humans outperforming the average LLM by over 35%. We attribute this gap to humans’ capacity to draw on multimodal knowledge [72, 73]: humans intuitively recall the pronunciation of words [37], naturally integrating implicit phonological rules with textual input. LLMs, by contrast, are trained exclusively on text and lack exposure to spoken language, which fundamentally limits their ability to process phonological alternations.

Morphology. An interesting reversal emerges in the Morphology category: humans tend to underperform in the `Part-of-Speech` subcategory, where LLMs achieve their highest scores within Morphology. This may reflect differences in how each processes language. Humans rely on intuitive contextual understanding [74] rather than explicit grammatical classification, while LLMs detect part-of-speech patterns through data-driven training, giving them a relative advantage.



The picture reverses in the Morpheme subcategory, where the human-LLM gap is the widest within Morphology. Humans instinctively segment words into root morphemes and inflectional affixes [75], whereas LLMs process words as whole units or subword tokens, making them less adept at handling morphological variation.

Syntax. LLMs and humans display broadly similar tendencies in Syntax, both performing better on Syntactic Features than on Sentence Structure. This pattern likely reflects the relatively flexible word order of Korean compared to English [76]. For instance, both “나는 밥을 먹는다_{I rice eat}” and “나는 먹는다 밥을_{I eat rice}” are grammatically acceptable in Korean. Such flexibility resists simple rule-based codification and demands the kind of intuitive understanding acquired through real-world experience. Syntactic Features, on the other hand, involves well-defined grammatical rules such as tense, voice, and honorifics, which are more amenable to systematic learning

Semantics. The Semantics category reveals particularly striking contrasts across subcategories. The Pragmatics subcategory shows the largest gap within Semantics, with an average difference of 32.85% between humans and LLMs, while Vocabulary shows the smallest, at 14.66%. This contrast is largely explained by the nature of each task. Vocabulary items primarily test definitional knowledge and correct word usage, which can be acquired through memorization. Pragmatics, in contrast, requires sensitivity to speaker intent, the relationship between interlocutors, and the communicative purpose of an utterance [77], all of which are grounded in real-world conversational experience that LLMs cannot directly access.

Norms. Both humans and LLMs perform relatively poorly in the Norms category compared to other areas. Despite receiving standardized instruction on linguistic norms throughout high school, Koreans frequently deviate from standard language regulations in practice [78], and LLMs appear to have limited exposure to texts that explicitly describe these norms. Among the Norms subcategories, the largest human-LLM gap, approximately 34.49%, was observed in Standard Pronunciation, echoing the pattern seen in Phonology and suggesting that phonological factors, not merely normative ones, underlie this disparity.



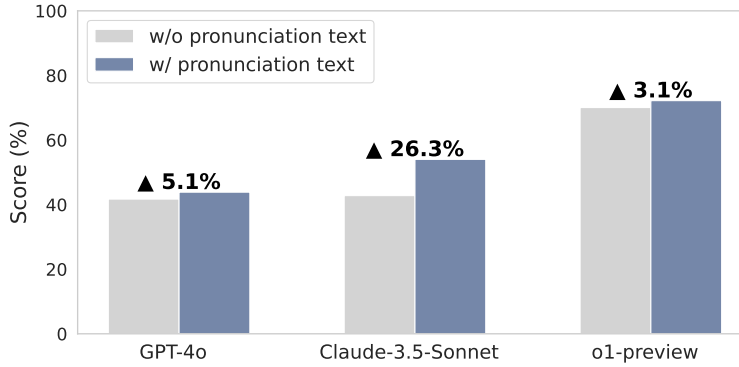


Figure 5.3: Performance comparison in Phonological Alternation with and without pronunciation text.

Summary:

- LLMs demonstrate a relatively narrow performance gap with humans on tasks that primarily require memorization and pattern recognition.
- However, they fall considerably short of human performance on tasks that demand experiential knowledge, such as Phonological Alternation, Morpheme, and Pragmatics, revealing a substantial gap between LLM and human linguistic competence.
- These findings suggest a fundamental limitation in the capacity of LLMs to ground linguistic knowledge in real-world experience.

5.5.3 Impact of Experiential Knowledge

Building on the subcategory-level analysis in the previous section, we identified subcategories that are relatively accessible to humans yet pose considerable challenges for LLMs. We examine whether incorporating experiential knowledge typically available to humans, such as pronunciation information for Phonological Alternation and morpheme information for Morpheme, can help bridge the performance gap between LLMs and humans.



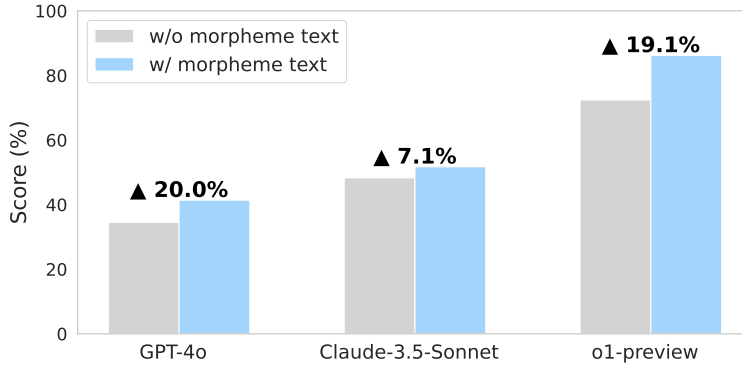


Figure 5.4: Performance comparison in Morpheme with and without morpheme text.

Phonological Alternation

To examine the role of experiential knowledge in the Phonological Alternation subcategory, we draw on the phenomenon of human subvocalization, whereby individuals mentally rehearse the pronunciation of a word upon reading it [79, 72]. To simulate this, we use g2pK⁴ to convert Korean graphemes into phonemes, generating a *pronunciation text* that is appended to each choice in the prompt. For example, “오늘은 하늘이 맑습니다 Today, the sky is clear” is rendered as “오느른 하느리 막쌌습니다 To-day, the skai iz kli-er”.

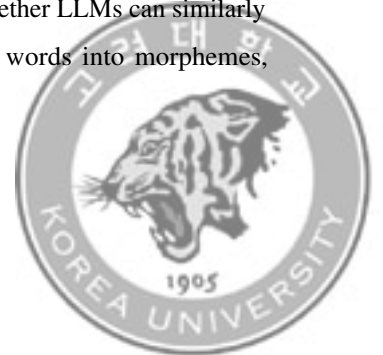
As shown in Figure 5.3, providing *pronunciation text* yields performance gains ranging from 3.1% to 26.3%, demonstrating that phonological information can substantially enhance LLMs’ competence in phonological tasks. That said, it remains unclear whether these gains reflect the activation of latent pronunciation knowledge already present in the model, or simply the provision of information the model otherwise lacks, a question we leave for future investigation.

Morpheme

In morphological processing, humans intuitively decompose words into their constituent morphemes, deriving meaning from root forms [80, 81]. To examine whether LLMs can similarly benefit from explicit morphological cues, we use Kiwi⁵ to segment words into morphemes,

⁴<https://github.com/Kyubyong/g2pK>

⁵<https://github.com/bab2min/Kiwi>



appending the resulting *morpheme text* to each choice in the prompt. For instance, “오늘은 하늘이 맑습니다 Today, the sky is clear” is segmented as { 오늘_{today} / 은_(auxiliary particle) / 하늘_{sky} / 이_{is} / 맑_{clear} / 습니다_(present progressive ending) }.

As shown in Figure 5.4, adding *morpheme text* leads to performance improvements of 7.1% to 20.0%, indicating that morphological awareness can meaningfully boost LLM performance on morphologically complex tasks. The magnitude of these gains further suggests that there remains considerable room for improvement in LLMs’ understanding of Korean morphology.

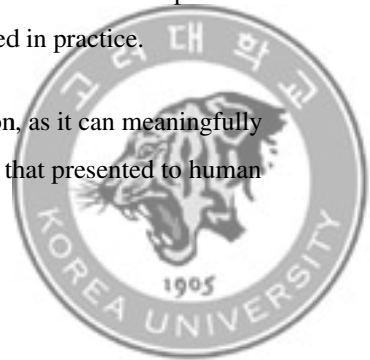
5.5.4 Limitations and Future Work

Throughout this section, we have examined the linguistic competence of LLMs and humans in Korean across five grammatical categories defined by theoretical linguistics, and further analyzed performance at the subcategory level, uncovering several findings not previously reported in the literature. Despite demonstrating the value of this work through experimental results and analysis, we acknowledge several limitations of the present study and discuss directions for future research.

Multilingual Extension. First, although KoGEM is built upon a fine-grained grammar taxonomy, it is currently limited to the Korean language. Extending our approach to other languages would require incorporating the unique linguistic knowledge of each target language. Nevertheless, since the core linguistic categories are universally applicable, the benchmark construction pipeline proposed in this work can serve as a foundation for multilingual extensions in future research.

Grammar Scope. Second, we categorized grammatical knowledge based on theoretical linguistics and standardized prescriptive grammar, and evaluated the linguistic competence of LLMs and humans accordingly. However, since prescriptive grammar focuses on how language ought to be used, a gap may exist between prescriptive norms and actual language use. Investigating this gap is important for understanding the linguistic competence of LLMs in processing real-world user input. Future work could address this by developing evaluation frameworks that incorporate a descriptive grammar perspective, capturing how language is actually used in practice.

Prompt Design. Third, prompt design is a potential source of limitation, as it can meaningfully affect model behavior. In this work, we aligned the prompt format with that presented to human



participants to ensure consistency in comparison. However, given prior findings that LLM performance is sensitive to prompt design, further exploration of prompt configurations that better elicit model capabilities may be warranted. Future work could investigate whether alternative prompt designs yield greater robustness or deeper insights into model performance.

Data Contamination. Finally, there is a concern that the performance advantage observed in definitional tasks may stem from data contamination between our benchmark and LLMs’ training corpora. Due to the lack of transparency in LLM training data, the possibility of contamination cannot be entirely ruled out. To mitigate this concern, we evaluated 27 diverse models to identify general trends across architectures and datasets, and the consistent performance patterns observed suggest that the results are unlikely to be driven solely by data contamination. Nevertheless, future work should apply quantitative contamination detection methods to verify this more rigorously.



5.6 Summary

In this chapter, we explored how linguistic knowledge can be leveraged to evaluate the linguistic competence of language models. Specifically, we constructed KoGEM, a Korean grammar evaluation benchmark, and used it to illuminate the strengths and weaknesses of LLMs in linguistic competence through systematic comparison with humans. Experiments on 27 open-source and closed-source LLMs revealed that while LLMs generally perform well on tasks requiring straightforward memorization or syntactic rule application, they struggle in categories that demand experiential knowledge grounded in the real world. Humans, by contrast, demonstrated relatively stronger performance in areas where LLMs fall short, such as phonological alternation and morpheme analysis, likely owing to the intuitive linguistic understanding and experiential knowledge accumulated through real-world interactions.

Further in-depth analysis confirmed that explicitly incorporating experiential knowledge can substantially enhance LLM performance in these challenging areas. Taken together, these findings underscore both the importance of fine-grained benchmarks like KoGEM for evaluating and advancing the multifaceted linguistic competence of LLMs, and the fundamental limitations of models trained solely on textual data. We hope that the insights offered by this work, through careful fine-grained comparison with humans, will contribute to advancing LLMs toward genuine linguistic competence.



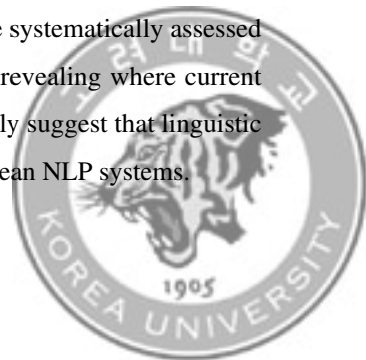
Chapter 6

Conclusion

Since the phonological system of Korean is different from that of Chinese, the Chinese characters that describe both Chinese and Korean cannot be used in communication. Accordingly, there are many among the general public who cannot express themselves in Chinese characters even though they have something they want to say. Feeling sorry for this, I have newly created twenty-eight letters because I want our people to learn them easily and use them conveniently everyday. [82]

— The preface of 『*Hunminjeongeum*』, King Sejong the Great, 1446

King Sejong recognized that Korean could not be fully captured by borrowed writing systems, and created Hangeul to give it a system of its own. This dissertation is grounded in a similar recognition: that Korean, as a language with its own unique structure and principles, requires NLP systems built with those principles in mind. Across three lines of studies, we have shown that the creation principles of Hangeul can serve as a productive foundation for word representation learning, that the phonemic nature of Korean provides a valuable signal for out-of-vocabulary word handling, and that the linguistic competence of LLMs in Korean can be systematically assessed through an evaluation framework grounded in theoretical linguistics, revealing where current models fall short of human-like performance. These findings collectively suggest that linguistic grounding is not merely beneficial, but essential for building robust Korean NLP systems.



6.1 Summary of Dissertation

In this section, we summarize the key findings from each chapter, with a focus on how linguistic grounding contributed to advancing Korean NLP systems.

Chapter 3. We introduced BTS units for Korean word embeddings, grounded in the creation principles of Hangeul, and demonstrated their effectiveness across various tasks.

- BTS-based embeddings achieved consistent performance gains across all evaluation tasks, including word analogy (Section 3.5.1), providing empirical support for incorporating the creation principles of Hangeul into the learning of word representations.
- Further analysis (Section 3.5.2) revealed that BTS units contribute to both semantic and syntactic aspects of word meaning, and prove particularly effective in handling previously unseen Korean OOV words, suggesting that they capture deeper linguistic regularities beyond character-level surface patterns.
- To our knowledge, this is the first study to draw on the creation principles of Hangeul as a foundation for Korean NLP system design, and the findings have since informed a line of follow-up research [35].

Chapter 4. While Chapter 3 demonstrated that the creation principles of Hangeul can serve as a productive foundation for word representation learning, Korean possesses another distinctive linguistic property that has been overlooked in Korean NLP community.

- Korean is a phonemic script in which letters and phonemes correspond systematically. Building on this insight, we proposed KOPL, a framework that improves Korean OOV word representations by explicitly incorporating the phonological information.
- Experiments across five Korean NLP benchmarks demonstrated consistent improvements over prior methods, suggesting that phonological grounding remains valuable yet largely underutilized for Korean OOV representation (Section 4.5.1). We further showed that KOPL integrates seamlessly with both static and contextual embedding models in a plug-and-play manner (Section 4.5.2).



- Taken together, these findings reinforce the argument that language-specific linguistic knowledge should be treated as a meaningful basis for NLP system design.

Chapter 5. Having established the value of linguistic grounding for Korean NLP systems in Chapters 3 and 4, we turn to the question of whether current LLMs actually possess such linguistic competence in Korean, and where they fall short.

- We constructed KoGEM, a Korean linguistic competence benchmark grounded in theoretical linguistics, comprising QA samples across five main categories and 16 subcategories based on prescriptive grammar, and used it to systematically evaluate 27 LLMs in comparison with human performance.
- The results revealed a consistent pattern. LLMs perform relatively well on tasks requiring memorization or syntactic rule application, but struggle in categories that demand experiential knowledge, precisely the areas where humans demonstrated stronger performance, suggesting a fundamental limitation of models trained solely on textual data.
- We further showed that explicitly providing linguistic cues such as phoneme or morpheme information can substantially improve LLM performance in challenging categories, though a persistent gap remains, pointing to the need to go beyond linguistic grounding, as discussed in Section 6.2.

6.2 Future Direction

In the previous section, we summarized the key findings of this dissertation, confirming that grounding NLP systems in the linguistic properties of Korean leads to meaningful improvements. Nevertheless, the findings also reveal that there remains much to be done. In this section, I discuss future directions that extend the scope of this work.

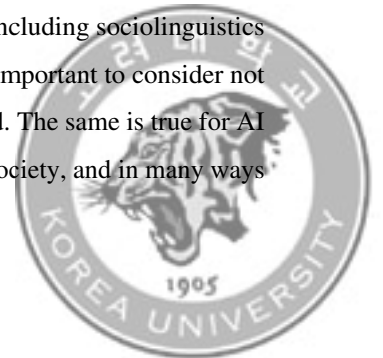
Linguistically Aware Model Architecture. The linguistic grounding approaches proposed in this dissertation have primarily focused on word representation and evaluation. However, exploring how these approaches can be applied to current LLMs and language agent is also



an important direction for future research. To this end, I believe that foundational research on developing tokenizers and language model architectures better suited for Korean should be pursued first. Specifically, unlike current subword tokenizers such as BPE that decompose units based on frequency, I consider the development of a Korean-specific tokenizer that jointly incorporates BTS units and morphological information to be a priority for future work. Furthermore, as demonstrated in this dissertation, phonological knowledge plays a crucial role in language understanding as evidenced by its effectiveness in OOV word handling, and thus further exploration of the relationship between phonological knowledge and tokenization [83] is required. Since advances in LLMs and agentic AI are built upon such foundational technologies, I intend to first explore these roots as the next step in my research.

Beyond Linguistic Grounding. The experimental results of KoGEM (Chapter 5) revealed that most LLMs perform considerably worse than humans in categories such as phonological alternation. One fundamental reason for this gap is that humans naturally integrate not only text but also phonetic elements such as sound when processing language. Models trained exclusively on text are inherently limited in this regard. This suggests that achieving true linguistic competence requires NLP systems to process language by incorporating information from multiple modalities, including acoustic signals. In this direction, future work will explore how text and speech can be jointly represented and modeled as a unified source of information, as well as how language-specific prosodic features such as rhythm, stress, and tone influence the performance of language models on linguistic tasks. I am particularly interested in investigating the capacity of AI to produce naturalistic prosody, and in building language-specific prosody datasets to support this line of research.

Aligning AI in Human Society This dissertation has primarily approached Korean NLP from the perspective of theoretical linguistics, which focuses on the systematic study of language itself. Linguistics, however, is a much broader field. It encompasses applied areas such as text linguistics and systemic functional linguistics, as well as interdisciplinary fields including sociolinguistics and psycholinguistics. To truly understand and process language, it is important to consider not only language itself, but also the people and societies in which it is used. The same is true for AI systems, which are becoming an increasingly important part of human society, and in many ways



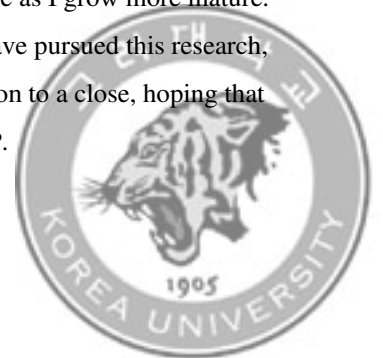
already are. Improving the linguistic performance of AI systems is not enough. It is essential to ground them in the implicit dimensions of human society that are expressed through language, including cultural norms, values, and preferences. I believe that research aimed at making these implicit dimensions explicit through natural language, and enabling AI to understand them, is of great importance. To this end, I am actively engaged in research on the design and collection of datasets for developing AI systems that are aligned with Korean social and cultural values, and I intend to contribute further to this direction.

6.3 Closing Remarks

The three studies presented in this dissertation, BTS, KOPL, and KoGEM, share a common motivation rooted in the unique structure and principles of Korean. Throughout my doctoral studies, I was frequently asked why Korean NLP deserves its own dedicated research, and why Korean linguistic knowledge should be incorporated into NLP systems. While I do not believe there is a single definitive answer, I would like to share my own perspective.

King Sejong recognized that Korean was fundamentally distinct from Chinese, and that this distinction demanded a writing system of its own. Had he lacked that understanding, Koreans might still be writing their language in borrowed characters. This would not necessarily have been wrong, yet something would have been lost. Certain aspects of Korean can be fully expressed solely through Hangeul. I believe the same principle applies to the machines we build to process language. A system designed for another language may inevitably miss something that a system built exclusively for Korean can capture. Every language has something of its own that deserves to be processed on its own terms, and doing so well requires systems built with that language in mind.

In this dissertation, I have proposed linguistic grounding as one way of moving toward that goal. My doctoral journey taught me that most research questions do not have one fixed right answer, and that my views on what I now believe to be true may change as I grow more mature. Perhaps the very fact that no clear answer exists is itself the reason I have pursued this research, and the reason I will continue to do so. With that, I bring this dissertation to a close, hoping that this is not an ending, but the beginning of a new chapter in Korean NLP.



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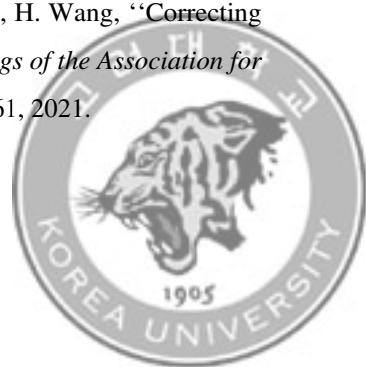
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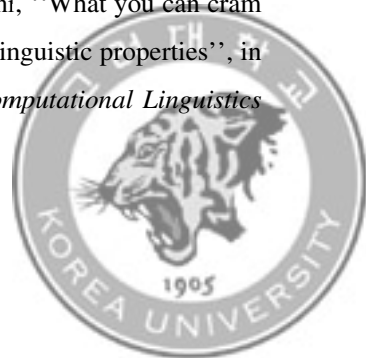
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Appendix A

Appendix: Subword Informed Word Representations for Korean

A.1 Full Comparison of jamo-level and BTS-level Decomposition

Table A.1 presents a detailed comparison of the jamo- and BTS-level subword decomposition for each Korean jamo letter.

A.2 Dataset Details

A.2.1 Corpus

We provide detailed information on three sources comprising the training corpus.

2020 Newspaper Corpus. The 2020 newspaper corpus comprises roughly 0.7M articles gathered from 35 news outlets, spanning eight domains: political, economic, social, life, IT/science, entertainment, sports culture, and beauty/health. In total, the corpus encompasses approximately 133M words and 9M sentences.





Consonant Letters		Vowel Letters	
jamo-level	BTS-level	jamo-level	BTS-level
ㄱ	ㄱ	ㅏ	ㅏ
ㄴ	ㄴ	ㅑ	ㅑ
ㄷ	ㄷ	ㅓ	ㅓ
ㄹ	ㄹ	ㅕ	ㅕ
ㅁ	ㅁ	ㅗ	ㅗ
ㅂ	ㅂ	ㅛ	ㅛ
ㅅ	ㅅ	ㅜ	ㅜ
ㅇ	ㅇ	ㅠ	ㅠ
ㅈ	ㅈ	ㅡ	ㅡ
ㅊ	ㅊ	ㅣ	ㅣ
ㅌ	ㅌ		
ㅍ	ㅍ		
ㅎ	ㅎ		
ㄱ	ㄱ	ㅏ	ㅏ
ㄴ	ㄴ	ㅑ	ㅑ
ㄷ	ㄷ	ㅓ	ㅓ
ㄹ	ㄹ	ㅕ	ㅕ
ㅁ	ㅁ	ㅗ	ㅗ
ㅂ	ㅂ	ㅛ	ㅛ
ㅅ	ㅅ	ㅜ	ㅜ
ㅇ	ㅇ	ㅠ	ㅠ
ㅈ	ㅈ	ㅡ	ㅡ
ㅊ	ㅊ	ㅣ	ㅣ
ㅌ	ㅌ		
ㅍ	ㅍ		
ㅎ	ㅎ		
ㅏ	ㅏ	ㅏ	ㅏ
ㅑ	ㅑ	ㅑ	ㅑ
ㅓ	ㅓ	ㅓ	ㅓ
ㅕ	ㅕ	ㅕ	ㅕ
ㅗ	ㅗ	ㅗ	ㅗ
ㅛ	ㅛ	ㅛ	ㅛ
ㅜ	ㅜ	ㅜ	ㅜ
ㅠ	ㅠ	ㅠ	ㅠ
ㅡ	ㅡ	ㅡ	ㅡ
ㅣ	ㅣ	ㅣ	ㅣ

Table A.1: Full decomposition mapping of all Korean letters at the jamo- and BTS- levels.

Korean Wikipedia. The Korean Wikipedia corpus consists of 0.4M documents with a total of 69M words and 7M sentences.

21st Century Sejong Corpus. The 21st Century Sejong Corpus was constructed between 1998 and 2007 as part of the national initiative known as the "21st Century Sejong Project." It encompasses written and spoken modern Korean texts, along with parallel corpora (e.g., Korean–English, Korean–Japanese), sourced from newspapers, magazines, books, and monologue/dialogue transcripts. We employ a filtered subset of this corpus comprising 31M words and 3M sentences.

A.2.2 Word Analogy

The Korean word analogy evaluation dataset [10] is organized into 10 categories, five targeting semantic features and five targeting syntactic features. Each category comprises 1K entries, with every entry consisting of four words designed to assess the semantic and syntactic properties captured by the learned word representations. Below, we describe the word analogy dataset introduced by Park et al. [10] by providing a representative example for each category.

Semantic :

- Capital-Country (Capt): Pairs of a country name and its capital city

베를린Berlin : 독일Germany = 서울Seoul : 대한민국Korea

- Male-Female (Gend): Pairs of corresponding male-female words

고모부uncle : 고모aunt = 남편husband : 아내wife

- Name-Nationality (Name): Pairs of a celebrity name and nationality

고흐Gogh : 네덜란드Netherlands = 링컨Lincoln : 미국USA

- Country-Language (Lang): Pairs of a country name and official language :

브라질Brazil : 포르투갈어Portuguese language = 일본Japan : 일본어Japanese language

- Miscellaneous (Misc): Various semantic features not included in the above four categories

개dog : 강아지puppy = 닭chicken : 병아리chick



Syntactic :

- Case: Pairs of a noun and the noun with case marker attached

여정_{journey} : 여정을_{journey+case(을)} = 마술사_{magician} : 마술사를_{magician+case(를)}

- Tense: Pairs of the present tense and past tense of a verb

돕다_{help} : 도왔다_{helped} = 사용하다_{use} : 사용했다_{used}

- Voice: Pairs of active voice and passive voice of a verb

거절했다_{rejected} : 거절당했다_{was/were rejected} = 먹었다_{ate} : 먹혔다_{was/were eaten}

- Verb ending form (Verb): Pairs of a verb and the verb with the ending form attached

사다_{buy} : 사면서_{buy+form(면서)} = 잡다_{grab} : 잡으면서_{grab+form(으면서)}

- Honorific (Honr): Pairs of morphological variation for verbs in Korean.

갔다_{went} : 가셨다_{went+honorific} = 이끌고_{lead} : 이끌고_{lead+honorific}

A.2.3 Word Similarity

The Korean adaptation of the WS-353 dataset [33] was made publicly available by [10]. It comprises 353 word pairs, each annotated by 14 native Korean speakers to evaluate the degree of similarity between the given word pairs.

A.2.4 Sentiment Analysis

The Naver Sentiment Movie Corpus (NSMC) dataset is composed of 150K training samples and 50K test samples, collected from the Naver Movies platform. Each sample includes a Korean-language movie review along with a binary sentiment label (0 for negative, 1 for positive). We construct our splits by sampling 100K training, 25K validation, and 25K test instances, ensuring a balanced label distribution across all partitions. Given that the original data source is inherently prone to noise, we apply a preprocessing step that retains only Korean characters, English letters, and numerical digits.



A.3 Word Analogy Results by N-gram Range

We report the experimental results of word analogy by n-gram range.

Single Decomposition Models



Figure A.1: SISG(stroke)

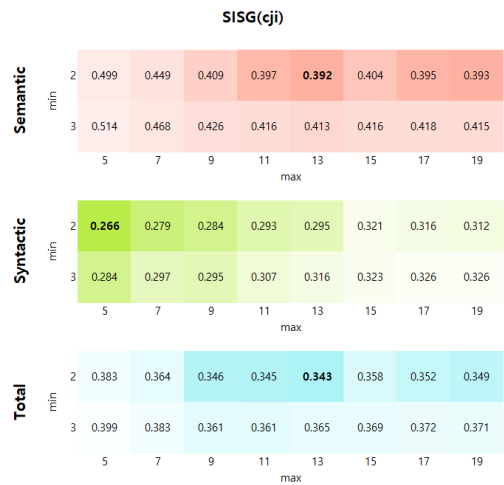


Figure A.2: SISG(cji)



Figure A.3: SISG(BTS)



Jamo-Integrated Models

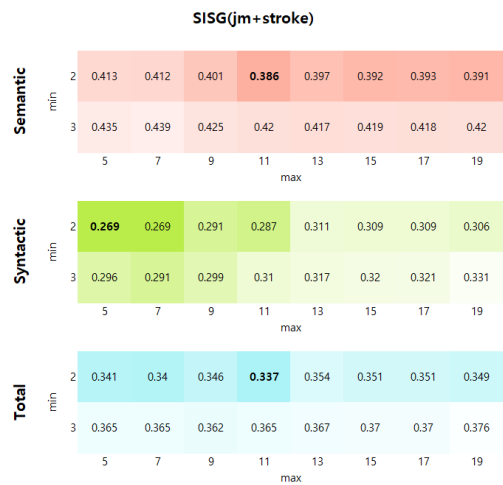


Figure A.4: SISG(jm+stroke)

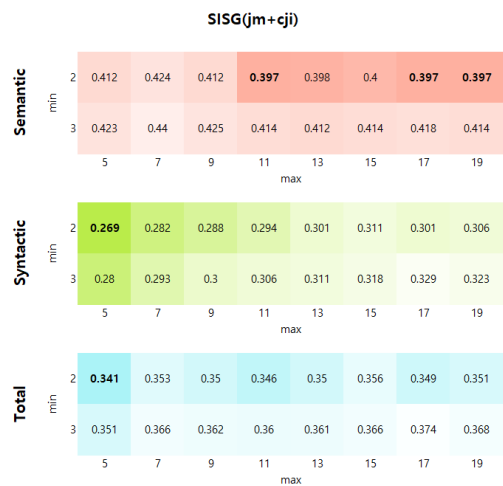


Figure A.5: SISG(jm+cji)



Figure A.6: SISG(jm+BTS)



Character(1-4)-Integrated Models



Figure A.7: SISG(ch4+stroke)



Figure A.8: SISG(ch4+cji)

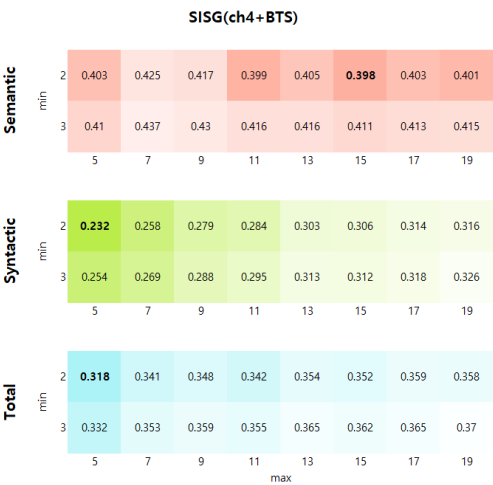


Figure A.9: SISG(ch4+BTS)



Character(1-6)-Integrated Models

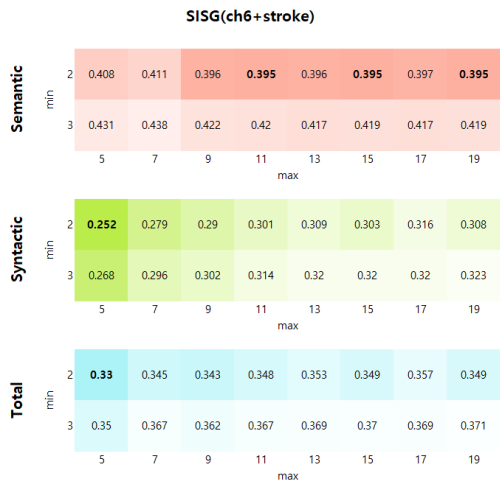


Figure A.10: SISG(ch6+stroke)



Figure A.11: SISG(ch6+cji)

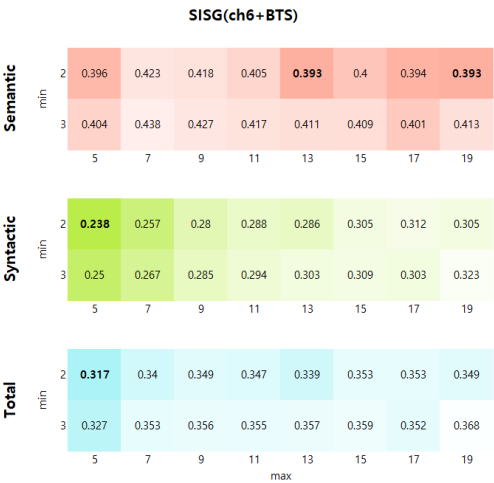


Figure A.12: SISG(ch6+BTS)



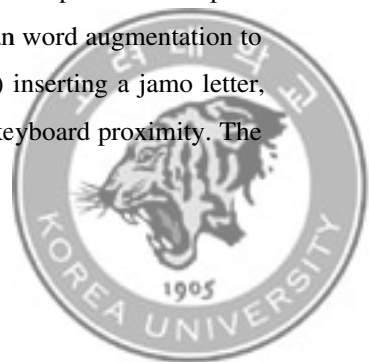
Appendix B

Phoneme-Guided Out-of-Vocabulary Word Representations for Korean

B.1 Details about the Baselines

We describe the implementation details of Korean-adapted versions of OOV models originally developed for English. A primary distinction between the Korean and non-Korean variants of the baselines lies in the input processing strategy for handling words across different languages. For MIMICK [38], each word is decomposed into its constituent jamo letters, and the resulting jamo sequence is used as the model input. For LOVE [39], we additionally incorporate morpheme-based subword tokenization [12] to obtain subword units for each word. For GRM [48], we substitute English WordPiece tokenization [84] with Korean morpheme-based subword tokenization [12] to segment Korean words into sub-units. This substitution is motivated by the agglutinative nature of Korean, in which words are formed through the combination of multiple morphemes.

A further distinction concerns the augmentation strategy used to construct positive word pairs for contrastive learning. Following [39], we adopt four types of Korean word augmentation to enrich training examples: (1) permuting the order of jamo letters, (2) inserting a jamo letter, (3) deleting a jamo letter, and (4) substituting a jamo letter based on keyboard proximity. The



synonym replacement and hard negative generation strategies proposed in [39] are excluded due to the absence of a large-scale lexical resource comparable to WordNet and a general-purpose text embedding model such as E5 [85], which are required for their implementation. Finally, we utilize pre-trained Korean word embeddings (i.e., $\mathbf{v}_w$ as defined in Section 4.3.1), derived from SISG_{jm} [10], a widely adopted Korean word embedding model, as the target representations for mimicking. All reported results are averaged over 5 independent runs with distinct random seeds, and all remaining experimental configurations follow the published settings of the respective baselines.



Appendix C

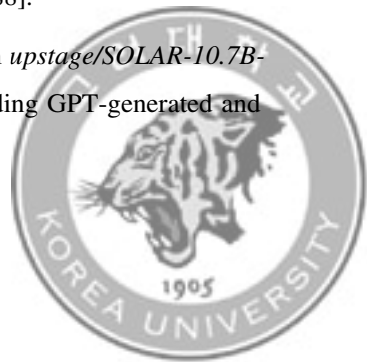
Evaluating Linguistic Competence of Language Models in Korean

C.1 Details of LLM Baselines

The zero-shot evaluation on KoGEM is conducted using a total of 27 LLMs, comprising eight Korean-centric model series trained primarily on Korean data and eleven English-centric model series trained primarily on English data, each described in turn below.

- **Korean-centric LLMs**

1. Bllossom-8B [86]: A bilingual language model for Korean and English built upon the open-source LLaMA3 architecture, with a focus on bridging knowledge representation across the two languages.
2. SOLAR-v1.0-10.7B-Instruct [87]: A 10.7 billion parameter LLM refined through a combination of supervised fine-tuning and direct preference optimization [88].
3. KULLM-3-10.7B [89]: An instruction-tuned model derived from *upstage/SOLAR-10.7B-v1.0* [87], trained on a mixture of Korean instruction data including GPT-generated and hand-crafted examples [90].



4. EEVE-v1.0-10.8B-Instruct [91]: A DPO-tuned variant of *yanolja/EEVE-Korean-10.8B-v1.0*,¹ which extends the Korean vocabulary of *upstage/SOLAR-10.7B-v1.0* [87].²
5. EXAONE-3.5-7.8B-Instruct [92]: A bilingual instruction-tuned model for English and Korean released by LG AI Research, designed to make effective use of system prompts.³
6. EXAONE-3.5-32B-Instruct [92]: A larger variant sharing the same architecture as EXAONE-3.5-7.8B-Instruct, differing only in the number of parameters.
7. HyperCLOVA-HCX-DASH-001 [93]: A speed- and cost-optimized derivative of HyperCLOVA-HCX-003, suited for less demanding tasks while retaining solid overall performance.
8. HyperCLOVA-HCX-003 [93]: The flagship model in NAVER’s HyperCLOVA X lineup, targeting complex and demanding tasks with high-quality output.

• English-centric LLMs

1. Gemma-2-9B-Instruct [94]: A 9 billion parameter decoder-only model from Google’s Gemma series, released with open weights for both pre-trained and instruction-tuned variants, suitable for tasks such as question answering, summarization, and reasoning.
2. Gemma-2-27B-Instruct [94]: The largest model in Google’s Gemma series with 27 billion parameters, achieving strong performance across NLP tasks through its scale and training approach.
3. Qwen2.5-7B-Instruct [66]: A 7 billion parameter multilingual model from Alibaba’s Qwen2.5 series, supporting a broad range of language understanding and generation tasks across English, Chinese, and other languages.
4. Qwen2.5-14B-Instruct [66]: A mid-scale variant of the Qwen2.5 series with 14 billion parameters, offering improved language capabilities and support for context lengths of up to 128,000 tokens.
5. Qwen2.5-32B-Instruct [66]: A 32 billion parameter model from Alibaba’s Qwen2.5 series, handling complex multilingual tasks with a context window of up to 128,000 tokens. All Qwen2.5 models are released under the Apache 2.0 license.

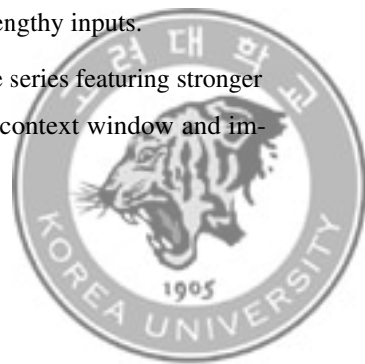
¹<https://huggingface.co/yanolja/EEVE-Korean-10.8B-v1.0>

²<https://github.com/axolotl-ai-cloud/axolotl>

³<https://huggingface.co/LGAI-EXAONE/EXAONE-3.5-7.8B-Instruct>



6. DeepSeek-R1-Distill-Qwen-14B [95]: A 14 billion parameter distilled model from DeepSeek, built on Qwen2.5-14B and fine-tuned on reasoning data produced by DeepSeek-R1 to strengthen performance on reasoning-intensive tasks.
7. DeepSeek-R1-Distill-Qwen-32B [95]: A 32 billion parameter distilled model from DeepSeek based on Qwen2.5-32B, fine-tuned with DeepSeek-R1 reasoning data and achieving competitive results across multiple benchmarks.
8. s1-32B [71]: A reasoning-focused model from Stanford and the University of Washington, built by fine-tuning Qwen2.5-32B-Instruct on 1,000 carefully curated samples using a budget forcing strategy, surpassing OpenAI's o1-preview on select benchmarks.
9. Llama-3.1-8B-Instruct [96]: An 8 billion parameter instruction-tuned model from Meta AI's Llama-3.1 series, optimized for a range of natural language understanding and generation tasks.
10. Llama-3-70B [96]: A 70 billion parameter model from Meta AI pre-trained on approximately 15 trillion tokens, with multilingual and multimodal capabilities and strong performance in coding and reasoning.
11. Llama-3.1-405B [96]: The largest model in Meta's Llama series at 405 billion parameters, demonstrating strong general knowledge and reasoning capabilities across benchmarks including MMLU-Pro and MMLU-redux.
12. Gemini-1.5-flash [97]: A multimodal model from Google DeepMind capable of processing text, images, audio, and video, developed for real-time interaction scenarios and integrated into multiple Google products.
13. Gemini-2.0-flash-exp [97]: An experimental release in the Gemini series offering improved speed and multimodal capabilities, including a live API for real-time audio and video, enhanced spatial understanding, and native tool integration.
14. Claude-3-haiku [98]: An Anthropic model designed for complex conversational tasks, supporting a context window of up to 200,000 tokens for processing lengthy inputs.
15. Claude-3.5-Sonnet [99]: An updated member of Anthropic's Claude series featuring stronger language understanding and generation capabilities, with a large context window and improved alignment with human preferences.



16. GPT-3.5-turbo [100]: An OpenAI model offering improved response accuracy over earlier GPT-3.5 variants, widely deployed for natural language understanding and generation applications.
17. GPT-4o-mini [101]: A compact and cost-efficient variant of GPT-4o from OpenAI supporting text, image, and audio inputs, optimized for low-latency real-time applications.
18. GPT-4o [101]: A multimodal model from OpenAI capable of processing and generating text, images, and audio, with fast response times and enhanced performance across non-English languages.
19. o1-preview [102]: An OpenAI model designed for complex problem-solving through extended internal reasoning prior to response generation, excelling in competitive programming, mathematics, and scientific tasks.

C.2 LLM Evaluation Details

We provide a comprehensive account of the evaluation setup adopted for our experiments, covering hardware configurations, hyperparameter choices, and the structure of the prompts fed into each zero-shot evaluation run.

C.2.1 Device Settings

Experiments involving open-source models are conducted using the Hugging Face library⁴ on local GPU infrastructure. This covers all Korean-centric LLMs, with the exception of the HyperCLOVA X series, together with several English-centric models including the Gemma, Qwen2.5, and DeepSeek-R1-Distill series, as well as Llama-3.1-8B. Depending on model size, inference is performed on either two RTX A6000 GPUs for models exceeding 20B parameters, or a single RTX A6000 GPU for smaller models. Closed-source models are queried through their respective official APIs.

⁴<https://huggingface.co/models>



C.2.2 Hyperparameters

All experiments are run with the temperature set to 0 or an effectively equivalent value (e.g., 1e-10) to promote reproducibility. Two exceptions apply: GPT-3.5-turbo does not expose temperature as a configurable parameter [100], and o1-preview similarly does not support this option. The random seed is held constant at 42 throughout all runs. Output length is capped at 200 tokens to control API costs. We verify that this constraint does not result in truncated responses, given that the prompts themselves already impose a 100-character limit on the expected output.

C.2.3 Prompt Designs for Main Results

The prompts employed in the main experiments fall into four categories, designated T1 through T4, each defined by a distinct combination of input fields drawn from *passage*, *question*, *paragraph*, and *choices*. To aid interpretation, a full English translation of the T1 prompt is included at the end of the examples below.

C.2.4 Statistics of Prompt Lengths

The average prompt length across all types is 534 characters, though a substantial proportion of instances exceed 1,000 characters owing to the inclusion of complete passages. This variation in length ensures that models are tested on their capacity to handle and integrate extended linguistic context.

- **T1** {*question+choices*}

[system]

다음은 한국어 언어 이해에 대한 객관식 문제입니다. 주어진 질문에 대한 정답으로 올바른 번호를 선택지에서 고르고, 그에 맞는 해설을 100자 내로 설명하십시오.

[user]

질문: 다음 선택지 1 부터 {4 or 5} 중 {*question*}

선택지: {*choices*}



- **T2** {*question+paragraph+choices*}

[system]

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[user]

설명: {*paragraph*}

질문: 다음 선택지 1 부터 {4 or 5} 중 {*question*}

선택지: {*choices*}

- **T3** {*passage+question+choices*}

[system]

다음은 한국어 언어 이해에 대한 객관식 문제입니다. 주어진 지문을 보고, 질문에 대한 정답으로 올바른 번호를 선택지에서 고르고, 그에 맞는 해설을 100자 내로 설명하시오.

[user]

지문: {*passage*}

질문: 다음 선택지 1 부터 {4 or 5} 중 {*question*}

선택지: {*choices*}

- **T4** {*passage+question+paragraph+choices*}

[system]

다음은 한국어 언어 이해에 대한 객관식 문제입니다. 주어진 지문과 설명을 보고, 질문에 대한 정답으로 올바른 번호를 선택지에서 고르고, 그에 맞는 해설을 100자 내로 설명하시오.

[user]

지문: {*passage*}

설명: {*paragraph*}

질문: 다음 선택지 1 부터 {4 or 5} 중 {*question*}

선택지: {*choices*}



- Translation of **T4**

[system]

The following is a multiple-choice question about Korean language comprehension. Based on the given passage and explanation, choose the correct number from the choices as the answer to the question and provide a corresponding explanation within 100 characters.

[user]

Passage: {*passage*}

Paragraph: {*paragraph*}

Question: Choose from options 1 to {4 or 5} for {*question*}

Choices: {*choices*}

C.2.5 Prompt Design for Addition of Pronunciation Experiment

Section 5.5.2 presents additional experiments on the phonological alternation task, in which pronunciation information is incorporated into the answer choices within the prompt. A representative example from this experiment is shown in Figure C.4.

C.3 Details of KoGEM

This section presents detailed information about KoGEM, covering the sources from which data were drawn, the preprocessing procedures applied, and the subcategories comprising the Korean grammar taxonomy.

C.3.1 Source Data

Below we describe the characteristics of each data source included in KoGEM, along with relevant copyright and licensing information.



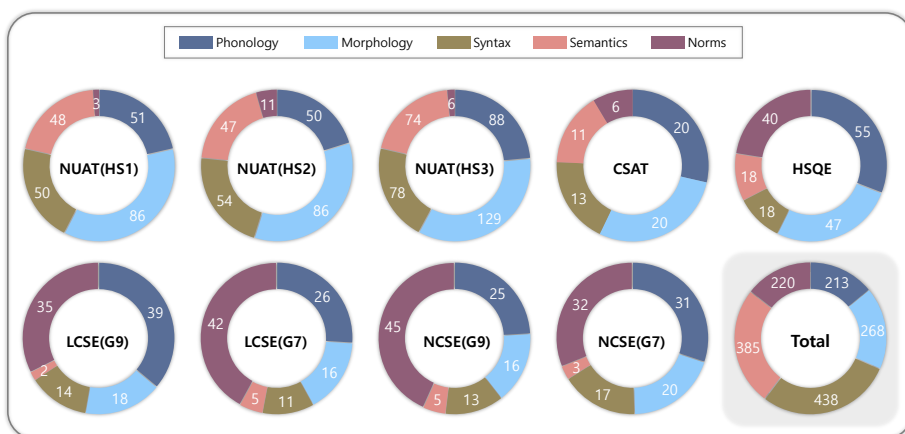


Figure C.1: Specific Statistics for Each Source Exam.

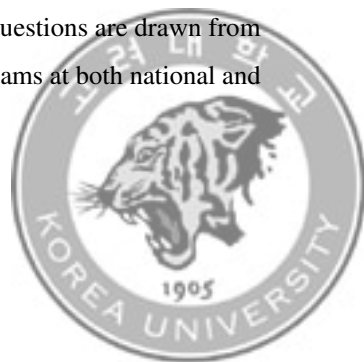
Detailed Description

- The College Scholastic Ability Test (CSAT) is a standardized examination administered by the Korea Institute for Curriculum and Evaluation, used as a criterion for university admission selection. It is held once annually.⁵
- The National United Achievement Test (NUAT) is a practice examination modeled on the CSAT format, organized by the Seoul Education Research and Information Institute. It is taken regularly by high school students across all grade levels, with four administrations per year for first- and second-year students and six for third-year students.⁶
- The High School Qualification Exam (HSQE) is an examination conducted by the Korea Institute for Curriculum and Evaluation to verify whether candidates meet the requirements for high school graduation. Only the Korean language section is used in this study. The exam takes place twice a year.⁷
- The Civil Service Exam (CSE) is administered by the Ministry of Personnel Management for the recruitment of national and local government employees. Grammar questions are drawn from the Korean language sections of the 7th- and 9th-grade recruitment exams at both national and

⁵<https://www.suneung.re.kr>

⁶<https://www.jinhak.or.kr>

⁷<https://www.kice.re.kr>



local levels. While additional types of CSEs exist, only those for which copyright clearance has been obtained are included. The exam is held once per year.⁸

License

The source examinations used in this study are made publicly available under the Public Copyright Free Use Permit Standard, known as the Korean Open Government License (KOGL) Type 1. This license permits unrestricted and free use of public works, including for commercial purposes, as well as modification and adaptation for the creation of derivative works.⁹ In addition to relying on this license, we have directly contacted each institution and received explicit permission for research use.

C.3.2 Data Preprocessing

Textual content is extracted from exam images using HyperCLOVA X OCR¹⁰, a process that frequently introduces errors including typos, fragmented characters, and inconsistent formatting. To address these issues and establish a uniform representation, we adopt HTML-based formatting conventions. The rules applied are as follows:

- **Underline.** Underlined text is wrapped in the `<u>` tag. Where the underlined portion contains a specific symbol such as ㉠, a corresponding symbol tag is used instead, e.g., `<㉠>`.
- **Boldface.** Text that requires emphasis is enclosed within the `<b>` tag.
- **Box.** Text intended to appear inside a box is surrounded by the `<box>` tag.
- **Section.** Content belonging to a common section is grouped under an appropriate section tag, such as `<가>`, `<A>`, `<예>`, or `<보기>`.
- **Table.** Individual rows are delimited by newline characters (`\n`). Within each row, the header column is marked with a colon (:), and remaining columns are separated by a slash (/).

⁸<https://www.gosi.kr>

⁹<https://www.kogl.or.kr/info/license.do>

¹⁰<https://clova.ai/hyperclova>



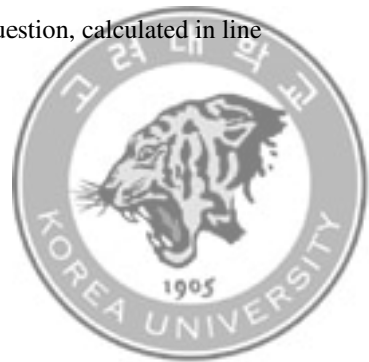
Exam	years	No. QA pairs
CSAT	1994-2023	70
NUAT	2006-2024	861
QE	2001-2024	178
CSE	2006-2023	415
Total		1,524

Table C.1: Source Data Collection Period

- **Abbreviated Expression.** Sequences of more than three dots are normalized to a fixed three-dot representation (. . .).
- **Bullet Point.** Original bullet symbols from the source image are preserved where possible. For visually similar symbols such as hollow middle circles or lower hollow circles, a single representative symbol is assigned to ensure consistency.
- **Text with Circle.** Text that appears circled in the original image is represented using parentheses.
- **Text with Rectangle.** Text enclosed within a rectangle in the original is represented using square brackets.

C.3.3 Data Categorization

Three annotators with backgrounds in Korean linguistics independently assigned each pre-processed item to one of 16 subcategories. The annotation process proceeded in two steps: each annotator first identified the overarching linguistic category, then selected the most appropriate subcategory. Label assignment was finalized through majority vote, and cases where consensus was not reached were settled via discussion among all three annotators. To maintain the focus of each item on a single linguistic phenomenon, questions requiring knowledge from more than three categories were discarded. Compensation was set at roughly \$0.15 per question, calculated in line with the minimum hourly wage standard.



Linguistic Category	Subcategory	Description
Phonology	Phonological System (PHS)	The system of Korean consonants and vowels and the principles of syllable formation.
	Phonological Alternation (PHA)	Phonological insertion, deletion, replacement, and combination phenomena in Korean.
Morphology	Part-of-Speech (POS)	The nine parts of speech in Korean (e.g., numeral, determiner, adverb, particle, exclamation).
	Morpheme (MOR)	The concept and types of morphemes (e.g., lexical morphemes, functional morphemes).
	Word Formation (WOF)	Word formation process such as compounding and derivation.
Syntax	Sentence Structure (STS)	The sentence structure and the components of a sentence (e.g., predicate, SVO word order, etc.).
	Syntactic Features (STF)	Various syntactic elements in sentences (e.g., tense, particles, negation, voice, honorifics, etc.).
Semantic	Vocabulary (VOC)	Rote knowledge of the dictionary definition of words.
	Lexical Semantics (LES)	Understanding semantic relationships of words (e.g., homonym, antonym, polysemy, etc.).
	Pragmatics (PRA)	Understanding discourse (e.g., context, reference, intentions of speakers, etc.).
Norms	Orthography (ORT)	Accurate rules of Korean spelling.
	Standard Language (STL)	Accurate knowledge of standard Korean.
	Standard Pronunciation (STP)	Accurate rules of Korean pronunciation.
	Loanword Orthography (LWO)	Korean orthography rules for loanwords.
	Romanization (ROM)	Rules for transcribing Korean into Roman script.
	Cross-Category (CRC)	Questions asking about two or more subcategories within the Norms category.

Table C.2: An overview of the five main categories, their subcategories, and corresponding descriptions. Abbreviations for each subcategory are shown in parentheses (e.g., PHS, PHA).



Language	Type	Model	Phonology		Morphology			Syntax		Semantics			Norms						Total
			PHS	PHA	POS	MOR	WOF	STS	STF	VOC	LES	PRA	ORT	STL	STP	LWO	ROM	CRC	
Korean	Open	llama-3-Korean-Blossom-8B	42.31	21.93	25.29	31.03	23.68	19.31	27.70	37.72	30.91	35.85	33.33	29.63	22.22	27.78	22.22	14.29	27.10
		SOLAR-10.7B-Instruct-v1.0	30.77	24.06	24.14	31.03	25.66	24.83	30.41	38.32	28.48	20.75	31.58	14.81	16.67	16.67	33.33	14.29	27.36
		KULLM3	15.38	22.46	31.03	31.03	23.03	25.86	26.35	34.13	26.67	22.64	32.46	29.63	13.89	33.33	22.22	28.57	26.64
		EEVE-Korean-Instruct-10.8B-v1.0	26.92	21.93	26.44	37.93	25.66	26.90	29.73	50.30	32.12	37.74	29.82	25.93	19.44	22.22	33.33	42.86	30.25
		EXAONE-3.5-7.8B-Instruct	38.46	22.99	34.48	31.03	27.63	28.97	38.51	50.30	38.79	37.74	35.09	29.63	22.22	38.89	22.22	28.57	33.60
		EXAONE-3.5-32B-Instruct	38.46	25.67	35.63	51.72	35.53	32.76	43.24	58.08	46.06	41.51	37.72	44.44	25.00	50.00	33.33	42.86	38.98
	Closed	HyperClova-HCX-DASH-001	46.15	20.86	32.18	31.03	30.92	26.90	22.97	44.91	36.97	32.08	35.09	37.04	25.00	22.22	22.22	28.57	30.77
		HyperClova-HCX-003	46.15	30.48	48.28	31.03	40.13	38.97	45.27	64.07	53.94	32.08	52.63	66.67	25.00	50.00	33.33	57.14	44.62
English	Open	Gemma-2-9B-Instruct	46.15	21.39	22.99	34.48	35.53	30.69	31.76	52.10	35.15	32.08	29.82	25.93	19.44	44.44	22.22	57.14	32.68
		Gemma-2-27B-Instruct	34.62	26.20	31.03	24.14	29.61	37.24	35.81	50.30	43.03	50.94	29.82	25.93	13.89	38.89	44.44	42.86	35.70
		Qwen2.5-7B-Instruct	42.31	20.32	29.89	37.93	30.26	31.03	39.86	52.10	38.18	39.62	30.70	25.93	11.11	22.22	16.67	28.57	33.27
		Qwen2.5-14B-Instruct	53.85	26.20	36.78	48.28	37.50	38.28	41.22	54.49	50.30	50.94	37.72	22.22	19.44	38.89	38.89	57.14	40.22
		Qwen2.5-32B-Instruct	42.31	24.06	44.83	24.14	33.55	42.41	40.54	64.67	64.85	50.94	38.60	37.04	16.67	55.56	27.78	42.86	43.04
		DeepSeek-R1-Distill-Qwen-14B	53.85	25.67	31.03	20.69	34.87	39.31	39.19	47.90	50.91	37.74	32.46	44.44	25.00	50.00	16.67	14.29	37.73
		DeepSeek-R1-Distill-Qwen-32B	65.38	32.09	48.28	31.03	46.71	39.31	42.57	67.07	61.21	45.28	28.95	25.93	27.78	50.00	38.89	0.00	44.55
		s1-32B	53.85	37.97	40.23	44.83	44.74	44.14	50.00	65.87	61.82	54.72	44.74	33.33	33.33	33.33	50.00	14.29	48.03
		Llama-3.1-8B-Instruct	42.31	21.93	17.24	31.03	26.97	23.79	24.32	43.11	29.09	37.74	27.19	22.22	25.00	33.33	27.78	28.57	27.62
		Llama-3-70B	38.46	22.46	35.63	24.14	32.24	31.03	41.22	50.30	42.42	41.51	30.70	29.63	19.44	22.22	27.78	28.57	34.58
	Closed	Llama-3.1-405B	57.69	34.76	49.43	44.83	39.47	47.24	46.62	65.27	63.64	49.06	32.46	44.44	19.44	44.44	55.56	42.86	47.18
		Gemini-1.5-flash	50.00	35.83	47.13	41.38	33.55	41.38	50.68	69.46	56.97	45.28	43.86	37.04	11.11	44.44	33.33	0.00	45.34
		Gemini-2.0-flash-exp	65.38	3.32	56.32	37.93	48.03	54.83	60.81	70.06	6.36	56.60	47.37	51.85	27.78	50.00	50.00	1.43	56.04
		Claude-3-haiku	46.15	17.65	36.78	31.03	33.55	33.79	39.19	48.50	43.64	35.85	30.70	37.04	16.67	50.00	33.33	28.57	34.97
		Claude-3.5-Sonnet	80.77	42.78	59.77	48.28	49.34	63.10	66.89	76.65	75.15	66.04	54.39	44.44	25.00	66.67	27.78	42.86	59.97
		GPT-3.5-turbo	30.77	19.79	25.29	27.59	28.95	25.86	30.41	34.73	27.27	33.96	22.81	29.63	22.22	27.78	44.44	14.29	27.30
		GPT-4o-mini	50.00	29.95	29.89	20.69	40.13	35.17	39.19	57.49	48.48	47.17	42.98	33.33	25.00	61.11	38.89	14.29	39.96
		GPT-4o	65.38	41.71	57.47	34.48	51.32	53.45	59.46	78.44	70.91	54.72	57.89	66.67	50.00	66.67	61.11	57.14	57.87
		o1-preview	84.62	70.05	81.61	72.41	79.61	79.66	81.08	91.02	88.48	86.79	80.70	81.48	69.44	83.33	72.22	100.00	81.04
		LLMs Avg.	47.72	29.06	38.48	35.38	36.60	37.64	41.67	56.20	48.96	43.61	38.21	36.90	23.97	42.39	35.18	34.92	40.24
		Human	65.13	66.93	50.67	62.28	59.19	64.12	65.88	70.86	69.35	76.46	55.30	50.25	58.46	60.20	41.68	51.58	63.04

Table C.3: Zero-shot evaluation results for the whole main linguistic category and subcategory.



다음은 한국어 문법에 관한 객관식 문제입니다.
주어진 지문과 설명을 보고, 질문에 대한 정답으로 올바른 것을 선택지에서 고르세요.
그리고 정답을 고른 근거(해설)를 간략히 작성하세요.

Translation
This is a multiple-choice question about Korean grammar.
Read the given passage and explanation and select the correct answer from the choices provided
Also, briefly write the reason (explanation) for choosing your answer.

※근거 작성 지침
질문에 대한 정답을 선택지에서 고른 후, 정답을 고른 근거를 작성해야 합니다.
근거를 작성할 때는 모호한 표현을 피하고, 자신이 정답을 선택한 근거를 작성해야 합니다.

아래는 근거 작성의 올바른 예시와 잘못된 예시입니다.

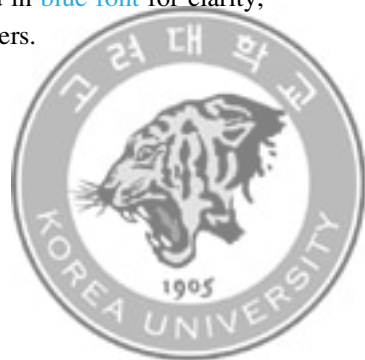
Translation
※ Guidelines for Writing Reasons
After selecting the answer to the question from the choices, you must write the reason for choosing
When writing the reason, avoid ambiguous expressions and write the basis for choosing your answer.
Below are examples of correct and incorrect ways to write reasons.

질문: 밑줄 친 단어의 표기가 바르지 않은 것은? 1) 놀잡 대처로 큰 화를 자초했다. 2) 나의 바램은 교사가 되는 것이다. (V) 3) 삼수갑산에 갈망정 이 일은 해야겠다. 4) 양반은 얼어 죽어도 껌불은 찌지 않는다.	Translation Question: Which underlined word is spelled incorrectly? 1. Due to a <u>delayed</u> response, a major problem occurred. 2. My <u>*dreame</u> is to become a teacher. (V) 3. I must do this <u>no matter what</u>. 4. A nobleman would rather freeze than warm up with <u>embers</u>.
잘못된 근거 작성 예시: - 바램은 틀린 표기이기 때문에 (X) - 왠지 그럴 것 같아서 (X) - 특별한 이유 없음, 그냥 (X)	Incorrect explanations: - *dreame is incorrect spelling. (X) - It just feels right. (X) - No particular reason, just because. (X)
올바른 근거 작성 예시: - 바램이 아니라 '바람'이 올바른 표기이기 때문이다. (O) 2번이 바른 표기가 아니라고 생각한 근거를 작성해야 합니다.	Correct explanations: - The <u>correct spelling is 'dream'</u>, not *dreame. You must provide a justification for <u>why the answer is incorrect</u>.

⑦ 실문관련 문의

다음 >

Figure C.2: The instructions given to crowdworkers prior to the human evaluation describe the overall evaluation procedure, the requirements of the task, and the conditions under which responses would be deemed invalid. English translations are provided in blue font for clarity, though these were not included in the materials presented to crowdworkers.



질문46. 문장 성분이 자연스럽게 호응하는 것은?

Translation

Question 46. Which sentence components naturally align?

- ☐ 우리는 각자의 꿈을 결코 이루어 낼 거야.
- ☐ 비록 힘들지라도 부디 용기를 잃지 말아라.
- ☐ 너희는 이번 경기에서 절대로 승리해야 해.
- ☐ 나는 오늘 학교 축제에서 노래와 춤을 추었어.

Translation

We will achieve our respective dreams.

Even if it's tough, don't lose courage.

You must absolutely win in this game.

I sang and danced at today's school festival.

질문46-1. 정답을 고른 근거(해설)를 10자 이상으로 간략히 작성하세요.

Translation

Question 46-1. Provide a brief rationale(explanation) of at least 10 characters for selecting your answer.

⑦ 설문관련 문의

다음 >

Figure C.3: A representative example from the human evaluation illustrates the format and structure of the administered test items. Each item consists solely of a *question* and *choices*, corresponding to prompt type T4 as defined in Section C.2.3. English translations have been appended in blue font to aid comprehension, though these were not visible to crowdworkers during the evaluation.



Example in Phonological System

Question: <보기>의 음운 카드를 활용하여 학습한 내용으로 적절하지 않은 것은? (Which of the following is incorrect regarding the phonemes in the paragraph?)

Paragraph:

- ☉ 음운: 말의 뜻을 구별해 주는 소리의 가장 작은 단위 ‘ㄱ’, ‘ㅇ’, ‘ㅏ’, ‘ㅑ’, ‘ㅓ’

(Phoneme: The smallest unit of sound that distinguishes meaning in a language. ‘ㄱ’/g/ or /k/, ‘ㅇ’/m/, ‘ㅏ’/eo/, ‘ㅓ’/o/)

Answer Choices:

- ① ‘ㅇ’, ‘ㅏ’, ‘ㅑ’, ‘ㅓ’를 차례로 사용하면 ‘먹_{ink}’이라는 단어를 만들 수 있군.
(By using ‘ㅇ_{mi}’, ‘ㅏ_{jeo}’, and ‘ㅓ_{sa}’ in sequence, you can create the word ‘먹_{meok}’.)
- ② ‘먹_{ink}’의 가운데소리인 ‘ㅏ’ 대신 ‘ㅓ’를 사용하면 새로운 단어가 되는 군.
(If you replace the medial vowel ‘ㅏ_{jeo}’ in ‘먹_{meok}’ with ‘ㅓ_{sa}’, it forms a new word.)
- ③ ‘목_{mok}’에서 보면 첫소리가 끝소리에, 끝소리가 첫소리에도 쓰일 수 있군.
(From ‘목_{mok}’ : 곱_{gom}’, we can observe that the initial sound can also serve as a final sound and vice versa.)
- ④ ‘먹_{ink}’ : 목_{neck}처럼 가운데소리는 첫소리의 오른쪽에 써야 하는군.
(Like in ‘먹_{meok}’ : 목_{neck}’, the medial vowel must be placed to the right of the initial consonant.)
- ⑤ ‘목_{neck}’ / 먹_{ink} / 곱_{bear} / 검_{sword}처럼 음운의 결합에 따라 의미가 다른 여러 단어를 만들 수 있군.
(Words such as ‘목_{neck}’ / 먹_{meok} / 곱_{gom} / 검_{geom}’ demonstrate that different combinations of phonemes can create words with different meanings.)

Example in Phonological Alternation

Question: <보기>를 참고할 때 동화의 양상이 다른 것은? (Based on the paragraph, which one shows a different type of assimilation?)

Paragraph:

- ☉ 순행동화: 뒤의 음운이 앞의 음운의 영향을 받아 그와 비슷하거나 같게 소리 나는 현상.
(Progressive assimilation : A phenomenon where a following phoneme is influenced by a preceding phoneme and becomes similar to or identical to it.)
예) 칼날[칼랄] / kalnal [kallal], 강릉[강농] / gangneung [gangneung]
- ☉ 역행동화: 앞의 음운이 뒤의 음운의 영향을 받아 그와 비슷하거나 같게 소리 나는 현상.
(Regressive assimilation : A phenomenon where a preceding phoneme is influenced by a following phoneme and becomes similar to or identical to it.)
예) 편리[펼리] / pyeoli [pyeolli], 까막눈 [까망눈] / kkamaknun [kkamangnun]

Answer Choices:

- ① 종로 (Jongno) ② 작년 (Jangnyeon) ③ 신라 (Silla) ④ 법률 (Beommul) ⑤ 국민 (Gukmin)

Figure C.4: Examples in the *Phonology* Category



Example in Part-of-Speech

Question: <보기>의 밑줄 친 단어를 바르게 분류한 것은? (What is the correct classification of the underlined words in the paragraph?)

Paragraph:

형용사와 관형사를 구별하는 기준의 하나로 '서술하는 기능'이 있다. 예를 들어, '동물원에는 큰 사자가 있다.' 에서 '큰' 은 '사자가 크다' 처럼 주어인 '사자가' 를 서술하는 기능을 하므로 형용사이다. 그러나 관형사는 그런 기능을 하지 못한다.
(One of the criteria for distinguishing adjectives and determiners is their 'descriptive function.' For instance, in the sentence 'There is a big lion in the zoo', 'big' functions descriptively by saying 'The lion is big', which means it is an adjective. However, determiners do not perform such functions.)

- a. 정원에 아름다운 꽃이 피었다. (A beautiful flower bloomed in the garden.)
- b. 웬 말이 그렇게 많은지 모르겠다. (I don't know why there's so much talking.)
- c. 수리를 하고 나니 새 가구가 되었다. (After the repair, it became a new piece of furniture.)
- d. 모여 있던 모든 사람들이 일제히 나를 쳐다봤다. (All the people gathered suddenly stared at me.)
- e. 그의 빠른 일처리가 사람들을 만족스럽게 하였다. (His quick work satisfied the people.)

Answer Choices:

	형용사(adjective)	관형사(determiners)
①	a, c	b, d, e
②	a, e	b, c, d
③	b, d	a, c, e
④	a, c, e	b, d
⑤	b, c, d	a, e

Example in Morphemes

Question: 의존 형태소이면서 실질 형태소인 것만으로 묶인 것은? (Which of the following consists only of dependent morphemes that are also lexical morphemes?)

Paragraph:

- ㉠ 영희는 책을 집에 놓고 학교에 갔다. (Yeonghee put the book at home and went to school)

Answer Choices:

- ① 놓-, 가- (put, go) ② -고, -ㅁ (and, past tense) ③ 영희, 책, 집 (Yeonghee, book, home) ④ -는, -을, -에 (subject marker, object marker, locative marker)

Example in Word Formation

Question: 어휘의 구성이 나머지와 다른 것은? (Which of the following has a different lexical composition from the others?)

Answer Choices:

- ① 참숯 (charcoal) ② 헛웃 (cast-off) ③ 풋과일 (unripe fruit) ④ 개살구 (wild apricot)

Figure C.5: Examples in the *Morphology* Category



Example in Sentence Structure

Question: <보기>의 (a)~(c)를 이해한 내용으로 적합하지 않은 것은? (Which sentence has proper grammatical concord?)

Paragraph:

- (a) 그는 위기를 좋은 기회로 삼았습니다. (He considered the crisis a good opportunity.)
- (b) 바다가 눈이 부시게 파랗다. (The sea is blue to the extent that it blinds the eyes.)
- (c) 동주는 반짝이는 별을 응시했다. (Dongju gazed at the twinkling stars.)

Answer Choices:

- ① (a)의 '삼았다'는 주어 이외에도 두 개의 문장 성분을 필수적으로 요구하는군. ('삼았다'_{considered}' in (a) requires two more sentence elements in addition to the subject.)
- ② (b)의 '바다가'와 '눈이'는 각각 다른 서술어의 주어이군. ('바다가'_{the sea}' and '눈이'_{eyes}' in (b) are the subjects of different predicates.)
- ③ (c)의 '별'은 안긴문장의 목적어이면서 안은문장의 목적어이군. ('별'은'_{stars}' in (c) is the object of the embedded sentence and the object of the enclosed sentence.)
- ④ (a)의 '좋은'과 (c)의 '반짝이는'은 안긴문장의 서술어이군. ('좋은'_{good}' in (a) and '반짝이는'_{twinkling}' in (c) are predicate phrases in the embedded sentence.)
- ⑤ (b)의 '눈이 부시게'와 (c)의 '반짝이는'은 수식의 기능을 하는군. ('눈이 부시게'_{blinds the eyes}' in (b) and '반짝이는'_{twinkling}' in (c) function as modifiers.)

Example in Syntactic Features

Question: 밑줄 친 말의 쓰임이 옳지 않은 것은? (Which of the following usages of the underlined word is incorrect?)

Answer Choices:

- ① 그는 아까운 능력을 쓰히고 있다. (He is wasting his precious talent.)
- ② 음식물 쓰레기를 쓰혀서 거름으로 만들었다. (Food waste was decomposed to make compost.)
- ③ 나는 이제까지 부모님 속을 쓰혀 본 적이 없다. (I have never been wasted my parents' peace of mind.)
- ④ 그들은 새로 구입한 기계를 창고에서 쓰히고 있다. (They are wasting the newly purchased machine in the warehouse.)

Figure C.6: Examples in the *Syntax* Category



Example in Vocabulary

Question: <보기>의 빈 칸에 공통적으로 들어갈 단어로 적절한 것은?
(Which word is appropriate to fill in the blanks in all the sentences below?)

Paragraph:

- 사람들은 그를 천재라고 (). / (People _____ him as a genius.)
- 복은 또 다른 복을 (). / (Luck _____ another stroke of luck.)
- 그는 속으로 패재를 (). / (He _____ inwardly with delight.)

Answer Choices:

- ① 말하다 (speak) ② 부르다 (call) ③ 여기다 (consider) ④ 외치다 (shout) ⑤ 생각하다 (think)

Example in Lexical Semantics

Question: 밑줄 친 '의'의 앞말과 뒷말의 의미 관계를 잘못 파악한 것은?
(Which of the following incorrectly identifies the semantic relationship between the underlined "of" and the words it connects?)

Answer Choices:

- ① 어머니의 신발이 많이 낡았다. / [소유주와 소유물] (The shoes of my mother are very worn out. / [Owner and Possession])
 ② 오늘 나의 짝은 선행상을 받았다. / [속성과 대상] (The partner of mine received the Good Conduct Award today. / [Attribute and Object])
 ③ 이 일은 선생님의 충고를 따라야겠다. / [주체와 행위] (This matter requires following the advice of the teacher. / [Subject and Action])
 ④ 졸업한 선배들에게 축하의 박수를 보내자. / [목표와 수단] (Let's give a round of applause of congratulations to the graduates. / [Goal and Means])
 ⑤ 우리 국민의 절반이 축구 중계방송을 시청했다. / [전체와 부분] (Half of the people of our nation watched the soccer broadcast. / [Whole and Part])

Example in Pragmatics

Question: <보기1>의 밑줄 친 부분의 예를 <보기2>에서 고른다고 할 때, 가장 적절한 것은?
(When selecting an example of the underlined part of <Paragraph 1> from <Paragraph 2>, which one is the most appropriate?)

Paragraph 1:

발화는 발화자의 어떤 의도를 담고 있다. 따라서 발화자가 상대방(청자)에게 무엇인가를 요구할 때, 일반적으로 명령문을 사용하여 발화자의 의도를 직접 드러낸다. 하지만 담화 상황에 따라 발화자가 요구하는 바를 평서문을 통해 상대방에게 간접적으로 표현하거나 의문문을 통해 상대방에게 간접적으로 표현할 수도 있다.
(An utterance carries a certain intention of the speaker. Therefore, when the speaker makes a request to the listener, a command sentence is typically used to express the speaker's intention directly. However, depending on the discourse situation, the speaker may indirectly express the request through a declarative sentence or indirectly express it through an interrogative sentence.)

Paragraph 2:

- 모임에서 만나 둘이 이야기를 하는 상황 (A situation where two people are talking at a meeting)
 남자 A: (a)저는 ○○ 고등학교에 다닙니다. (Man A: I attend ○○ High School.)
 남자 B: 그 학교는 어디에 있나요? (Man B: Where is that school located?)
- 병원에서 의사가 환자를 진료하는 상황 (A situation where two people are talking at a meeting)
 의사: (b)예전보다 많이 좋아지셨네요. (Doctor: You've improved a lot compared to before.)
 환자: 전부 의사 선생님 덕분입니다. (Patient: It's all thanks to you, doctor.)
- 개학 후 교사가 학생들을 처음 대면한 상황 (A situation where a teacher meets students for the first time after the semester begins)
 교사: (c)여러분, 많이 보고 싶었어요. (Teacher: Everyone, I've missed you a lot.)
 학생: 선생님, 저희도 그래요. (Students: We've missed you too.)
- 귀가한 아들이 어머니에게 말하는 상황 (A situation where a son speaks to his mother after returning home)
 아들: (d)엄마, 배가 너무 고파요. (Son: Mom, I'm so hungry.)
 엄마: 그래, 금방 차려 줄게. (Mom: Okay, I'll prepare something right away.)
- 여행객이 아름다운 경치를 보고 있는 상황 (A situation where a traveler is admiring a beautiful view)
 여행객 A: (e)이곳은 정말 아름답습니다. (Traveler A: This place is truly beautiful.)
 여행객 B: 그래요, 정말 아름답네요. (Traveler B: Yes, it really is stunning.)

Answer Choices:

- ① (a) ② (b) ③ (c) ④ (d) ⑤ (e)

Figure C.7: Examples in the *Semantics* Category



Example in Orthography

Question: 다음 중 한글 맞춤법에 맞는 단어들로만 묶인 것은? (Which group contains words that all adhere to Korean orthography rules?)

Answer Choices:

- ① 머릿방 - 핏기 - 섯방 - 공간 (Attic - Glow on the face - Small rented room - Storeroom)
- ② 넓따랄다 - 높따랄다 - 굵직하다 - 짹막하다 (Wide - Tall - Thick - Short)
- ③ 넘어지다 - 허여지다 - 흩어지다 - 버러지다 (Fall down - Turn pale - Scatter - Break apart)
- ④ 합격률 - 규율 - 선율 - 실패율 (Pass rate - Discipline - Melody - Failure rate)

Example in Standard Language

Question: <보기>의 설명에 해당하지 않는 것은? (Which of the following does not match the explanation in the passage?)

Paragraph: 우리말에서는 뜻이 같으면서 형태가 다른 낱말들이 있을 때, 그 쓰임의 범위 차이가 크게 나지 않는다면 모두 표준어로 삼고 있다. 가령, ‘신’과 ‘신발’은 쓰임의 범위가 비슷하므로 모두 표준어이다. 이를 가리켜, ‘복수 표준어’라 한다. (In Korean language, when there are words with the same meaning but different forms, if the range of their usage does not differ significantly, both are recognized as standard words. For example, ‘shoe’ and ‘shoes’ have similar usage ranges, so both are considered standard. This is referred to as ‘multiple standard words.’)

Answer Choices:

- ① 천둥 / 우레 (Thunder) ② 나귀 / 당나귀 (Donkey) ③ 옥수수 / 강냉이 (Corn) ④ 자물쇠 / 자물통 (Lock) ⑤ 선머슴 / 풋머슴 (Tomboy)

Example in Standard Pronunciation

Question: 밑줄 친 부분의 발음이 표준 발음법에 맞는 것은? (Which of the underlined parts matches the standard pronunciation rules?)

Answer Choices:

- ① 보리의 생장에는 겨울철 밭기[밭끼]가 중요하다. (The growth of barley requires winter trampling.)
- ② 한글 자모 순서에서 ‘ㄴ’ 다음에는 ‘ㄷ’이 [디그디] 온다. (In the order of Korean consonants, ‘ㄴ’ is followed by ‘ㄷ’.)
- ③ 당시 학계에는 일원론보다는 이원론 [이: 원논]이 우세하였다. (In the academic field at the time, dualism was more dominant than monism.)
- ④ 오전에 맑다[말따]가 오후에 차차 흐려져 밤늦게 비가 오겠습니다. (In the morning, it will be sunny and gradually become cloudy in the afternoon and rain late at night.)

Example in Romanization

Question: 다음 중 로마자 표기법에 맞는 것은? (Which one is the correct roman script?)

Answer Choices:

- ① 북음밥 fried rice / Bokkeumbap ② 동래 Dongnae (The name of local place) / Dongrae ③ 벚꽃 cherry blossoms / beotkkot ④ 식혜 sweet rice punch / shikhye

Example in Loanword Orthography

Question: 밑줄 친 외래어의 표기가 바른 것은? (Which of the underlined foreign words is correctly written?)

Answer Choices:

- ① 신나는 음악을 듣고 있으니 엔돌핀이 용솟음치는 듯하다. (Listening to exciting music feels like endorphins are surging.)
- ② 축제 기간에 진행되는 행사들은 팜플렛을 통해 소개되었다. (Events during the festival were introduced through pamphlets.)
- ③ 이번 여행에서 가장 기억에 남는 것은 야외에서의 바베큐 파티였다. (The most memorable part of this trip was the outdoor barbecue party.)
- ④ 자동차에 문제가 있는 것 같아 갓길에 정차하여 차의 보닛을 열어 보았다. (Thinking there might be an issue with the car, I stopped on the street and opened the car's bonnet.)

Example in Cross-Category

Question: 한글 맞춤법 및 표준어 규정에 맞게 쓴 것은? (Which one is correctly written according to Korean orthography and standard language rules?)

Answer Choices:

- ① 철수가 문제의 답을 잘못 알아맞했다. (Chulsoo guessed the answer to the question incorrectly.)
- ② 작년에 이 학교는 취업률이 매우 높았다. (Last year, this school had a very high employment rate.)
- ③ 교배를 시키려고 튼튼한 숫놈을 사왔다. (They bought a strong male to breed.)
- ④ 그건 막을래야 막을 수가 없는 재난이었다. (That was a disaster that couldn't be stopped even if one tried.)

Figure C.8: Examples in the *Norms* Category

